PolyKybd
RoHS Compliance Appendix

Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

CE Technical File — Restriction of Hazardous Substances evidence for all populated components and the bare printed circuit board.

Prepared: 2026-07-12

Parts → Certificate Reference

Sec.	Designators	Part / Value	Qty	Manufacturer	Certificate document
1	C1_1-C1_36	2.2uF 0603 25V	36	Samsung	2304140030_Samsung-Electro-Mechanics-CL05A105KA5NQNC_C52923.pdf
	C2,C3_1-C3_36,C5_1-C5_36,C6_1-C6_36,C24,C25,C30-C32	1uF 0402 16V	114	Samsung	
	C2_1-C2_36	2.2uF 0603 16V	36	Samsung	
	C3,C8	27pF 0402 16V	2	Samsung	
	C4,C10	4.7uF 0805 25V	2	Samsung	
	C5,C11	6.8pF 0603 50V	2	Samsung	
	C6,C12-C14,C26	10uF 0805 16V	5	Samsung	
	C9,C15-C23	100nF 0402 16V	10	Samsung	
2	C4_1-C4_36,C33,C34	4.7uF 3216 25V Tantalum	38	Vishay	vishay-rohs-20250901.pdf
3	D1	Red 0805 C84256	1	NationStar	2409272203_Foshan-NationStar-Optoelectronics-NCD0805R1_C84256.pdf
4	D1_1-D1_36,D12	1N4148WSX	37	Shenzhen JingYang	2405091035_Shenzhen-JingYang-1N4148WSX_C6423741.pdf
5	D2	1N5819WS	1	Hottech	ROHS3HOTTECH.pdf
6	D5	Green 0805 C2297	1	Hubei KENTO	C2297.pdf
7	D6	Yellow 0805 C2296	1	Hubei KENTO	1806151129_Hubei-KENTO-Elec-KT-0805Y_C2296.pdf
8	FB1,FB2	GZ2012D601TF	2	Sunlord	2310301640_Sunlord-GZ2012D601TF_C1017.pdf
9	J1-J36	FH34SRJ-14S-0.5SH	36	JUSHUO	FH34SRJ-14S-0.5SH(50)_CL0580-1252-8-50_SpecSheet_0000414509.pdf
10	J37	FH34SRJ-12S-0.5SH	1	Hirose	FH34SRJ-12S-0.5SH(99)_CL0580-1253-0-99_SpecSheet_0000414526.pdf
11	J39	FH12-30S-0.5SH	1	Hirose	FH12-30S-05SH.pdf
12	L1,L2	2.2uH	2	SXN	2108131530_SXN-Shun-Xiang-Nuo-Elec-SMMS0420-2R2M_C133191.pdf
13	LED1-LED36	XL-3030RGBC	36	Xinglight	2402181502_XINGLIGHT-XL-3030RGBC-WS2812B_C5349958.pdf
14	R1,R2	22 0603	2	Yageo	2304140030_YAGEO-RC0603JR-0710ML_C141675.pdf
	R1_1-R1_36,R15,R20,R25	10k 0402	39	Yageo	
	R3,R4,R7,R9	5.1K 0402	4	Yageo	
	R5,R6,R16,R19,R21	1k 0402	5	Yageo	
	R8	5.6K 0402	1	Yageo	
	R10,R11,R30	390k 0603 1%	3	Yageo	
	R12,R17	150k 0603	2	Yageo	
	R13	12k 0603	1	Yageo	
	R14	680k 0603	1	Yageo	
	R18	10M 0603	1	Yageo	
	R22,R23	27 0603	2	Yageo	
	R24	2M2 0603	1	Yageo	
15	SW2,SW3	EVQPUC02K	2	Panasonic	panasonic_rohs_2ma_2000379.pdf
16	SW_K_1-SW_K_36	Kailh CPG151101S11-16	36	Kailh	2208291600_Kailh-CPG151101S11-16_C5156480.pdf
17	U1,U2	TLV62569DBVR	2	TI	TexasTLV62569DBVR.pdf
18	U3	SY6280AAC	1	Silergy	ROHS-2024-SilergyCorp.pdf
19	U4-U8	74HC595BQ	5	Nexperia	Nexperia - Statement on RoHS 20241008.pdf
	U12-U25	74AHC1G125GW,125	14	Nexperia	
20	U9	W25Q64JVXGIM	1	Winbond	winbond-W25Q64JV-rohs.png
21	U10	RP2040	1	Raspberry Pi	rp2040-rohs.png
22	U11,U26	TPD4E05U06DQAR	2	MSKSEMI	C2836386-TPD4E05U06DQAR-MS.pdf
23	USB1,USB2	TYPE-C-31-M-12	2	Hroparts	2410010003_Korean-Hroparts-Elec-TYPE-C-31-M-12_C165948.pdf
24	Y1	7M12000044	1	TXC	TXC-7M12000044.pdf

Sec.	Designators	Part / Value	Qty	Manufacturer	Certificate document
25	PCB / bare board	Printed circuit board (ENIG & lead-free HASL finish); incl. exposed copper of mounting holes & test points	-	JLCPCB	rohs-certificate-of-conformity.pdf

Table of Contents

Section 1
Samsung

Designators	Part / Value	Qty
C1_1-C1_36	2.2uF 0603 25V	36
C2,C3_1-C3_36,C5_1-C5_36,C6_1-C6_36,C24,C25,C30-C32	1uF 0402 16V	114
C2_1-C2_36	2.2uF 0603 16V	36
C3,C8	27pF 0402 16V	2
C4,C10	4.7uF 0805 25V	2
C5,C11	6.8pF 0603 50V	2
C6,C12-C14,C26	10uF 0805 16V	5
C9,C15-C23	100nF 0402 16V	10

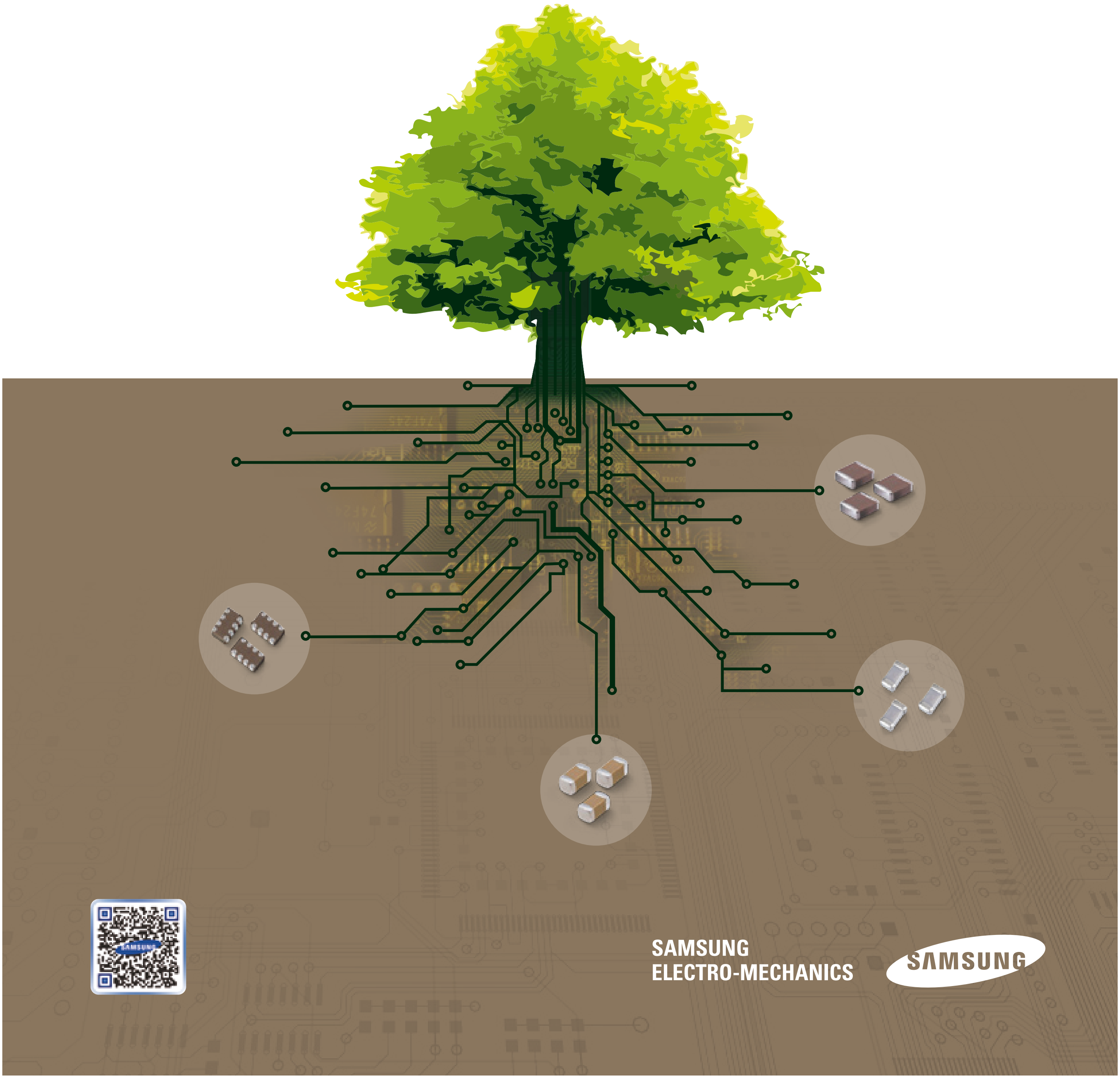
RoHS evidence document: **2304140030_Samsung-Electro-Mechanics-CL05A105KA5NQNC_C52923.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 13, 19, 25 of 84 — RoHS-relevant pages only; the complete document is retained in the project repository.

November 2015

Carbon Emission 1.86g CO_2
MLCC(02A104M) 1piece

RoHS+Halogen Compliant
ecoparts

MULTILAYER CERAMIC CAPACITORS



SAMSUNG ELECTRO-MECHANICS

SAMSUNG

We declare that all our MLCCs are produced in accordance with EU ROHS and REACH Directive.

1. RoHS Compliance and restriction of Br

The following restricted materials are not used in packaging materials as well as products in compliance with the law and restriction.
- Cd, Pb, Hg, Cr6+, As, Br and the compounds, PCB, asbestos

2. No use of materials breaking Ozone layer

The following ODS materials are not used in our fabrication process.
- ODS material : Freon, Haron, 1-1-1 TCE, CCl4, HCFC

If you want more detailed Information,Please Visit Samsung Electro-mechanics Website
[http://www.semlcr.com]

Please, see the last page of this catalog for our environ mental certification list.

SAMSUNG
ELECTRO-MECHANICS

Product Lineup (Standard & High Capacitors-C0G)

	Part Number	Size L x W (mm)	Capacitance	Rated Voltage (Vdc)	Capacitance Tolerance	Thickness Max.(mm)
1	CL05C0R5CB5NNN □	1.00×0.50	0.5pF	50	±0.25pF	0.55
2	CL05CR75CB5NNN □		0.75pF	50	±0.25pF	0.55
3	CL05C010CB5NNN □		1.0pF	50	±0.25pF	0.55
4	CL05C1R2CB5NNN □		1.2pF	50	±0.25pF	0.55
5	CL05C1R5CB5NNN □		1.5pF	50	±0.25pF	0.55
6	CL05C1R8CB5NNN □		1.8pF	50	±0.25pF	0.55
7	CL05C020CB5NNN □		2.0pF	50	±0.25pF	0.55
8	CL05C2R2CB5NNN □		2.2pF	50	±0.25pF	0.55
9	CL05C2R4CB5NNN □		2.4pF	50	±0.25pF	0.55
10	CL05C2R5CB5NNN □		2.5pF	50	±0.25pF	0.55
11	CL05C2R7CB5NNN □		2.7pF	50	±0.25pF	0.55
12	CL05C030CB5NNN □		3.0pF	50	±0.25pF	0.55
13	CL05C3R3CB5NNN □		3.3pF	50	±0.25pF	0.55
14	CL05C3R5CB5NNN □		3.5pF	50	±0.25pF	0.55
15	CL05C3R6CB5NNN □		3.6pF	50	±0.25pF	0.55
16	CL05C3R9CB5NNN □		3.9pF	50	±0.25pF	0.55
17	CL05C040CB5NNN □		4.0pF	50	±0.25pF	0.55
18	CL05C4R3CB5NNN □		4.3pF	50	±0.25pF	0.55
19	CL05C4R7CB5NNN □		4.7pF	50	±0.25pF	0.55
20	CL05C050DB5NNN □		5.0pF	50	±0.5pF	0.55
21	CL05C5R6DB5NNN □		5.6pF	50	±0.5pF	0.55
22	CL05C060DB5NNN □		6.0pF	50	±0.5pF	0.55
23	CL05C6R2DB5NNN □		6.2pF	50	±0.5pF	0.55
24	CL05C6R8DB5NNN □		6.8pF	50	±0.5pF	0.55
25	CL05C070DB5NNN □		7.0pF	50	±0.5pF	0.55
26	CL05C080DB5NNN □		8.0pF	50	±0.5pF	0.55
27	CL05C8R2DB5NNN □		8.2pF	50	±0.5pF	0.55
28	CL05C090DB5NNN □		9.0pF	50	±0.5pF	0.55
29	CL05C9R1DB5NNN □		9.1pF	50	±0.5pF	0.55
30	CL05C100JB5NNN □		10pF	50	±5%	0.55
31	CL05C110JB5NNN □		11pF	50	±5%	0.55
32	CL05C120JB5NNN □		12pF	50	±5%	0.55
33	CL05C130JB5NNN □		13pF	50	±5%	0.55
34	CL05C150JB5NNN □		15pF	50	±5%	0.55
35	CL05C160JB5NNN □		16pF	50	±5%	0.55
36	CL05C180JB5NNN □		18pF	50	±5%	0.55
37	CL05C200JB5NNN □		20pF	50	±5%	0.55
38	CL05C220JA5NNN □		22pF	25	±5%	0.55
39	CL05C220JB5NNN □		22pF	50	±5%	0.55
40	CL05C240JB5NNN □		24pF	50	±5%	0.55
41	CL05C270JB5NNN □		27pF	50	±5%	0.55
42	CL05C270JA5NNN □		27pF	25	±5%	0.55
43	CL05C300JB5NNN □		30pF	50	±5%	0.55
44	CL05C330JB5NNN □		33pF	50	±5%	0.55
45	CL05C360JB5NNN □		36pF	50	±5%	0.55
46	CL05C390JB5NNN □		39pF	50	±5%	0.55
47	CL05C430JB5NNN □		43pF	50	±5%	0.55
48	CL05C470JB5NNN □		47pF	50	±5%	0.55
49	CL05C510JB5NNN □		51pF	50	±5%	0.55
50	CL05C560JB5NNN □		56pF	50	±5%	0.55

※ □ mark means packaging code. If you want to learn the code or quantity in detail, please see p74.

SAMSUNG ELECTRO-MECHANICS

Product Lineup (Standard & High Capacitors-X5R)

	Part Number	Size L x W (mm)	Capacitance	Rated Voltage (Vdc)	Capacitance Tolerance	Thickness Max.(mm)
1	CL05A223KO5NNN□		22 nF	16	±10%	0.55
2	CL05A104KA5NNN□		0.1 μF	25	±10%	0.55
3	CL05A104KO5NNN□		0.1 μF	16	±10%	0.55
4	CL05A104KP5NNN□		0.1 μF	10	±10%	0.55
5	CL05A224KA5NNN□		0.22 μF	25	±10%	0.55
6	CL05A224KP5NNN□		0.22 μF	10	±10%	0.55
7	CL05A334KA5NNN□		0.33 μF	25	±10%	0.55
8	CL05A334KP5NNN□		0.33 μF	10	±10%	0.55
9	CL05A474KA5NNN□		0.47 μF	25	±10%	0.55
10	CL05A474KO5NNN□		0.47 μF	16	±10%	0.55
11	CL05A474KP5NNN□		0.47 μF	10	±10%	0.55
12	CL05A474KQ5NNN□		0.47 μF	6.3	±10%	0.55
13	CL05A474KR5NNN□		0.47 μF	4	±10%	0.55
14	CL05A105KA5NQN□		1 μF	25	±10%	0.60
15	CL05A105KO5NNN□		1 μF	16	±10%	0.55
16	CL05A105KO3LQN□		1 μF	16	±10%	0.33
17	CL05A105KP5NNN□		1 μF	10	±10%	0.55
18	CL05A105KP3LNN□		1 μF	10	±10%	0.33
19	CL05A105KQ5NNN□		1 μF	6.3	±10%	0.55
20	CL05A105KQ3LNN□	1.00×0.50	1 μF	6.3	±10%	0.33
21	CL05A105KR5NNN□		1 μF	4	±10%	0.55
22	CL05A105KR3LNN□		1 μF	4	±10%	0.33
23	CL05A225MA5NUN□		2.2 μF	25	±20%	0.70
24	CL05A225KO5NQN□		2.2 μF	16	±10%	0.60
25	CL05A225MP5NSN□		2.2 μF	10	±20%	0.57
26	CL05A225KP3LRN□		2.2 μF	10	±10%	0.33
27	CL05A225MQ5NNN□		2.2 μF	6.3	±20%	0.55
28	CL05A225KQ3LRN□		2.2 μF	6.3	±10%	0.33
29	CL05A225MR5NNN□		2.2 μF	4	±20%	0.55
30	CL05A225KR3LRN□		2.2 μF	4	±10%	0.33
31	CL05A475MO5NUN□		4.7 μF	16	±20%	0.70
32	CL05A475MP5NRN□		4.7 μF	10	±20%	0.65
33	CL05A475MQ5NRN□		4.7 μF	6.3	±20%	0.65
34	CL05A475MQ3LUN□		4.7 μF	6.3	±20%	0.35
35	CL05A106MP5NUN□		10 μF	10	±20%	0.70
36	CL05A106MQ5NUN□		10 μF	6.3	±20%	0.70
37	CL05A106MR5NRN□		10 μF	4	±20%	0.65
38	CL05A156MR5NUN□		15 μF	4	±20%	0.70
39	CL05A226MR5NZN□		22 μF	4	±20%	0.90
1	CL10A474KB8NNN□		0.47 μF	50	±10%	0.90
2	CL10A474KA8NNN□		0.47 μF	25	±10%	0.90
3	CL10A474KP8NNN□		0.47 μF	10	±10%	0.90
4	CL10A474KQ8NNN□		0.47 μF	6.3	±10%	0.90
5	CL10A474KR8NNN□		0.47 μF	4	±10%	0.90
6	CL10A105KB8NNN□		1 μF	50	±10%	0.90
7	CL10A105KA5LNN□	1.60×0.80	1 μF	25	±10%	0.50
8	CL10A105KA8NNN□		1 μF	25	±10%	0.90
9	CL10A105KO8NNN□		1 μF	16	±10%	0.90
10	CL10A105KO5LNN□		1 μF	16	±10%	0.50
11	CL10A105KP8NNN□		1 μF	10	±10%	0.90
12	CL10A105KP5LNN□		1 μF	10	±10%	0.50
13	CL10A105KQ8NNN□		1 μF	6.3	±10%	0.90
14	CL10A105KQ5LNN□		1 μF	6.3	±10%	0.50

※ □ mark means packaging code. If you want to learn the code or quantity in detail, please see p74.

Part Numbering System
Standard & High Capacitors
Super Small Size Capacitors
High-Q Capacitors
Medium-High Voltage Capacitors
Array Type Capacitors
Low ESL Capacitors
Reliability Test Condition
Premium Capacitors for Automotive Applications
Packaging Specification
Application Manual for Surface Mounting

SAMSUNG ELECTRO-MECHANICS

Product Lineup (Standard & High Capacitors-X7R, X7S)

	Part Number	Size L x W (mm)	Capacitance	Rated Voltage (Vdc)	Capacitance Tolerance	Thickness Max.(mm)
1	CL05B151KB5NNN □		150pF	50	±10%	0.55
2	CL05B181KB5NNN □		180pF	50	±10%	0.55
3	CL05B221KB5NNN □		220pF	50	±10%	0.55
4	CL05B271KB5NNN □		270pF	50	±10%	0.55
5	CL05B331KB5NNN □		330pF	50	±10%	0.55
6	CL05B391KB5NNN □		390nF	50	±10%	0.55
7	CL05B471KB5NNN □		470pF	50	±10%	0.55
8	CL05B561KB5NNN □		560pF	50	±10%	0.55
9	CL05B681KB5NNN □		680pF	50	±10%	0.55
10	CL05B102KB5NNN □		1nF	50	±10%	0.55
11	CL05B122KB5NNN □		1.2nF	50	±10%	0.55
12	CL05B152KB5NNN □		1.5nF	50	±10%	0.55
13	CL05B182KB5NNN □		1.8nF	50	±10%	0.55
14	CL05B222KB5NNN □		2.2nF	50	±10%	0.55
15	CL05B272KB5NNN □		2.7nF	50	±10%	0.55
16	CL05B332KB5NNN □		3.3nF	50	±10%	0.55
17	CL05B472KB5NNN □		4.7nF	50	±10%	0.55
18	CL05B562KB5NNN □	1.00×0.50	5.6nF	50	±10%	0.55
19	CL05B682KB5NNN □		6.8nF	50	±10%	0.55
20	CL05B822KB5NNN □		8.2nF	50	±10%	0.55
21	CL05B103KB5NNN □		10nF	50	±10%	0.55
22	CL05B123KA5NNN □		12nF	25	±10%	0.55
23	CL05B153KA5NNN □		15nF	25	±10%	0.55
24	CL05B223KA5NNN □		22nF	25	±10%	0.55
25	CL05B273KO5NNN □		27nF	16	±10%	0.55
26	CL05B333KO5NNN □		33nF	16	±10%	0.55
27	CL05B393KO5NNN □		39nF	16	±10%	0.55
28	CL05B473KO5NNN □		47nF	16	±10%	0.55
29	CL05B563KO5NNN □		56nF	16	±10%	0.55
30	CL05B683KO5NNN □		68nF	16	±10%	0.55
31	CL05B823KO5NNN □		82nF	16	±10%	0.55
32	CL05B104KO5NNN □		100nF	16	±10%	0.55
33	CL05B224KO5NNN □		220nF	16	±10%	0.55
34	CL05B474KP5NNN □		470nF	10	±10%	0.55
35	CL05B105KQ5NQN □		1㎌	6.3	±10%	0.60
1	CL05Y474KP5NNN □	1.00×0.50	470nF	10	±10%	0.55
37	CL10B101KB8NNN □		100pF	50	±10%	0.90
38	CL10B121KB8NNN □		120pF	50	±10%	0.90
39	CL10B151KB8NNN □		150pF	50	±10%	0.90
40	CL10B181KB8NNN □		180pF	50	±10%	0.90
41	CL10B201KB8NNN □		200pF	50	±10%	0.90
42	CL10B221KB8NNN □		220pF	50	±10%	0.90
43	CL10B271KB8NNN □		270pF	50	±10%	0.90
44	CL10B331KB8NNN □		330pF	50	±10%	0.90
45	CL10B391KB8NNN □	1.60×0.80	390pF	50	±10%	0.90
46	CL10B471KB8NNN □		470pF	50	±10%	0.90
47	CL10B561KB8NNN □		560pF	50	±10%	0.90
48	CL10B681KB8NNN □		680pF	50	±10%	0.90
49	CL10B751KB8NNN □		750pF	50	±10%	0.90
50	CL10B821KB8NNN □		820pF	50	±10%	0.90
51	CL10B102KB8NNN □		1nF	50	±10%	0.90
52	CL10B122KB8NNN □		1.2nF	50	±10%	0.90

※ □ mark means packaging code. If you want to learn the code or quantity in detail, please see p74.

Section 2
Vishay

Designators	Part / Value	Qty
C4_1-C4_36,C33,C34	4.7uF 3216 25V Tantalum	38

RoHS evidence document: **vishay-rohs-20250901.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

VISHAY

ROHS Compliance Statement

Directives 2002/95/EG, 2011/65/EU, 2015/863/EU
(RoHS I – III)

September 1st, 2025

Dear Valued Customer,

Vishay Intertechnology, Inc. is committed to be in compliance with all applicable global laws, directives and regulations.

Unless otherwise specified, Vishay Intertechnology, Inc. hereby certifies that all its products that are identified as RoHS compliant satisfy the requirements of the above-listed directives of the European Parliament respecting the restriction of the use of certain hazardous substances in electrical and electronic equipment.

Vishay Intertechnology, Inc. is closely following the various RoHS exemption renewal requests, which as of today, are still pending and awaiting a decision by the European Commission.

According to RoHS Article 5. 5, existing exemptions shall remain valid until a decision on the renewal application is taken by the European Commission and a new expiry date is set. Where necessary, further renewal requests will be issued at least 18 months before the new expiry date.

Aside from satisfying its regulatory obligations, Vishay is actively searching for possible alternatives and scrutinizing viable options based on performance, quality and cost. Currently concerned exemptions as per Annex II, 2011/65/EU are 6 a, b, c, 7 a, 7 c(I) and 7 c(II).

Please mail your request to VPECM.Surveys@Vishay.com for specific products RoHS compliance reports.

Roy Shoshani

Roy Shoshani
Executive Vice President
COO – Semiconductors and CTO
Vishay Intertechnology, Inc.

63 Lancaster Avenue, Malvern, PA, 19355, USA (610) 644-1300

Section 3
NationStar

Designators	Part / Value	Qty
D1	Red 0805 C84256	1

RoHS evidence document: **2409272203_Foshan-NationStar-Optoelectronics-NCD0805R1_C84256.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 8 — RoHS-relevant pages only; the complete document is retained in the project repository.

表单编号：FGH/BD-0053.01-15A

佛山市国星光电股份有限公司
FOSHAN NATIONSTAR OPTOELECTRONICS CO., LTD

产品规格书
SPECIFICATION

顾客名称 Customer		产品名称 Product	Chip LED
顾客型号 Customer Type		产品型号 Type	NCD0805R1
顾客部品号 Customer No.		版本号 Version NO	A 版

ISO14001 环境管理体系认证

ISO9001 质量管理体系认证

地址：广东省佛山市禅城区华宝南路 18 号
Add: NO.18 South Huabao Rd,Foshan,Guangdong,China
电话（Tel）：0757-82100219
传真（Fax）：0757-82100220
邮编（Zip）：528000
邮箱（Email）：chipLED@nationstar.com
http://www.nationstar.com

佛山市国星光电股份有限公司 研发中心

研 发 中 心 Research & Development Center			客户（加盖公章） Customer（Stamp）
制 定 DRAW	审 核 CHECK	批 准 APPROVE	确 认 CONFIRM
陆紫珊	李友民	朱明军	
发放日期 (Release Date)：2019-10-30			

保存期限：长期

表单编号：FGH/BD-0053.01-15A

国星光电
NATIONSTAR

片式发光二极管

NCD0805R1

Chip Light Emitting Diode

技术数据表 Technical Data Sheet

本产品主要作为信号指示及照明的电子元件广泛应用于各类使用表面贴装结构的电子产品中，如家用电器的开关指示灯、手机键盘灯、汽车仪表盘指示灯等。

This product is generally used as indicator and luminance for surface mounted electronic equipment, such as household appliance, communication equipment, and dashboard.

特性：
Features:

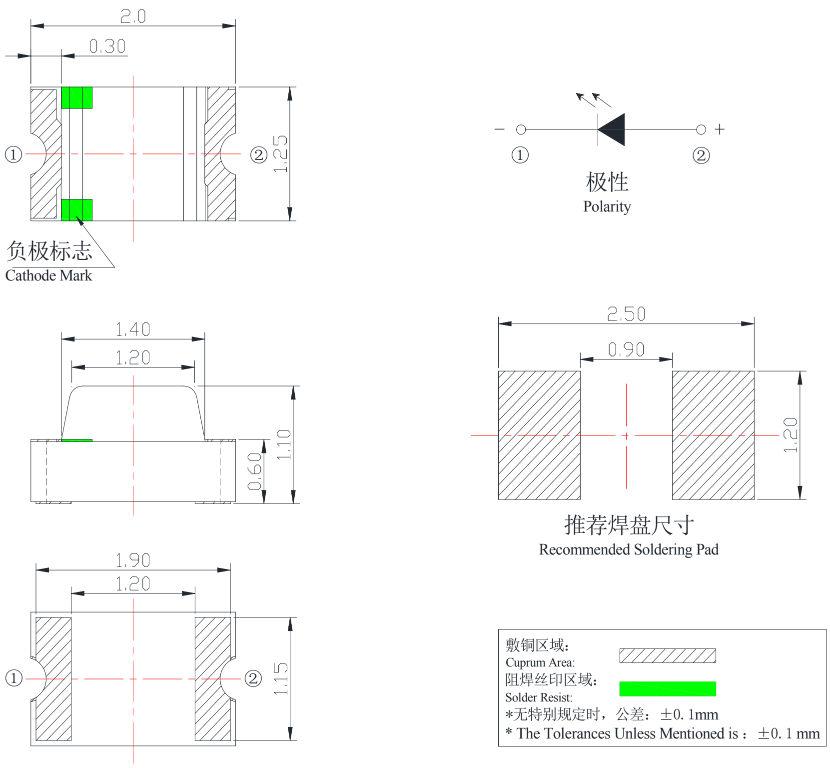
- 管芯材料：AlGaInP
  Material: AlGaInP
- 封装材料：环氧树脂
  Encapsulation: Epoxy Resin
- 焊接方法：无铅回流焊
  Soldering methods: Pb-Free reflow soldering
- 光强高，功耗低，可靠性好，寿命长
  High Luminous Intensity ,Low Power Dissipation, Good Reliability and Long Lifespan
- 符合欧盟公布的 ROHS 指令要求
  Complied With ROHS Directive

发光颜色：红色
Emitting Color: Red

* 产品规格如因工艺改进而有所改变，恕不另行通知。
* The specifications of the product may be modified for improvement without notice.

外形尺寸

Outline Dimension



保存期限：长期

表单编号：FGH/BD-0053.01-15A

国星光电 NATIONSTAR

片式发光二极管

光电参数（1）
Electro-Optical Characteristics(1)

✧ **极限参数**（温度=25℃）
Absolute Maximum Ratings (Temperature=25°C)

参数名称 Parameter	符号 Symbol	数值 Rating	单位 Unit
正向电流 Forward Current	I_F	25	mA
正向脉冲电流* Pulse Forward Current*	I_{FP}	50	mA
反向电压 Reverse Voltage	V_R	5	V
工作温度 Operating Temperature	T_{OPR}	-30 ～+85	℃
贮存温度 Storage Temperature	Tstg	-40 ～+100	℃
功耗 Power Dissipation	P_D	65	mW

* 注：脉冲宽度≤0.1ms，占空比≤1/10　　* Note: Pulse Width≤0.1ms, Duty≤1/10

✧ **光电参数**（温度=25℃）
Electro-Optical Characteristics (Temperature=25°C)

参数名称 Parameter	符号 Symbol	条件 Condition	最小值 Min.	典型值 Typ.	最大值 Max.	单位 Unit
反向电流 Reverse Current	I_R	V_R=5V	-	-	10	μA
视角度 View Angle	$2\theta_{1/2}$	-	-	130	-	deg.
正向电压 Forward Voltage	V_F	I_F=20mA	1.6	2.0	2.6	V
峰值波长 Peak Wavelength	λ_P		-	630	-	nm
主波长 Dominant Wavelength	λ_d		615	620	630	nm
半波宽度 Spectrum Radiation Bandwidth	$\Delta\lambda$		-	15	-	nm
光强 Luminous Intensity	I_V		67	115	195	mcd

* 注 1：光强偏差±15%；压降偏差±0.1V；(X,Y)坐标偏差±0.01；单色光波长偏差±1nm。
* Note1: Tolerance on each Luminous Intensity bin is ±15%; Tolerance on each Forward Voltage bin is ±0.1V;Tolerance on each Hue(X,Y) bin is ±0.01; Tolerance of Dominant Wavelength ±1nm.
* 注 2：以上参数仅供参考，请以实物标签为准。我司给出的参数均由国星测试系统测得。
* Note2: The parameters above are only for your reference. In case of any discrepancy, please adhere tothe label of our actual products. All parameters are tested by the standard testing system of NationStar.

保存期限：长期

Section 4
Shenzhen JingYang

Designators	Part / Value	Qty
D1_1-D1_36,D12	1N4148WSX	37

RoHS evidence document: **2405091035_Shenzhen-JingYang-1N4148WSX_C6423741.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

JYELECTRONICS®

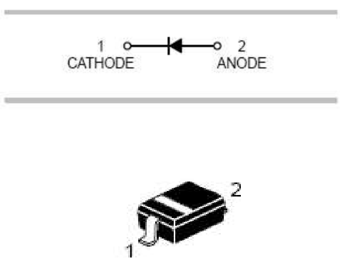
1N4148WSX

Silicon Epitaxial Planar Diode

FEATURES

- Fast Switching Speed:trr=4ns
- Surface Mount Package Ideally Suited For Automatic Insertion
- For General Purpose Switching Applications
- High Conductance
- Available in Lead Free Version
- ROHS compliant
- MSL: 3

Lead-free



SOD-323

APPLICATIONS

- Surface mount fast switching diode

ORDERING INFORMATION

Type No.	Marking	Package Code
1N4148WSX	T4	SOD-323

MAXIMUM RATING @ Ta=25℃ unless otherwise specified

Characteristic	**Symbol**	**Value**	**Unit**
Non-Repetitive Peak Reverse Voltage	V_{RM}	100	V
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Reverse Voltage	V_{RRM} V_{RWM} V_R	75	V
RMS Reverse Voltage	$V_{R(RMS)}$	53	V
Average Rectified Output Current	I_o	150	mA
Non-Repetitive Peak Forward Surge Current @t=1.0 μs @t=1.0 s	I_{FSM}	2.0 1.0	A
Power Dissipation	P_d	200	mW
Thermal Resistance Junction to Ambient Air	$R_{\theta JA}$	625	℃/W
Operating and Storage Temperature Rage	T_j,T_{STG}	-65 to+150	℃

JYELECTRONICS®

1N4148WSX

Silicon Epitaxial Planar Diode

ELECTRICAL CHARACTERISTICS @ Ta=25℃ unless otherwise specified

Characteristic	Symbol	Min	Max	Unit	Test Condition
Reverse Breakdown Voltage	$V_{(BR)R}$	75	-	V	I_R=1.0μA
Forward Voltage	V_F	-	0.715 0.855 1.0 1.25	V	I_F=1.0mA I_F=10mA I_F=50mA I_F=150mA
Reverse Current	I_R	-	1.0 25	μA nA	V_R=75V V_R=20V
Junction Capacitance	C_J	-	2.0	pF	V_R=0,f=1.0MHz
Reverse Recovery Time	t_{rr}	-	6.0	ns	$I_F=I_R$=10mA, $I_{rr}=0.1\times I_R$,R_L=100Ω

TYPICAL CHARACTERISTICS @ Ta=25℃ unless otherwise specified

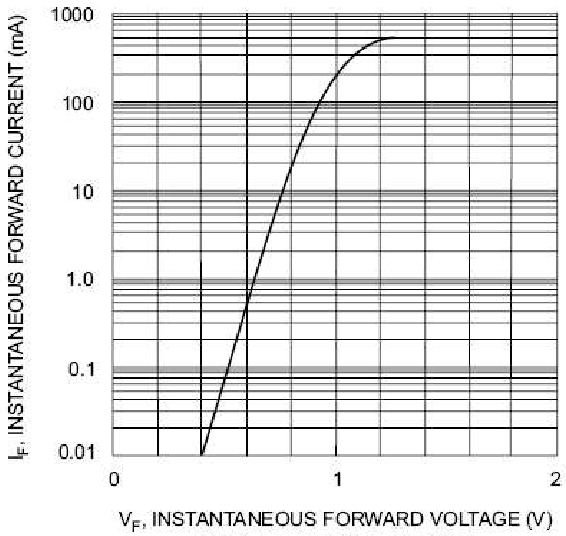


Fig. 1 Forward Characteristics

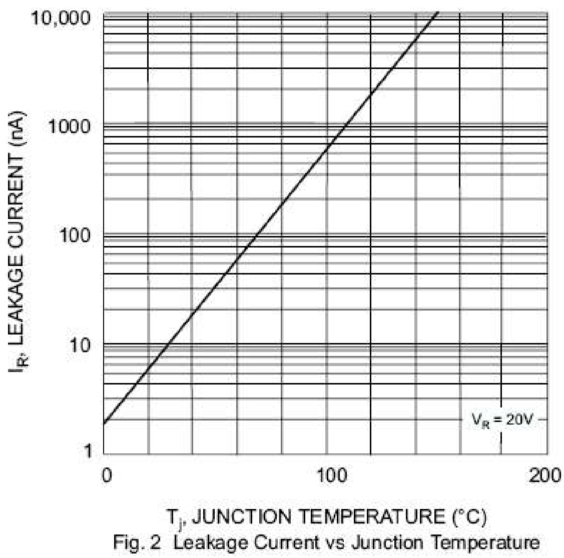


Fig. 2 Leakage Current vs Junction Temperature

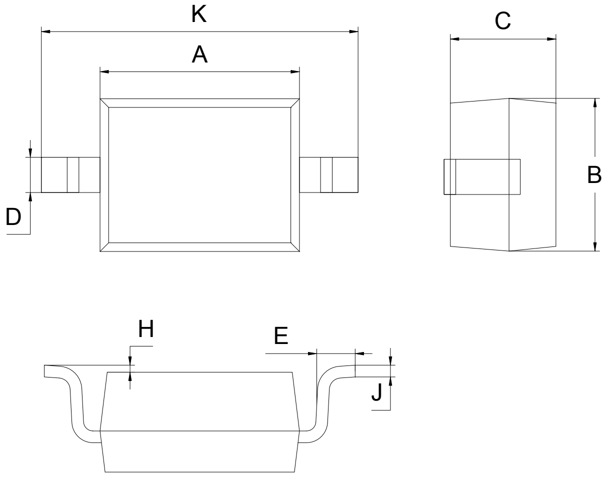
JYELECTRONICS®

1N4148WSX

Silicon Epitaxial Planar Diode

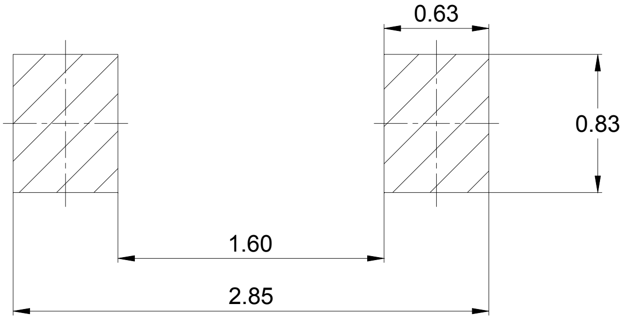
PACKAGE OUTLINE

Plastic surface mounted package SOD-323



SOD-323		
Dim	Min	Max
A	1.60	1.80
B	1.20	1.40
C	0.80	0.90
D	0.25	0.35
E	0.22	0.42
H	0.02	0.1
J	0.05	0.15
K	2.55	2.75
All Dimensions in mm		

SOLDERING FOOTPRINT



Unit ： mm

PACKAGE INFORMATION

Device	Package	Shipping
1N4148WSX	SOD-323	3000/Tape&Reel

Section 5
Hottech

Designators	Part / Value	Qty
D2	1N5819WS	1

RoHS evidence document: **ROHS3HOTTECH.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

Test Report

Report No: BSTDG200513827804CR Date: May 20, 2020 Page 1 of 6

Applicant : GUANGDONG HOTTECH INDUSTRIAL CO, LTD.

Address : No.183B, Puxinhu Bussiness District,Tangxia Town, Dongguan City, Guangdong Province,China

The following sample(s) was /were submitted and identified on behalf of the clients as：

Sample Name : Diode&Transistor/IC

Sample Model : SOT-23,323,523,723,363,223,23-5,23-6,89 TO-92,92L,126, 252,251,220,SOP8, TSSOP8 ,SOP14, DIP8, DIP14, SOD-123,

Sample Received Date : May 17, 2020

Testing Period : May 17, 2020 To May 20, 2020

Test Requested : Selected test (s) in the selected parts as requested by client with the RoHS 2 Directive 2011/65/EU Annex II (EU) 2015/863 as last amended by Directive (EU) 2017/2102.

Test Method : Please refer to next page(s).

Test Result : Please refer to next page(s).

Signed for and on behalf of

BST TESTING APPROVED

Tony Qian/ Approved Signatory

Dongguan BST Testing Co., Ltd.
A1201-1204 Xinsanqi of Dongbao Road, Dongcheng District, Dongguan, Guangdong, China
Tel: 400-8829628/ 800-9990305
Http://www.bst-lab.com E-mail: christina@bst-lab.com

BTDG-BG-H002

Test Report

Report No: BSTDG200513827804CR Date: May 20, 2020

Test Content:

Test Item(s)	Test Method	Reference	Unit	Limit	MDL
Cadmium(Cd)	IEC 62321-5:2013	ICP-OES	mg/kg	100	2
Lead(Pb)	IEC 62321-5:2013	ICP-OES	mg/kg	1000	2
Mercury(Hg)	IEC 62321-4:2013+AMD1:2017	ICP-OES	mg/kg	1000	2
Hexavalent Chromium(CrVI) (Metal)	IEC 62321-7-1:2015	UV-Vis	$\mu g/cm^2$	0.13	0.1
Hexavalent Chromium(CrVI) (Nonmetal)	IEC 62321-7-2:2017	UV-Vis	mg/kg	1000	8
PBBs (Next form)	IEC 62321-6:2015	GC-MS	mg/kg	1000	5
PBDEs (Next form)	IEC 62321-6:2015	GC-MS	mg/kg	1000	5
Dibutyl Phthalate(DBP)	IEC 62321-8:2017	GC-MS	mg/kg	1000	30
Butyl benzyl phthalate (BBP)	IEC 62321-8:2017	GC-MS	mg/kg	1000	30
Di-(2-ethylhexyl) Phthalate(DEHP)	IEC 62321-8:2017	GC-MS	mg/kg	1000	30
Diisobutyl phthalate (DIBP)	IEC 62321-8:2017	GC-MS	mg/kg	1000	30

PBBs		PBDEs	
Monobromobiphenyl	Hexabromobiphenyl	Monobromodiphenyl ether	Hexabromodiphenyl ether
Dibromobiphenyl	Heptabromobiphenyl	Dibromodiphenyl ether	Heptabromodiphenyl ether
Tribromobiphenyl	Octabromobiphenyl	Tribromodiphenyl ether	Octabromodiphenyl ether
Tetrabromobiphenyl	Nonabromobiphenyl	Tetrabromodiphenyl ether	Nonabromodiphenyl ether
Pentabromobiphenyl	Decabromobiphenyl	Pentabromodiphenyl ether	Decabromodiphenyl ether

Dongguan BST Testing Co., Ltd.

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BTDG-BG-H002

Test Report

Report No: BSTDG200513827804CR   Date: May 20, 2020   Page 3 of 6

Sample Description:

No.	Description	Name
1	Plastic	Ontology
2	Metal	Pin

Test Results:

Test Item(s)	No.1	No.2
Cadmium (Cd)	N.D.	N.D.
Lead (Pb)	N.D.	N.D.
Mercury (Hg)	N.D.	N.D.
Hexavalent Chrormium (CrVI)	N.D.	N.D.
PBBs	N.D.	--
PBDEs	N.D.	--
Dibutyl Phthalate (DBP)	N.D.	--
Butyl benzyl phthalate (BBP)	N.D.	--
Di-(2-ethylhexyl) Phthalate(DEHP)	N.D.	--
Diisobutyl phthalate (DIBP)	N.D.	--

Note:
1. mg/kg= ppm
2. N.D.= Not Detected(<MDL)
3. MDL = Method Detection Limit
4. -- = No Testing

Dongguan BST Testing Co., Ltd.
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BTDG-BG-H002

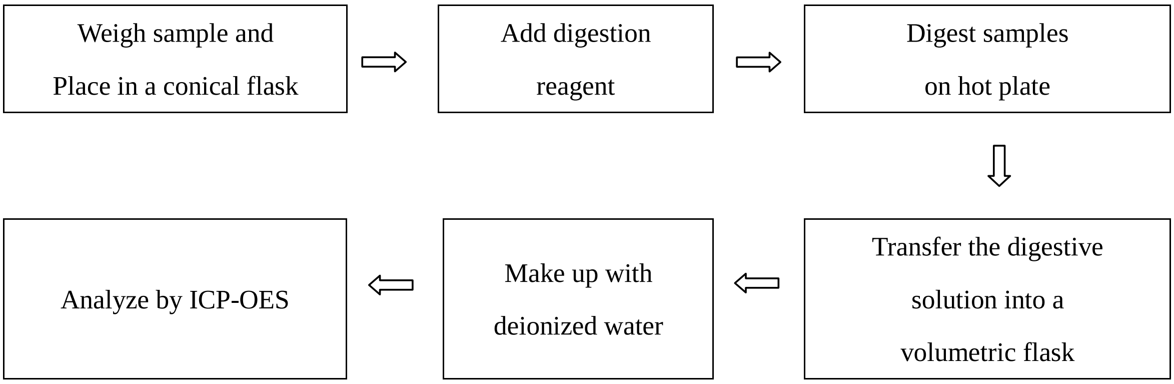
Test Report

Report No: BSTDG200513827804CR Date: May 20, 2020 Page 4 of 6

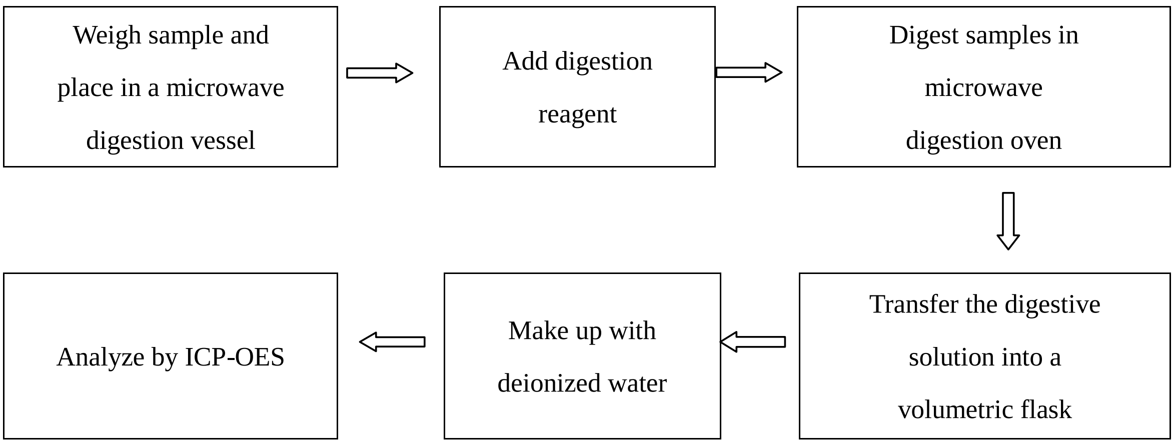
5. when Cr(VI) in a sample is detected below the 0.10 μg/cm^2 LOQ (limit of quantification), the sample is considered to be negative for Cr(VI). Since Cr(VI) may not be uniformly distributed in the coating even within the same sample batch, a "grey zone" between 0.10 μg/cm^2 and 0.13 μg/cm^2 has been established as "inconclusive" to reduce inconsistent results due to unavoidable coating variations. In this case, additional testing may be necessary to confirm the presence of Cr(VI). When Cr(VI) is detected above 0.13 μg/cm^2, the sample is considered to be positive for the presence of Cr(VI) in the coating layer. unavoidable coating variations may influence the determination Information on storage conditions and production date of the tested sample is unavailable and thus Cr(VI) results represent status of the sample at the time of testing.

Test Process:

1. Test for Cd/Pb Content



2. Test for Hg Content



Dongguan BST Testing Co., Ltd.

A1201-1204 Xinsanqi of Dongbao Road, Dongcheng District, Dongguan, Guangdong, China

Tel: 400-8829628/ 800-9990305
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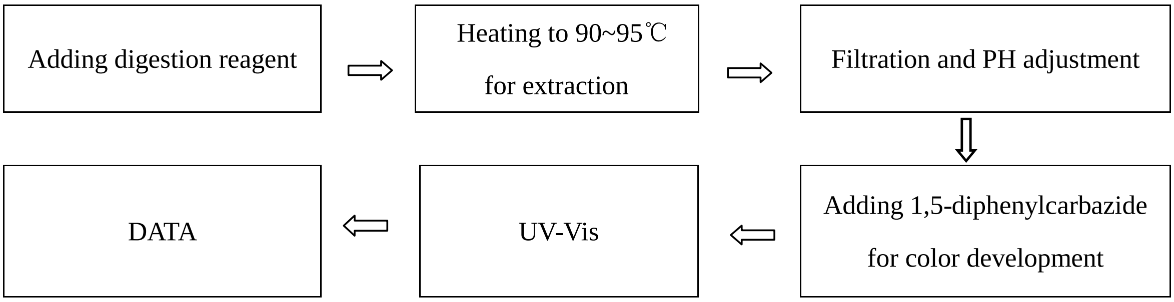
BTDG-BG-H002

Test Report

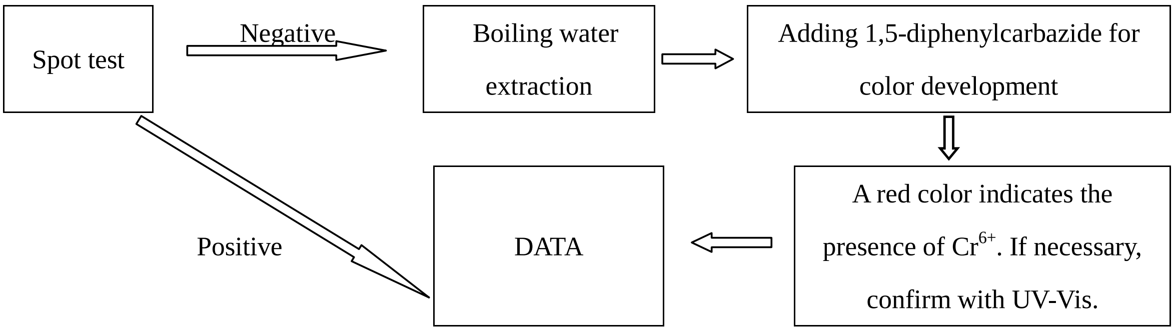
Report No: BSTDG200513827804CR Date: May 20, 2020

3. Test for Chromium (VI) Content

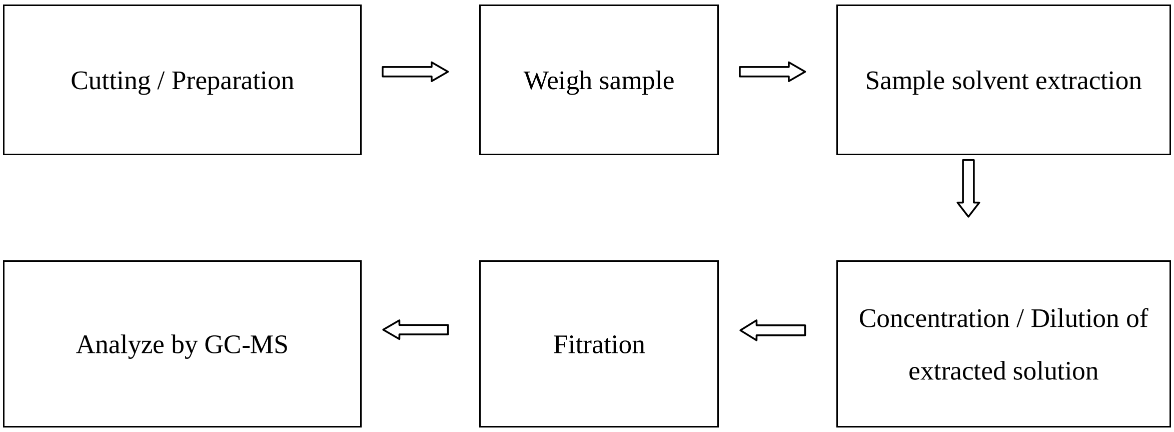
Nonmetal material



Metal material



4. Test for DBP, BBP, DEHP, DIBP, PBB, PBDE Content



Dongguan BST Testing Co., Ltd.
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BTDG-BG-H002

Test Report

Report No: BSTDG200513827804CR   Date: May 20, 2020   Page 6 of 6

Sample Photo:

*** End of Report ***

Dongguan BST Testing Co., Ltd.
A1201-1204 Xinsanqi of Dongbao Road, Dongcheng District, Dongguan, Guangdong, China
Tel: 400-8829628/ 800-9990305
Http://www.bst-lab.com E-mail: christina@bst-lab.com

BTDG-BG-H002

Section 6
Hubei KENTO

Designators	Part / Value	Qty
D5	Green 0805 C2297	1

RoHS evidence document: **C2297.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3, 8, 9, 10 of 11 — RoHS-relevant pages only; the complete document is retained in the project repository.

KENTO 湖北匡通电子股份有限公司

0805 翠绿

承 认 书

Specification for approval

客户名称(Customer name)：

经办者(Director)：

职称(title)：

客户料号(Customer part NO)：

版本(Revision)： A.0

发件日期(Issue date)： 2018-12-06

回文日期(Return date)：

一、谨致执事者：兹提供敝公司产品之有关详细规格及图面数据，敬请给予办理测试认定手续，同时敬请送返一份附有贵公司签认之测试认定后之样品认定书。

（We are please in sending you herewith our specification and drawings for your approval. ）

（Please return to us one copy “For Approval” with your approved signatures.）

二、附件(Accessory):

☐样品 ☐出货检验记录表 ☐封装尺寸图 ☐电气特性曲线

☐内部线路图 ☐焊性建议 ☐PAD 建议 ☐包装方式

三、客户意见栏(Customer’s Proposal)

☐ 同意(Agree)：(请于认可栏中签名)

☐ 不同意(Disagree)：

原因(Reason)：

客户认可签章(Customer Signature):

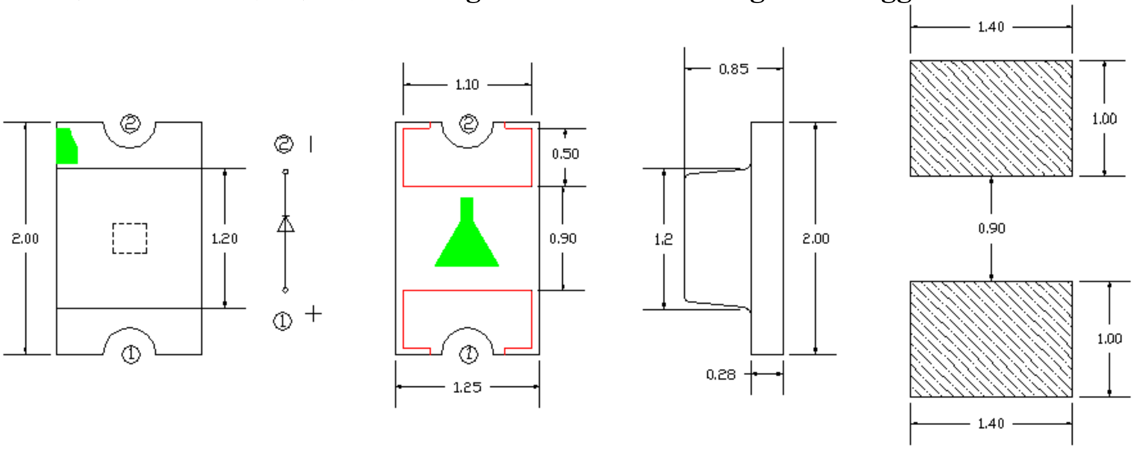
KENTO 湖北匡通电子股份有限公司

0805 翠绿

1. 产品描述/ Features

- 外观尺寸/ Package (L/W/H) :   2.0*1.25*0.85 mm
- 颜色/ Color: 翠绿 /   Green light
- 胶体/ Lens: 透明平面胶体/ Transparent planar colloid
- EIA规范标准包装/ EIA STD Package
- 环保产品，符合ROHS要求/ Meet ROHS, Green Product
- 适用于自动贴片机/ Compatible With SMT Automatic Equipment
- 适用于红外线回流焊制程/ Compatible With Infrared Reflow Solder Process

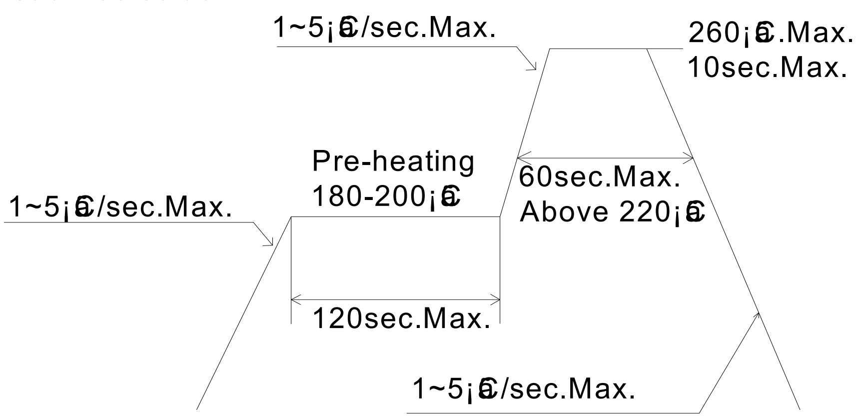
2 . 外形尺寸及建议焊盘尺寸/ Package Profile & Soldering PAD Suggested



注/ Notes: 1.   单位 ： 毫米（mm）/ All dimensions are in millimeters

2. 公差 ： 如无特别标注则为± 0.1 mm Tolerance is ± 0.10 mm unless otherwise noted

3. 建议焊接温度曲线 / Soldering Profile Suggested



KENTO 湖北匡通电子股份有限公司

0805 翠绿

4. 最大绝对额定值/ Absolute Maximum Ratings （Ta=25℃）

参 数/ Parameter	符号Symbol	最大额定值/ Rating	单 位/ Unit
消耗功率/ Power Dissipation	Pd	100	mW
最大脉冲电流/ Peak Forward Current (1/10占空比, 0.1ms脉宽)	IFP	60	mA
正向直流工作电流/ DC Forward Current	IF	30	mA
反向电压/Backward Voltage	VR	5	V
工作温度范围 Operating Temperature Range	Topr	-40°C ～ +85°C	
存储温度范围 Storage Temperature Range	Tstg	-40°C ～ +85°C	
焊接条件 Soldering Condition	Tsol	回流焊/ Reflow soldering : 260°C ， 10s 手动焊/ Hand soldering : 300°C ， 3s	
抗静电能力 Electrostatic Discharge	ESD		V

5. 光电参数/ Electrical Optical Characteristics （Ta=25℃）

参数 Parameter	符号 Symbol	最小值 Min.	代表值 Typ.	最大值 Max.	单位 Unit	测试条件 Test Condition
光强 Light Intensity	IV	175	--	430	mcd	IF = 5mA
半光强视角 Viewing Angle	2θ1/2	---	120	---	deg	IF = 5mA
主波长 Dominant Wavelength	λd	513		528	nm	IF = 5mA
峰值波长 Peak Wavelength	λp	516		525	nm	IF = 5mA
正向电压 Forward Voltage	VF	2.6		3.1	V	IF = 5mA
反向电压 Backward Voltage	IR	---	---	5	μA	VR = 5V
半波宽 Spectral Line Half-Width	Δλ		15		nm	IF = 5mA

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0805 翠绿

11. 注意事项/ Cautions：

11.1. 焊接/welding

11.1.1 SMD LED 灌封胶较软，外力易损坏发光面及塑料壳，焊接时要轻拿轻放。

SMD LED is soft and easy to damage the luminous surface and plastic shell by external force. It should be handled lightly when welding

11.1.2 建议使用易洗型的助焊剂，依照回流曲线条件回流焊接，回流次数最多两次，确保 LED 发光面干净，异物会影响发光颜色。

It is recommended to use soldering flux with tin wash type, reflow soldering according to the condition of reflux curve, reflow twice at most, ensure the LED luminous surface is clean, foreign matter will affect the luminous color。

11.1.3 只建议在修理和重工的情况下使用手工焊接; 最高焊接温度不应超过 300 度，且须在 3 秒内完成（手工焊接只可焊接一次）烙铁最大功率应不超过 25W。

Manual welding is only recommended for repair and heavy industry;The maximum welding temperature should not exceed 300 degrees, and must be completed within 3 seconds (manual welding can only be welded once) soldering iron maximum power should not exceed 25W.

11.1.4 焊接过程中，严禁在高温情况下碰触胶体； 焊接后，禁止对胶体施加外力，禁止弯折 PCB，避免元件受到撞击。

During the soldering process, do not touch the lens at high temperature，After soldering, any mechanical force on the lens or any excessive vibration shall not be accepted to apply, also the circuit board shall not be bent as well.

11.1.5 请不要将不同 BIN 级的 LED 使用于同一个产品上，否则可能会导致产品的严重色差。

Please do not use different BIN LED on the same product, otherwise it may cause serious color difference.

11.2. 清洗/cleaning

11.2.1 不能用超声波清洗,建议使用异丙醇（isopropyl alcohol）、纯酒精擦拭或浸渍(浸渍不超过 1 分钟)在室温下放置 15 分钟再使用；清洗后,确保 LED 发光面干净,异物会影响发光颜色。

/No ultrasonic cleaning. It is recommended to use isopropyl alcohol, pure alcohol to wipe or soak, not more than 1 minute, and leave at room temperature for 15 minutes before use. After cleaning, make sure the LED luminous surface is clean and the foreign matter will affect the luminous color。

11.2.2 应避免接触或污染天那水,三氯乙烯、丙酮、硫化物、氮化物、酸、碱、盐类，这些物质会损伤 LED.

Avoid touching or contaminating the water, trichloroethylene, acetone, sulfide, nitride, acid, alkali, and salts that can damage leds.

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11.3. 灌封/enbedment

11.3.1 挥发性物质会渗透到 LED 内部，在通电产生光子及热的条件下，会导致 LED 变色，进而造成严重光衰，严禁使用任何对 LED 器件的性能或者可靠性有害的物质或材料，针对特定的用途和使用环境，建议对所有的物质和材料进行相容性的测试。在贴装 LED 时候，不要使用能产生有机挥发性气体的粘结剂。

Volatile substances to leach into the LED inside, photons in electricity and heat conditions, will lead to the LED color, thus causing serious droop, it is forbidden to use any of the LED device performance or reliability of harmful substances or materials, for a specific purpose and use of the environment, advice on all the material and the material compatibility test.When attaching LED, do not use adhesive that can produce volatile organic gas.

11.3.2 使用正常灌封胶时,建议先以少量试验，常温点亮 168 小时，确定没有问题再作业。

It is recommended to light up for 168 hours at room temperature for a small amount of test before using normal filling and sealing glue。

11.4. 保存/save

11.4.1 打开包装前,LED 应存储在温度 30℃或以下,相对湿度在 RH60%以下,一年内使用。

Before opening the package, LED should be stored in a temperature 30 ℃ or below, under RH60 % relative humidity, used in a year。

11.4.2 LED 是湿度敏感元件,为避免元件吸湿,打开包装后,LED 应在温度 30℃或以下,相对湿度在 60%以内,使用时间 7 天。LED 吸潮后,回流焊时可能裂胶,影响发光颜色.对于未使用的散件,请去潮处理（卷装品：烘烤 60℃±5℃/24H；散装品：烘烤 105℃±5℃/1H）,然后再用铝箔袋密封后保存或者储存在氮气防潮柜内。

LED is humidity sensitive element, element to avoid moisture absorption, after open the packing, the LED should be in temperature 30 ℃ or below, within 60% relative humidity, using time 7 days. After moisture absorption, LED may crack when reflow soldering, influence the luminous color. For bulk is not used, please deal with the tide (for package product: bake 60 ℃ + / - 5 ℃ / 24 h.For bulk goods: baking 105 ℃ + 5 ℃, 1 hours), and then save after sealed with aluminum foil bag or stored in nitrogen moistureproof enclosure

11.4.3 保存环境中避免有酸、碱以及腐蚀气体存在，同时避免强烈震动及强磁场作用。

Avoid the presence of acid, alkali and corrosive gas in the preservation environment, and avoid strong vibration and strong magnetic field。

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11.5.静电/electrostatic

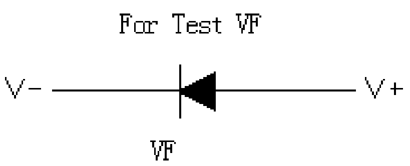
11.5.1 静电或峰值浪涌电压会损坏 LED,避免在开灯、关灯时产生瞬时电压。
Static electricity or peak surge voltage will damage the LED, avoiding instantaneous voltage when the lamp is turned on or off。

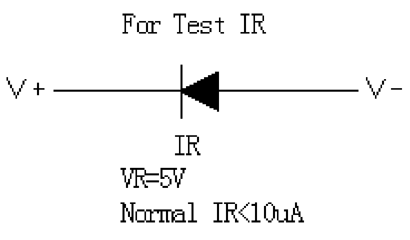
11.5.2 建议使用 LED 时佩戴防静电手腕带,防静电手套,穿防静电鞋,使用的设备、仪器正确接地。LED 损坏后,表现出漏电流明显增加,低电流正向电压变低，低电流点不亮等现象。
It is recommended to wear anti-static wrist bands, anti-static gloves and anti-static shoes when using LED. The equipment and instruments used are properly grounded. After the LED was damaged, the leakage current increased obviously, the forward voltage of low current became lower, and the low current point did not light, etc。

11.6 测试/test

11.6.1 LED 要在额定电流下驱动,同时电路中需要加限流电阻保护；否则,轻微的电压变化就会引起较大的电流变化,从而破坏 LED。
LED shall be driven at rated current, and shall be protected by current-limiting resistance in the circuit. Otherwise, slight voltage changes will cause large current changes, which will damage the LED.

11.6.2 在电路导通或关闭情况下,要避免瞬间浪涌电压的产生,否则 LED 将被烧坏。
When the circuit is on or off, avoid sudden surge voltage. Otherwise, the LED will be burnt out
请参照下图示检测 LED:/Please check the LED as shown





11.6.3 顺向电压 VF 过高或反向电压 VR 过高，均会损坏 LED.
If the forward voltage VF is too high or the reverse voltage VR is too high, the LED will be damaged.

11.6.4 点亮或测试 LED 时，加在 LED 两端的反向电压不得高于 5V，否则容易击伤 LED.
When lighting or testing the LED, the reverse voltage added on both ends of the LED shall not be higher than 5V, otherwise it is easy to damage the LED.

Section 7
Hubei KENTO

Designators	Part / Value	Qty
D6	Yellow 0805 C2296	1

RoHS evidence document: **1806151129_Hubei-KENTO-Elec-KT-0805Y_C2296.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 8 — RoHS-relevant pages only; the complete document is retained in the project repository.

承 认 书

Specification for approval

客户名称:
CUSTOMER NAME: ______________________________

经办者: 职称:
DIRECTOR: ______________ TITLE: ______________

客户料号:
CUSTOMER PART NO.: ______________________________

品名: 版本:
PART NUMBER: **KT -0805Y** REVISION: KT-0A

发件日期: 回文日期:
ISSUE DATE: ______________ RETURN DATE: ____/____/____

一、 谨致执事者：兹提供敝公司产品之有关详细规格及图面数据，
敬请给予办理测试认定手续。
同时敬请送返一份附有贵公司签认之测试认定后之样品认定书。
We are please in sending you herewith our specification and drawings for your approval.
Please return to us one copy “For Approval” with your approved signatures.

二、附件:

ACCESSORY：☐样品 ☐出货检验记录表 ☐封装尺寸图 ☐电气特性曲线
☐内部线路图 ☐焊性建议 ☐PAD 建议 ☐包装方式

三、客户意见栏 CUSTOMER’S PROPOSAL

☐ AGREE 同意 (请于认可栏中签名)
☐ DISAGREE 不同意

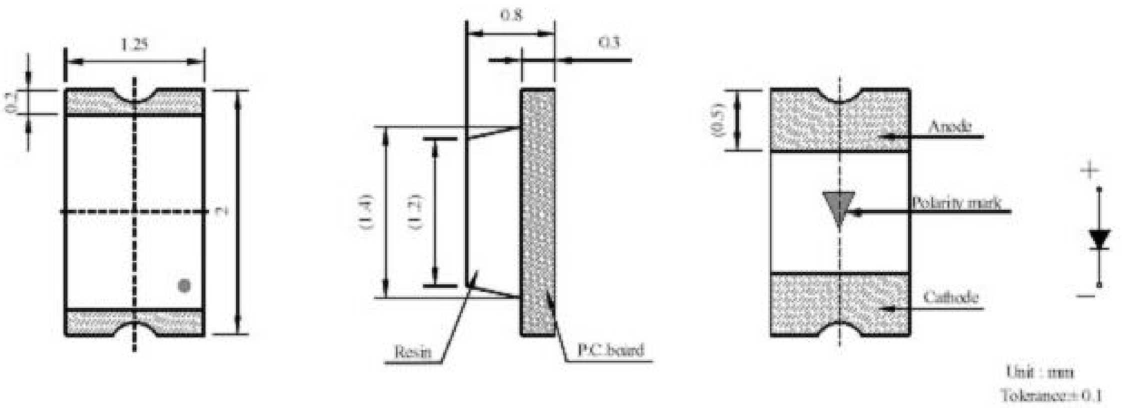
REASON 原因: ______________________________
____

客户认可签章:
CUSTOMER SIGNATURE: ______________

1、特性：

1.1 封装尺寸：2.0*1.2*0.8mm
1.2 发光颜色：黄色
1.3 发光类型：单色型
1.4 焊接方式：回流焊
1.5 符合 RoHS 标准

2、成品外观尺寸



备注：

1. 所有尺寸均以 mm 为单位
2. 在没有明确标注的情况下，公差均为 ±0. 10mm

Page：1 of 7

3、最大绝对标称值（环境温度=25℃）

参数	缩写	标称值	单位
顺向电流	I_F	20	mA
顺向峰值电流 *1	I_{FP}	100	mA
反向电压	VR	5	V
焊接温度	Tsol	回流焊：250 ˚C，8sec. 手工焊：300 ˚C，3sec.	
使用温度	Topr	-40˚C~+85	
储存温度	Tstg	-40˚C~+85	

*I_{FP} 条件：脉宽≤0.1msec，周期≤1/10

4、光电特性参数（环境温度=25˚C）：

参数	缩写	最小值	典型值	最大值	单位	条件
顺向电压	Vf	2.0		2.2	V	
亮度	Iv	95	-	113	mcd	IF=20mA
发光角度	2Θ1/2	-	120	-	deg	
反向电流	IR	-	-	1	μA	VR=5V
波长	WLD	592		594	nm	IF=20mA

备注：1．亮度偏差：±5%

2．电压偏差：±0.03V

3．波长偏差：±1nm

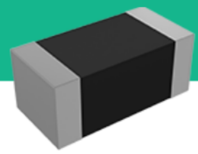
Page：2 of 7

Section 8
Sunlord

Designators	Part / Value	Qty
FB1,FB2	GZ2012D601TF	2

RoHS evidence document: **2310301640_Sunlord-GZ2012D601TF_C1017.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

Multilayer Chip Ferrite Bead – GZ Series



Operating temp. : -55℃ ~+125℃

FEATURES

- Internal silver printed layers and magnetic shielded structures to minimize crosstalk.
- Can be used in a wide range of frequency (from dozens of MHz to hundreds of MHz) to suppress EMI.
- Three types material and wide range of impedance values for various applications.

APPLICATIONS

- Noise suppression for low speed signal of electric equipments such as computers and peripheral devices, smart wearable device, LCD TVs, communication equipments, OA equipments, etc.

PRODUCT IDENTIFICATION

1	2	3	4	5	6	7
GZ	3216	D	121	T	F	(A99)

1

Type	
GZ	Chip Ferrite Bead for General Use

2

External Dimensions (L×W) (mm)	
0603 [0201]	0.6×0.3
1005 [0402]	1.0×0.5
1608 [0603]	1.6×0.8
2012 [0805]	2.0×1.25
3216 [1206]	3.2×1.6

3

Material Code
D, E, U

4

Nominal Impedance	
Example	Nominal Value
300	30Ω
121	120Ω
102	1000Ω

5

Packing	
T	Tape & Reel

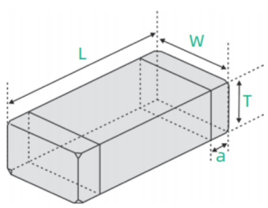
6

Hazardous Substance Free Products
F

7

Internal Code
A99

SHAPE AND DIMENSIONS



Unit: mm [inch]

Type	L	W	T	a
GZ0603 [0201]	0.6±0.05 [.024±.002]	0.3±0.05 [.012±.002]	0.3±0.05 [.012±.002]	0.15±0.05 [.006±.002]
GZ1005 [0402]	1.0±0.15 [.039±.006]	0.5±0.15 [.020±.006]	0.5±0.15 [.020±.006]	0.25±0.1 [.010±.004]
GZ1608 [0603]	1.6±0.15 [.063±.006]	0.8±0.15 [.031±.006]	0.8±0.15 [.031±.006]	0.3±0.2 [.012±.008]
GZ2012 [0805]	2.0 (+0.3, -0.1) [.079 (+.012, -.004)]	1.25±0.2 [.049±.008]	0.85±0.2 [.033±.008]	0.5±0.3 [.020±.012]
GZ3216 [1206]	3.2±0.2 [.126±.008]	1.6±0.2 [.063±.008]	1.1±0.2 [0.043±0.008]	0.5±0.3 [.020±.012]

Sunlord Industrial Park, Dafuyuan Industrial Zone, Guanlan, Shenzhen, China 518110 Tel: 0086-755-29832333 Fax: 0086-755-82269029 E-Mail: sunlord@sunlordinc.com

Shenzhen Sunlord Electronics Co., Ltd.

SPECIFICATIONS GZ0603 TYPE

Part Number	Impedance	Z Test Frequency	Max. DC Resistance	Max. Rated Current	Thickness
Units	Ω	MHz	Ω	mA	mm [inch]
Symbol	Z	Freq.	DCR	Ir	T
GZ0603D600TF	60±25%	100	0.40	200	0.3±0.05 [.012±.002]
GZ0603D750TF	75±25%	100	0.40	200	
GZ0603D800TF	80±25%	100	0.60	200	
GZ0603D121TF	120±25%	100	0.80	200	
GZ0603D241TF	240±25%	100	1.00	200	
GZ0603D471TF	470±25%	100	1.40	200	
GZ0603D601TF	600±25%	100	1.70	200	
GZ0603D102TF	1000±25%	100	2.50	100	
GZ0603U100TF	5~15	100	0.10	500	
GZ0603U700TF	70±25%	100	0.40	200	
GZ0603U800TF	80±25%	100	0.40	200	
GZ0603U121TF	120±25%	100	0.50	200	
GZ0603U241TF	240±25%	100	0.80	200	
GZ0603U601TF	600±25%	100	1.50	100	
GZ0603U102TF	1000±25%	100	2.50	100	

GZ1005 TYPE

Part Number	Impedance	Z Test Frequency	Max. DC Resistance	Max. Rated Current	Thickness
Units	Ω	MHz	Ω	mA	mm [inch]
Symbol	Z	Freq.	DCR	Ir	T
GZ1005D100TF	0~15	100	0.05	500	0.5±0.15 [.020±.006]
GZ1005D310TF	31±25%	100	0.20	300	
GZ1005D600TF	60±25%	100	0.30	200	
GZ1005D800TF	80±25%	100	0.35	200	
GZ1005D121TF	120±25%	100	0.40	200	
GZ1005D221TF	220±25%	100	0.45	150	
GZ1005D301TF	300±25%	100	0.50	100	
GZ1005D421TF	420±25%	100	0.60	100	
GZ1005D501TF	500±25%	100	0.80	100	
GZ1005D601TF	600±25%	100	0.90	100	
GZ1005D751TF	750±25%	100	1.00	100	
GZ1005D102TF	1000±25%	100	1.20	100	
GZ1005D152TF	1500±25%	100	1.60	100	
GZ1005D182TF	1800±25%	100	2.00	50	
GZ1005E800TF	80±25%	100	0.35	200	
GZ1005E121TF	120±25%	100	0.40	200	
GZ1005E241TF	240±25%	100	0.50	200	
GZ1005E601TF	600±25%	100	0.90	100	
GZ1005U100TF	0~15	100	0.05	500	
GZ1005U300TF	30±25%	100	0.20	300	
GZ1005U700TF	70±25%	100	0.30	200	
GZ1005U121TF	120±25%	100	0.40	200	
GZ1005U221TF	220±25%	100	0.50	100	
GZ1005U301TF	300±25%	100	0.60	100	
GZ1005U421TF	420±25%	100	0.80	100	
GZ1005U601TF	600±25%	100	0.90	100	
GZ1005U102TF	1000±25%	100	1.20	100	

Shenzhen Sunlord Electronics Co., Ltd.

SPECIFICATIONS GZ1608 TYPE

Part Number	Impedance	Z Test Frequency	Max. DC Resistance	Max. Rated Current	Thickness
Units	Ω	MHz	Ω	mA	mm [inch]
Symbol	Z	Freq.	DCR	Ir	T
GZ1608D110TF	0~15	100	0.05	2000	0.8±0.15 [.031±.006]
GZ1608D300TF	30±25%	100	0.05	2000	
GZ1608D600TF	60±25%	100	0.10	500	
GZ1608D800TF	80±25%	100	0.15	400	
GZ1608D101TF	100±25%	100	0.20	300	
GZ1608D121TF	120±25%	100	0.20	300	
GZ1608D221TF	220±25%	100	0.30	300	
GZ1608D301TF	300±25%	100	0.35	200	
GZ1608D471TF	470±25%	100	0.45	200	
GZ1608D601TF	600±25%	100	0.45	200	
GZ1608D751TF	750±25%	100	0.50	200	
GZ1608D102TF	1000±25%	100	0.60	200	
GZ1608D152TF	1500±25%	100	0.70	150	
GZ1608D182TF	1800±25%	100	0.90	100	
GZ1608D202TF	2000±25%	100	1.20	100	
GZ1608D222TF	2200±25%	100	1.20	100	
GZ1608D252TF	2500±25%	100	1.20	100	
GZ1608E121TF	120±25%	100	0.20	300	
GZ1608E181TF	180±25%	100	0.30	300	
GZ1608E601TF	600±25%	100	0.45	200	
GZ1608E102TF	1000±25%	100	0.60	200	
GZ1608U100TF	0~15	100	0.05	2000	
GZ1608U300TF	30±25%	100	0.05	2000	
GZ1608U600TF	60±25%	100	0.10	500	
GZ1608U121TF	120±25%	100	0.20	300	
GZ1608U221TF	220±25%	100	0.30	300	
GZ1608U301TF	300±25%	100	0.35	200	
GZ1608U471TF	470±25%	100	0.40	200	
GZ1608U601TF	600±25%	100	0.50	200	
GZ1608U102TF	1000±25%	100	0.60	200	

GZ2012 TYPE

Part Number	Impedance	Z Test Frequency	Max. DC Resistance	Max.Rated Current	Thickness
Units	Ω	MHz	Ω	mA	mm [inch]
Symbol	Z	Freq.	DCR	Ir	T
GZ2012D070TF	0~15	100	0.04	2000	0.85±0.2 [.033±.008]
GZ2012D190TF	19±25%	100	0.04	2000	
GZ2012D300TF	30±25%	100	0.05	1500	
GZ2012D800TF	80±25%	100	0.10	1000	
GZ2012D121TF	120±25%	100	0.15	800	
GZ2012D181TF	180±25%	100	0.18	700	
GZ2012D221TF	220±25%	100	0.20	600	
GZ2012D301TF	300±25%	100	0.20	500	
GZ2012D421TF	420±25%	100	0.30	500	
GZ2012D501TF	500±25%	100	0.30	500	
GZ2012D601TF	600±25%	100	0.30	500	
GZ2012D751TF	750±25%	100	0.35	500	
GZ2012D102TF	1000±25%	100	0.35	500	
GZ2012D152TF	1500±25%	100	0.40	500	

Multilayer Chip Ferrite Bead
Wire Wound Ferrite Bead
Multilayer Chip Common Mode Filter
Wire Wound Chip Common Mode Choke Coil for Signal Line

Sunlord expert in e components Specifications subject to change without notice. Please check our website for latest information. Revised 2023/06/01
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Shenzhen Sunlord Electronics Co., Ltd.

SPECIFICATIONS GZ2012 TYPE

Part Number	Impedance	Z Test Frequency	Max. DC Resistance	Max.Rated Current	Thickness
Units	Ω	MHz	Ω	mA	mm [inch]
Symbol	Z	Freq.	DCR	Ir	T
GZ2012D202TF	2000±25%	100	0.50	500	0.85±0.2 [.033±.008]
GZ2012D252TF	2500±25%	100	0.70	200	
GZ2012E800TF	80±25%	100	0.10	1000	
GZ2012E181TF	180±25%	100	0.20	600	
GZ2012E301TF	300±25%	100	0.20	500	
GZ2012E501TF	500±25%	100	0.30	500	
GZ2012E601TF	600±25%	100	0.30	500	
GZ2012E102TF	1000±25%	100	0.35	500	
GZ2012U100TF	0~15	100	0.04	2200	
GZ2012U170TF	17±25%	100	0.04	2000	
GZ2012U300TF	30±25%	100	0.05	1500	
GZ2012U700TF	70±25%	100	0.10	1000	
GZ2012U121TF	120±25%	100	0.15	800	
GZ2012U221TF	220±25%	100	0.20	600	
GZ2012U301TF	300±25%	100	0.20	500	
GZ2012U421TF	420±25%	100	0.25	500	
GZ2012U601TF	600±25%	100	0.30	500	
GZ2012U102TF	1000±25%	100	0.40	500	

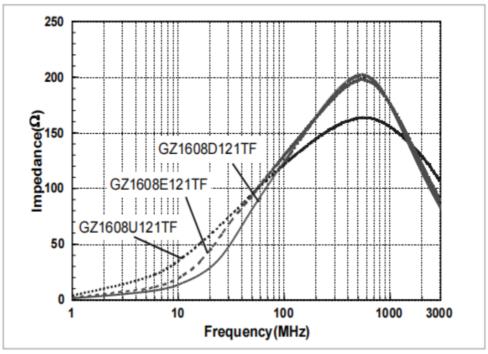
GZ3216 TYPE

Part Number	Impedance	Z Test Frequency	Max. DC Resistance	Max.Rated Current	Thickness
Units	Ω	MHz	Ω	mA	mm [inch]
Symbol	Z	Freq.	DCR	Ir	T
GZ3216D000TF	0~15	100	0.03	2200	0.85±0.2 [.033±.008]
GZ3216D310TF	31±25%	100	0.05	2000	
GZ3216D600TF	60±25%	100	0.10	1000	
GZ3216D301TF	300±25%	100	0.20	600	
GZ3216D100TFA99	0~15	100	0.05	2000	1.10±0.2 [0.043±0.008]
GZ3216D190TFA99	19±25%	100	0.05	2000	
GZ3216D800TFA99	80±25%	100	0.10	1000	
GZ3216D101TFA99	100±25%	100	0.10	1000	
GZ3216D121TFA99	120±25%	100	0.10	1000	
GZ3216D151TFA99	150±25%	100	0.15	800	
GZ3216D221TFA99	220±25%	100	0.20	600	
GZ3216D501TFA99	500±25%	100	0.30	600	
GZ3216D601TFA99	600±25%	100	0.30	600	
GZ3216D102TFA99	1000±25%	100	0.60	500	
GZ3216D122TFA99	1200±25%	100	0.60	300	
GZ3216U601TFA99	600±25%	100	0.30	600	

※: Products with other electrical characteristics can be provided upon customer's request. Please contact your local sales.

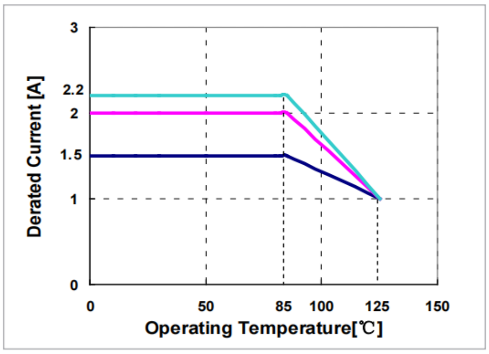
TYPICAL ELECTRICAL CHARACTERISTICS

D, E, U Material Comparison



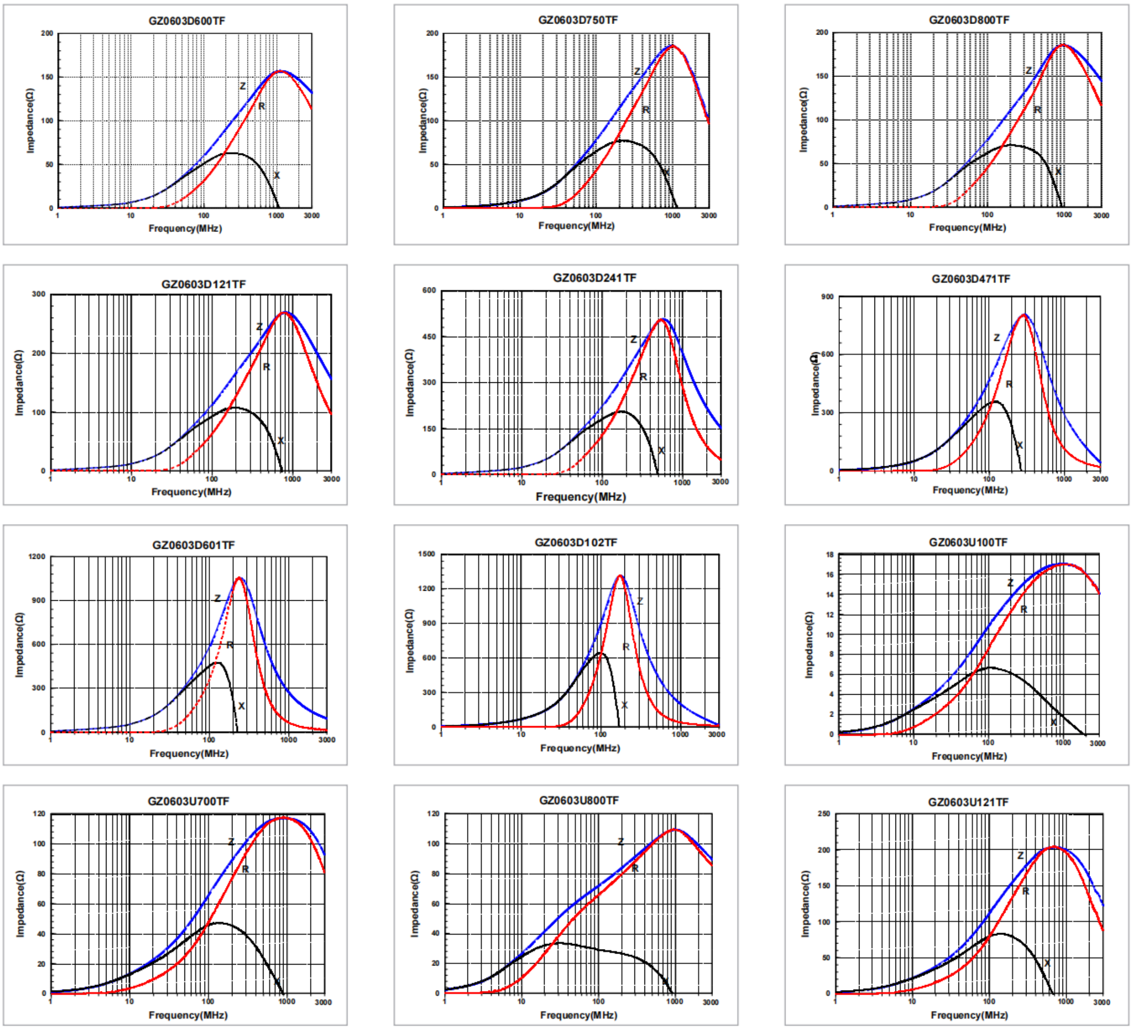
Rated Current

When operating temperatures exceeding +85°C , derating of current is necessary for chip ferrite beads for which rated current is 1000mA over. Please apply the derating curve shown in chart according to the operating temperature.



DETAIL ELECTRICAL CHARACTERISTICS

GZ0603 TYPE



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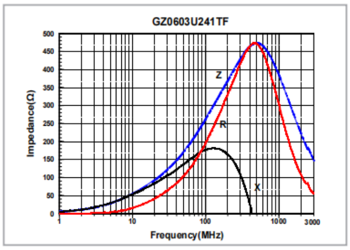
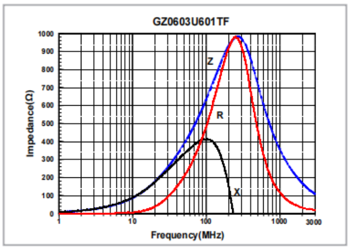
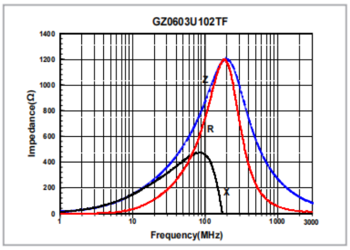
DETAIL ELECTRICAL CHARACTERISTICS

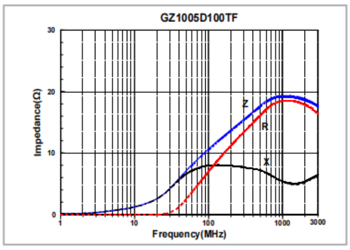
GZ0603 TYPE

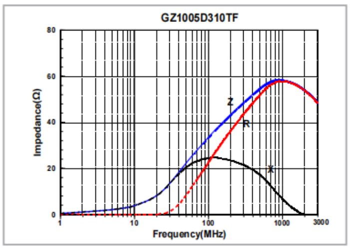


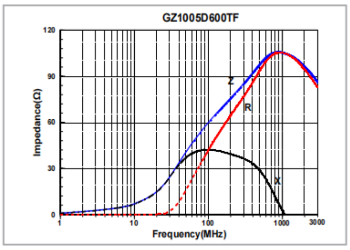


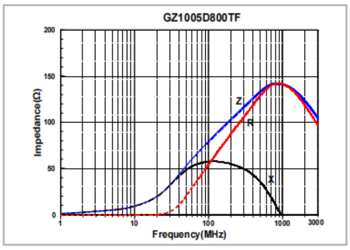


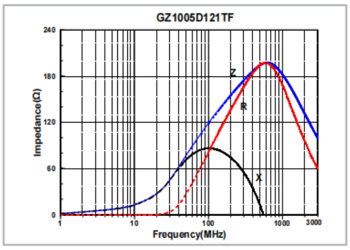
GZ1005 TYPE

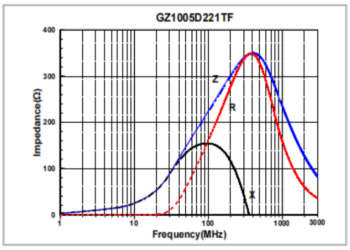


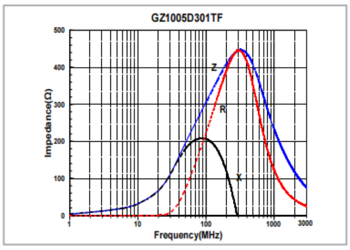


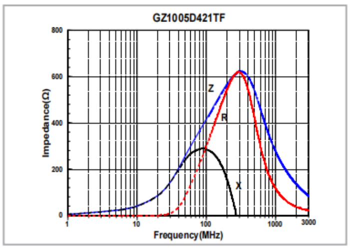


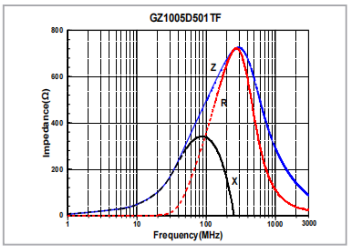


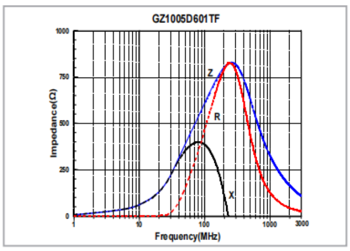


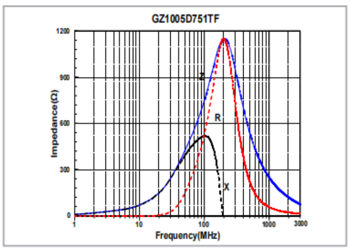


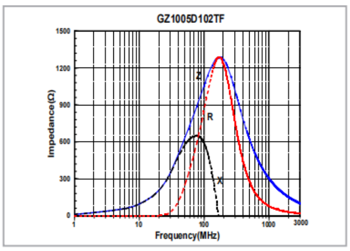






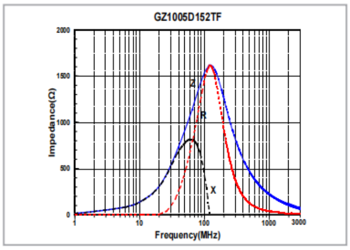


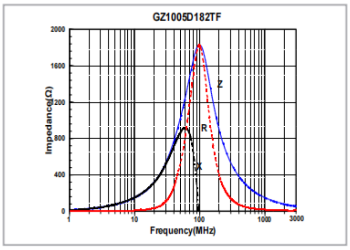


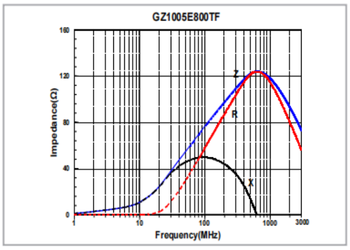


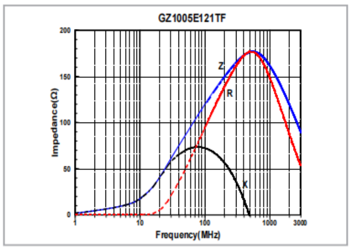
DETAIL ELECTRICAL CHARACTERISTICS

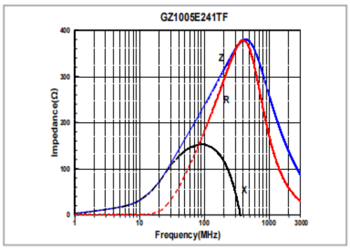
GZ1005 TYPE

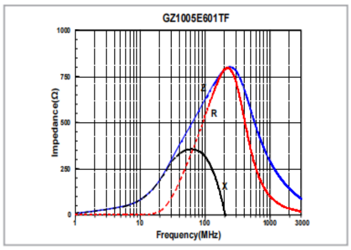


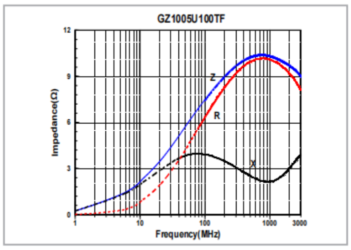


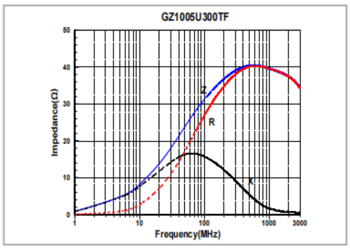


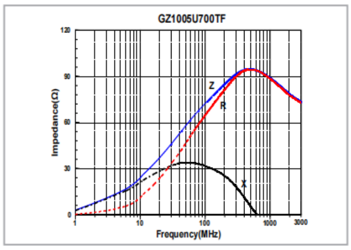


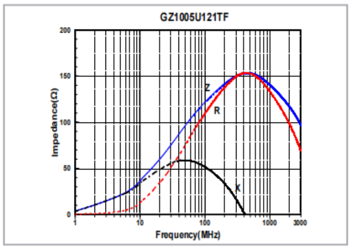


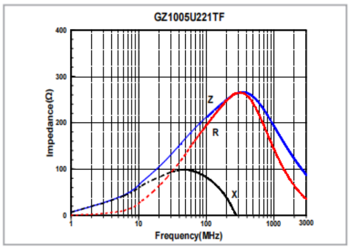


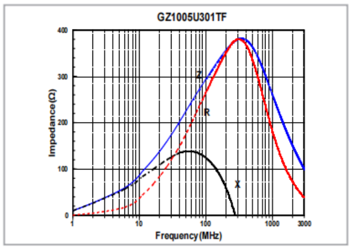


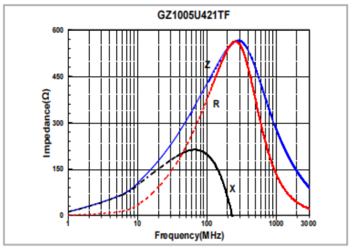


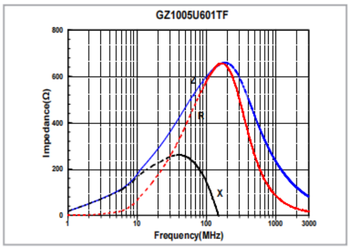


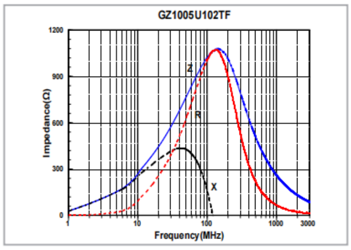












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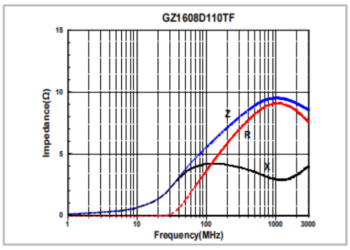
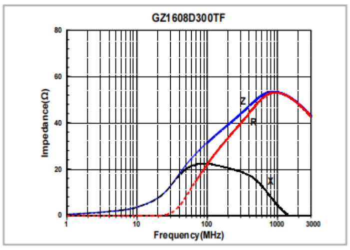
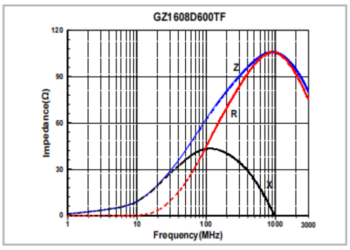
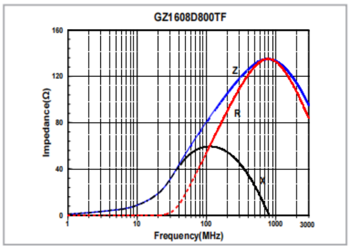
DETAIL ELECTRICAL CHARACTERISTICS

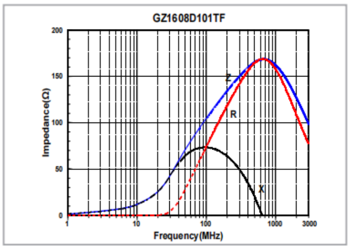
GZ1608 TYPE

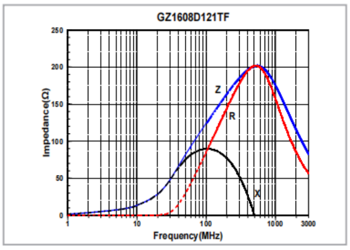


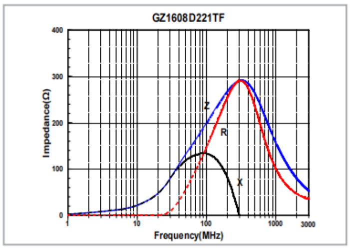


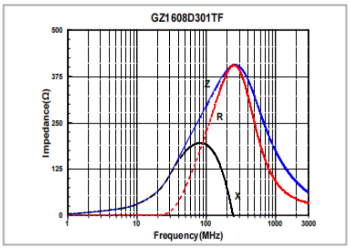


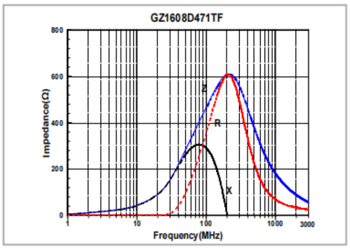


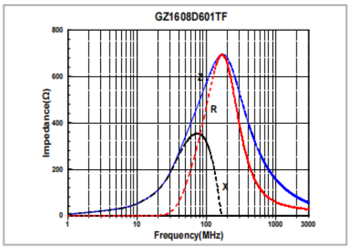


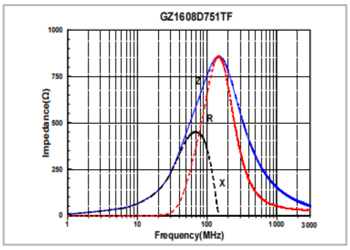


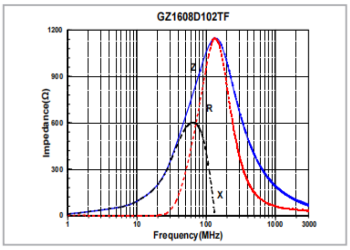


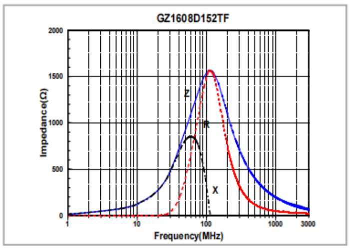


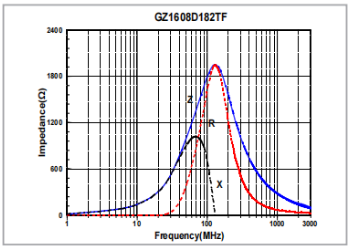


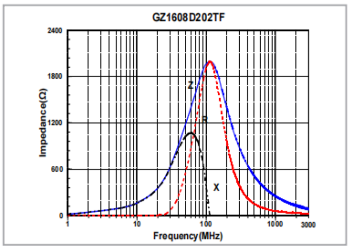


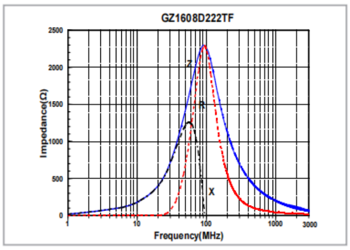


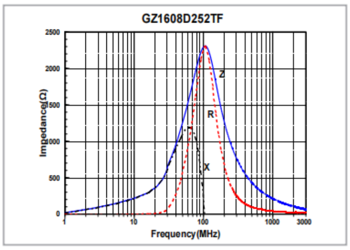


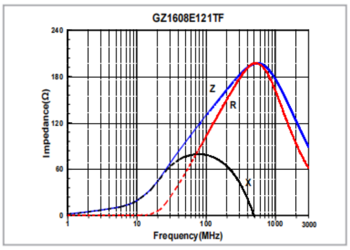






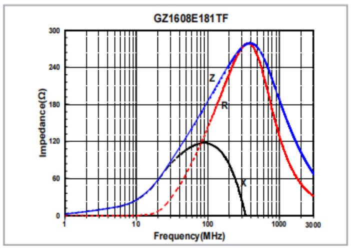


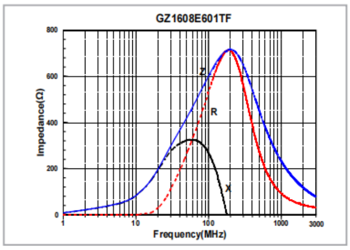


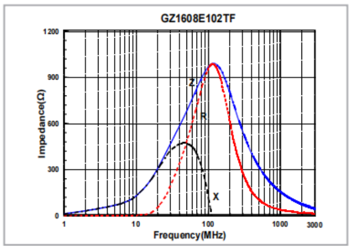


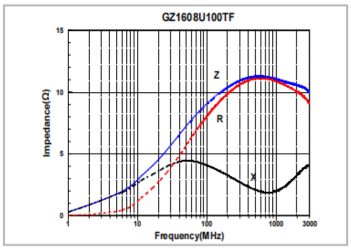
DETAIL ELECTRICAL CHARACTERISTICS

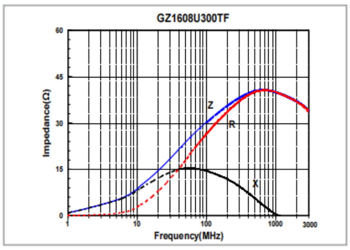
GZ1608 TYPE

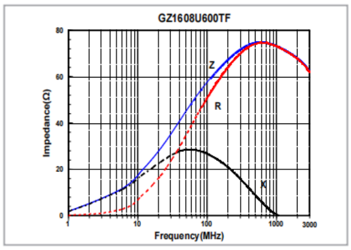


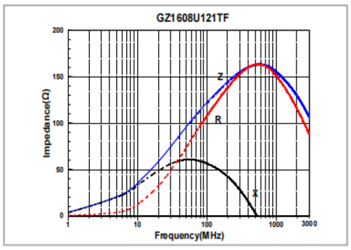


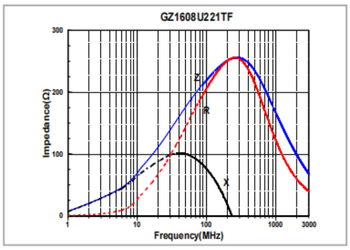


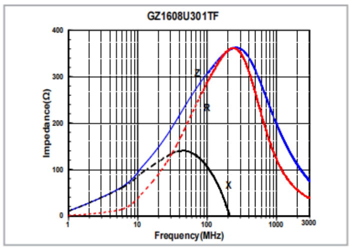


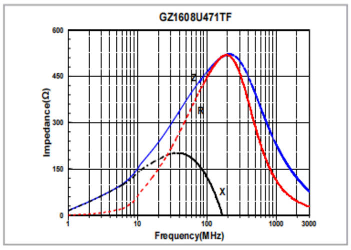


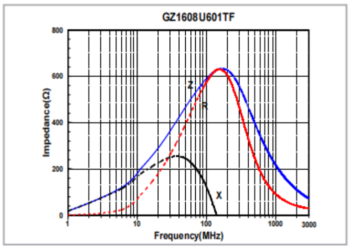


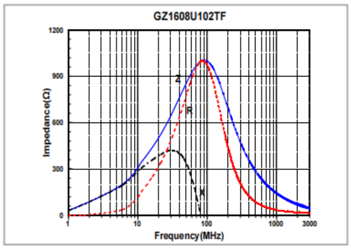




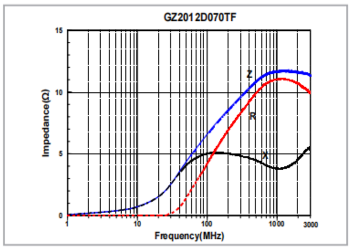


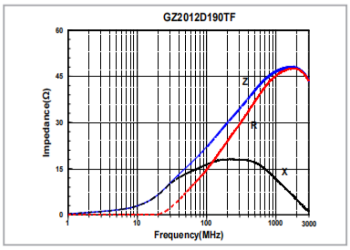


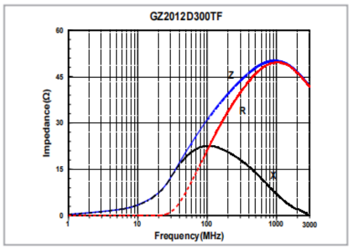




GZ2012 TYPE







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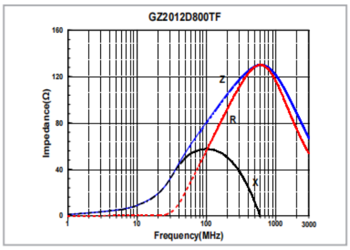
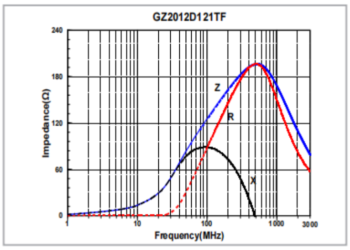
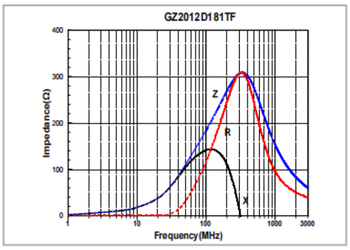
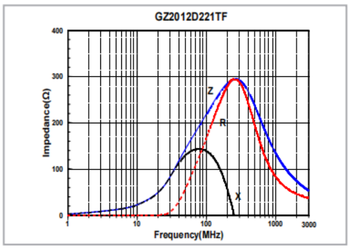
DETAIL ELECTRICAL CHARACTERISTICS

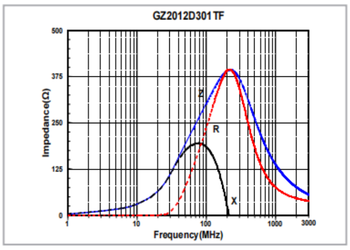
GZ2012 TYPE

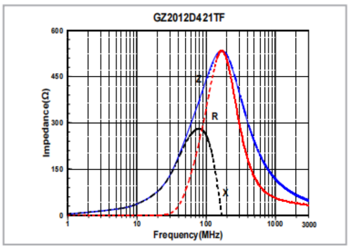


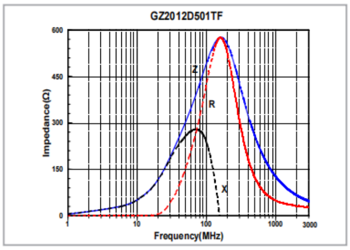


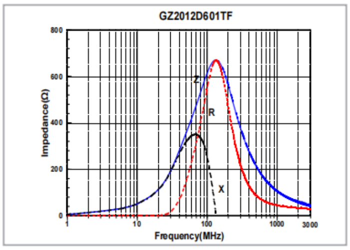


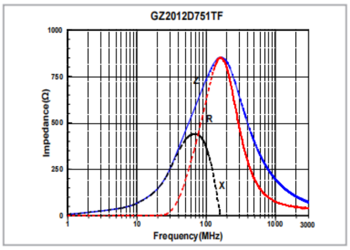


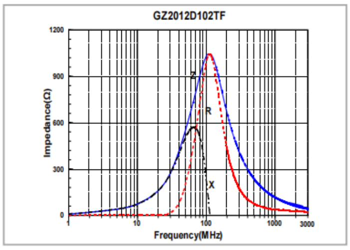


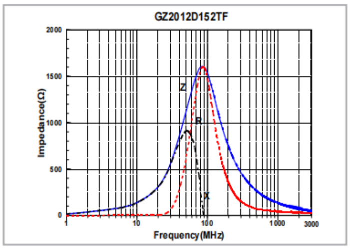


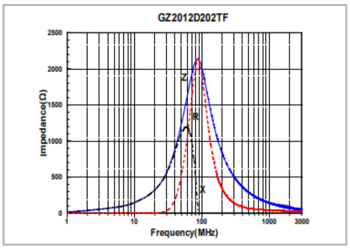


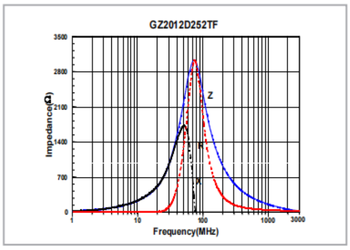


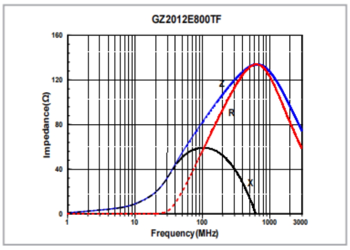


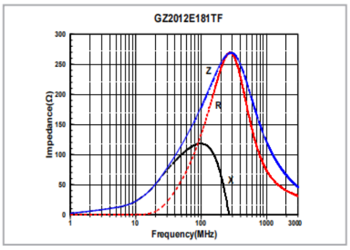


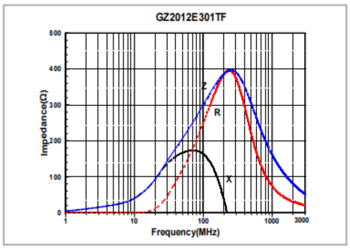


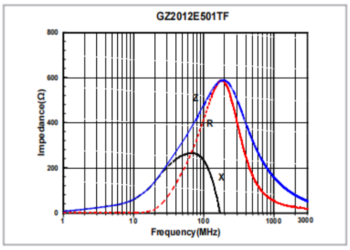


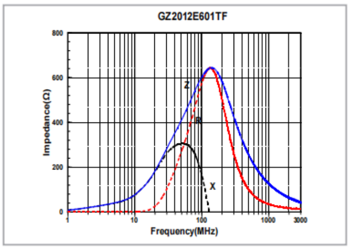






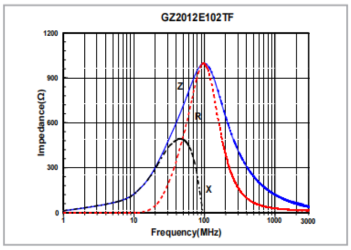


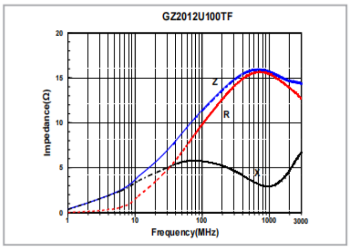


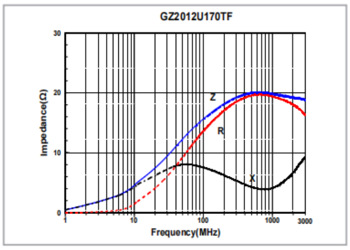


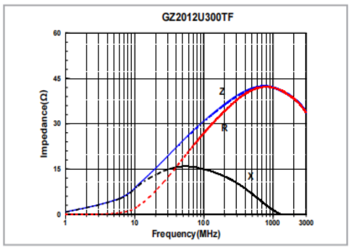
DETAIL ELECTRICAL CHARACTERISTICS

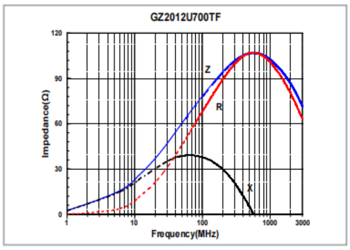
GZ2012 TYPE

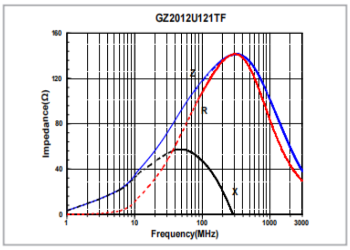


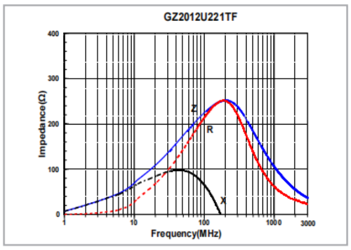


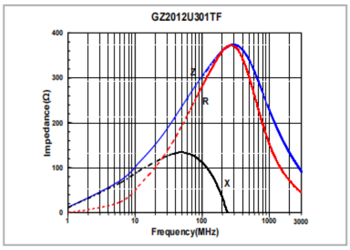


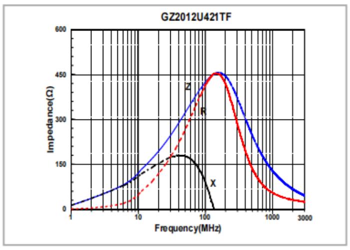


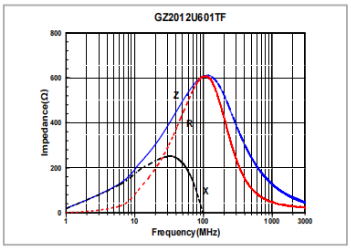


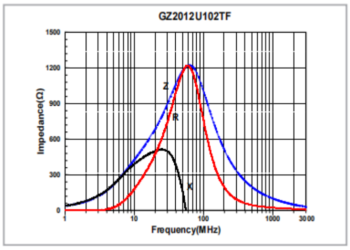




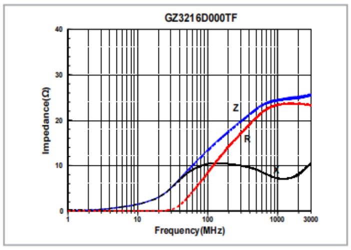


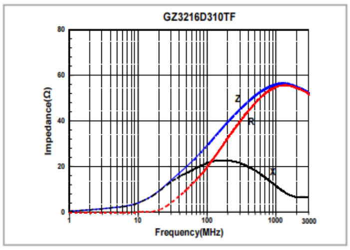


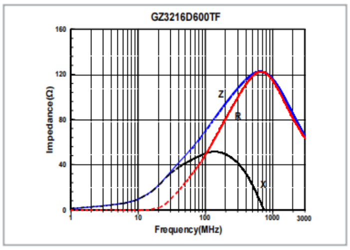




GZ3216 TYPE



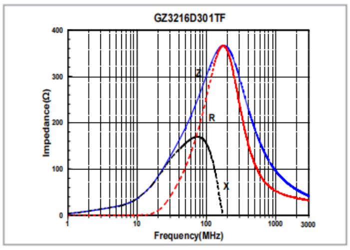


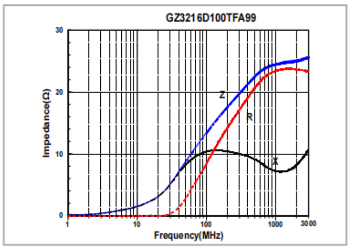


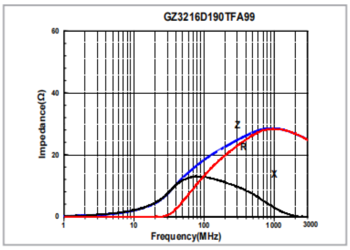
Shenzhen Sunlord Electronics Co., Ltd.

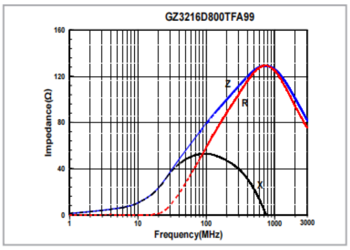
DETAIL ELECTRICAL CHARACTERISTICS

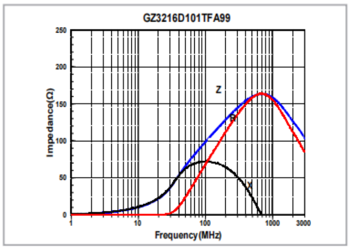
GZ3216 TYPE

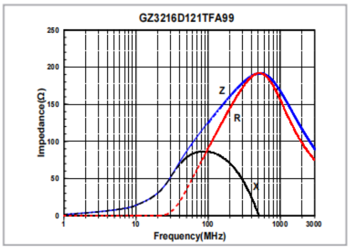


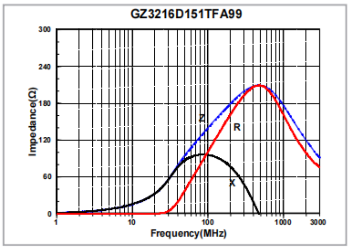


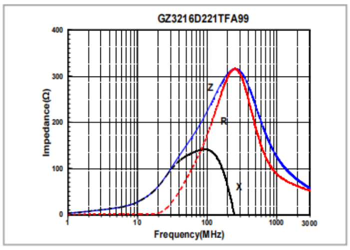


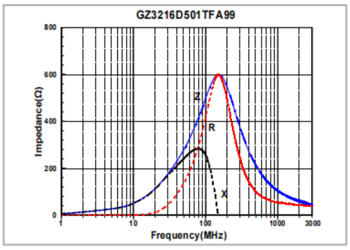


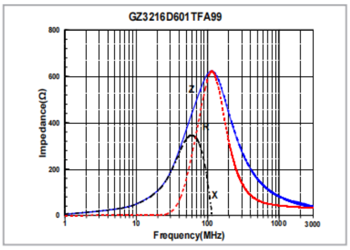


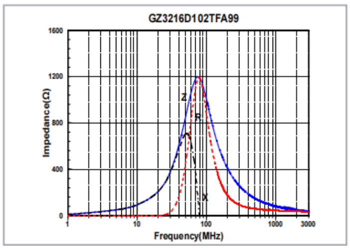


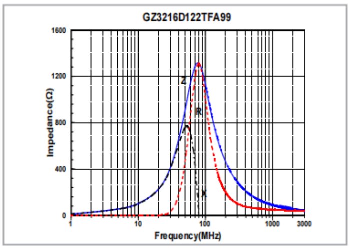


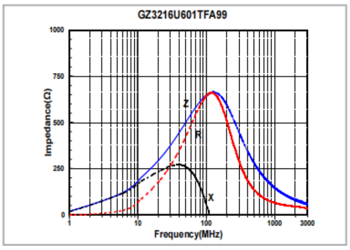












All information and data presented in this Catalog are for information only. You are requested to approve our product specification before your ordering.

Section 9
JUSHUO

Designators	Part / Value	Qty
J1-J36	FH34SRJ-14S-0.5SH	36

RoHS evidence document: **FH34SRJ-14S-0.5SH(50)_CL0580-1252-8-50_SpecSheet_0000414509.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

APPLICABLE STANDARD				
RATING	OPERATING TEMPERATURE RANGE	-55 °C TO 105 °C	STORAGE TEMPERATURE RANGE	−10 ℃ TO 50 ℃ (PACKED CONDITION)
	VOLTAGE	50 V AC / DC	OPERATING OR STORAGE HUMIDITY RANGE	RELATIVE HUMIDITY 90 % MAX (NOT DEWED)
	CURRENT	0.5 A	APPLICABLE CABLE	t=0.3±0.03mm, GOLD PLATING

SPECIFICATIONS

ITEM	TEST METHOD	REQUIREMENTS	QT	AT
CONSTRUCTION				
GENERAL EXAMINATION	VISUALLY AND BY MEASURING INSTRUMENT.	ACCORDING TO DRAWING.	×	×
MARKING	CONFIRMED VISUALLY.		×	×
ELECTRICAL CHARACTERISTICS				
VOLTAGE PROOF	250 V AC FOR 1 min.	NO FLASHOVER OR BREAKDOWN.	×	×
INSULATION RESISTANCE	100 V DC.	500 MΩ MIN.	×	×
CONTACT RESISTANCE	AC/DC 20 mV MAX (AC:1 KHz) , 1 mA .	100 mΩ MAX. INCLUDING FPC,FFC BULK RESISTANCE (L=8mm)	×	×
MECHANICAL CHARACTERISTICS				
VIBRATION	FREQUENCY 10 TO 55 Hz, HALF AMPLITUDE 0.75 mm, FOR 10 CYCLES IN 3 AXIAL DIRECTIONS.	① NO ELECTRICAL DISCONTINUITY OF 1 μs. ② CONTACT RESISTANCE: 100 mΩ MAX. ③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
SHOCK	981 m/s^2 , DURATION OF PULSE 6 ms AT 3 TIMES IN 3 BOTH AXIAL DIRECTIONS.		×	—
MECHANICAL OPERATION	20 TIMES INSERTIONS AND EXTRACTIONS.	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
FPC RETENTION FORCE	MEASURED BY APPLICABLE FPC. (THICKNESS OF FPC SHALL BE t=0.30mm AT INITIAL CONDITION.)	DIRECTION OF INSERTION : (TOP CONTACT) 0.2N × NUMBER OF CONTACTS MIN. (BOTTOM CONTACT) 0.3N × NUMBER OF CONTACTS MIN. (***note 1***)	×	—
ENVIRONMENTAL CHARACTERISTICS				
CORROSION SALT MIST	EXPOSED AT 35±2 ℃ , 5 % SALT WATER SPRAY FOR 96 h.	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. ③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR.	×	—
RAPID CHANGE OF TEMPERATURE	TEMPERATURE-55→+15TO+35→ +105→+15TO+35°C ⚠2 TIME 30→ 2 TO 3 → 30→ 2 TO 3 min UNDER 5 CYCLES.	① CONTACT RESISTANCE: 100 mΩ MAX. ② INSULATION RESISTANCE: 50 MΩ MIN. ③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
DAMP HEAT (STEADY STATE)	EXPOSED AT 40±2 °C, RELATIVE HUMIDITY 90 TO 95 %, 96 h.		×	—
DAMP HEAT,CYCLIC	EXPOSED AT -10 TO +65 °C, RELATIVE HUMIDITY 90 TO 96 %, 10 CYCLES,TOTAL 240 h.	① CONTACT RESISTANCE: 100 mΩ MAX. ② INSULATION RESISTANCE: 1 MΩ MIN. (AT HIGH HUMIDITY) ③ INSULATION RESISTANCE: 50 MΩ MIN. (AT DRY) ④ NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—

	COUNT	DESCRIPTION OF REVISIONS	DESIGNED	CHECKED	DATE
⚠2	3	DIS-F-00005614	SE. YOKOYAMA	HS. HIRAHARA	20200611

REMARK

This product is RoHS compliant.

Unless otherwise specified, refer to IEC 60512.

APPROVED	MO. ISHIDA	20131129
CHECKED	HS. SAKAMOTO	20131129
DESIGNED	YS. EBI	20131128
DRAWN	NM. SANPEI	20131128

Note QT:Qualification Test AT:Assurance Test X:Applicable Test

HRS	SPECIFICATION SHEET	DRAWING NO.	ELC4-159714-04		
		PART NO.	FH34SRJ-*S-0.5SH(50)		
	HIROSE ELECTRIC CO., LTD.	CODE NO.	CL580	⚠2	1/2

FORM HD0011-2-1

SPECIFICATIONS

ITEM	TEST METHOD	REQUIREMENTS	QT	AT
DRY HEAT	EXPOSED AT 105±2 °C, 96 h.	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
COLD	EXPOSED AT -55±3°C, 96 h.		×	—
SULPHUR DIOXIDE [JIS C 60068-2-42]	EXPOSED AT 40±2 °C , RELATIVE HUMIDITY 80±5% 25±5 ppm FOR 96 h.	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. ③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR.	×	—
HYDROGEN SULPHIDE [JIS C 60068-2-43]	EXPOSED AT 40±2 °C , RELATIVE HUMIDITY 80±5% , 10 TO 15 ppm FOR 96 h.		×	—
SOLDERABILITY	SOLDERED AT SOLDER TEMPERATURE, 235±5°C FOR IMMERSION DURATION, 2±0.5 sec.	A NEW UNIFORM COATING OF SOLDER SHALL COVER A MINIMUM OF 95 % OF THE SURFACE BEING IMMERSED.	×	—
RESISTANCE TO SOLDERING HEAT	1) REFLOW SOLDERING : PEAK TMP. 250 °C MAX . REFLOW TMP. OVER 230 °C WITHIN 60 sec. 2) SOLDERING IRONS : TMP. 350 ± 10 °C FOR 5±1 sec .	NO DEFORMATION OF CASE OF EXCESSIVE LOOSENESS OF THE TERMINALS.	×	—

(***note1***)

FASTEN FPC ON PCB OR SOMETHING FIXED IF FORCE IN VERTICAL DIRECTION SHALL BE PREDICTED.
DO NOT CLOSE THE ACTUATOR BEFORE INSERTING FPC EVEN AFTER THE CONNECTOR IS MOUNTED ONTO A PCB. CLOSING THE ACTUATOR WITHOUT FPC COULD MAKE THE CONTACT GAP SMALLER, WHICH INCREASES THE FPC INSERTION FORCE.
THIS CONNECTOR HAS CONTACTS ON THE BOTH TOP AND BOTTOM.
2 THERE'S A CASE WHICH FPC/FFC RETENTION FORCE DOESN'T FULFILL THE VALUE, BECAUSE FPC SPECIFICATION AFFECTS THE RESULT OF FPC/FFC RETENTION FORCE.

Note QT:Qualification Test AT:Assurance Test X:Applicable Test		DRAWING NO.	ELC4-159714-04
HRS	SPECIFICATION SHEET	PART NO.	FH34SRJ-*S-0.5SH(50)
	HIROSE ELECTRIC CO., LTD.	CODE NO	CL580 2 2/2

FORM HD0011-2-2

Section 10
Hirose

Designators	Part / Value	Qty
J37	FH34SRJ-12S-0.5SH	1

RoHS evidence document: **FH34SRJ-12S-0.5SH(99)_CL0580-1253-0-99_SpecSheet_0000414526.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

APPLICABLE STANDARD				
RATING	OPERATING TEMPERATURE RANGE	-55 °C TO 105 °C	STORAGE TEMPERATURE RANGE	−10 ℃ TO 50 ℃ (PACKED CONDITION)
	VOLTAGE	50 V AC / DC	OPERATING OR STORAGE HUMIDITY RANGE	RELATIVE HUMIDITY 90 % MAX (NOT DEWED)
	CURRENT	0.5 A	APPLICABLE CABLE	t=0.3±0.03mm, GOLD PLATING

SPECIFICATIONS

ITEM	TEST METHOD	REQUIREMENTS	QT	AT
CONSTRUCTION				
GENERAL EXAMINATION	VISUALLY AND BY MEASURING INSTRUMENT.	ACCORDING TO DRAWING.	×	×
MARKING	CONFIRMED VISUALLY.		×	×
ELECTRICAL CHARACTERISTICS				
VOLTAGE PROOF	250 V AC FOR 1 min.	NO FLASHOVER OR BREAKDOWN.	×	×
INSULATION RESISTANCE	100 V DC.	500 MΩ MIN.	×	×
CONTACT RESISTANCE	AC/DC 20 mV MAX (AC:1 KHz) , 1 mA .	100 mΩ MAX. INCLUDING FPC,FFC BULK RESISTANCE (L=8㎜)	×	×
MECHANICAL CHARACTERISTICS				
VIBRATION	FREQUENCY 10 TO 55 Hz, HALF AMPLITUDE 0.75 mm, FOR 10 CYCLES IN 3 AXIAL DIRECTIONS.	① NO ELECTRICAL DISCONTINUITY OF 1 μs. ② CONTACT RESISTANCE: 100 mΩ MAX. ③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
SHOCK	981 m/s^2 , DURATION OF PULSE 6 ms AT 3 TIMES IN 3 BOTH AXIAL DIRECTIONS.		×	—
MECHANICAL OPERATION	20 TIMES INSERTIONS AND EXTRACTIONS.	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
FPC RETENTION FORCE	MEASURED BY APPLICABLE FPC. (THICKNESS OF FPC SHALL BE t=0.30mm AT INITIAL CONDITION.)	DIRECTION OF INSERTION : (TOP CONTACT) 0.2N × NUMBER OF CONTACTS MIN. (BOTTOM CONTACT) 0.3N × NUMBER OF CONTACTS MIN. (***note 1***)	×	—
ENVIRONMENTAL CHARACTERISTICS				
CORROSION SALT MIST	EXPOSED AT 35±2 ℃ , 5 % SALT WATER SPRAY FOR 96 h.	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. ③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR.	×	—
RAPID CHANGE OF TEMPERATURE	TEMPERATURE-55→+15TO+35→ +105→+15TO+35°C △2 TIME 30→ 2 TO 3 → 30→ 2 TO 3 min UNDER 5 CYCLES.	① CONTACT RESISTANCE: 100 mΩ MAX. ② INSULATION RESISTANCE: 50 MΩ MIN. ③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
DAMP HEAT (STEADY STATE)	EXPOSED AT 40±2 °C, RELATIVE HUMIDITY 90 TO 95 %, 96 h.		×	—
DAMP HEAT,CYCLIC	EXPOSED AT -10 TO +65 °C, RELATIVE HUMIDITY 90 TO 96 %, 10 CYCLES,TOTAL 240 h.	① CONTACT RESISTANCE: 100 mΩ MAX. ② INSULATION RESISTANCE: 1 MΩ MIN. (AT HIGH HUMIDITY) ③ INSULATION RESISTANCE: 50 MΩ MIN. (AT DRY) ④ NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—

	COUNT	DESCRIPTION OF REVISIONS	DESIGNED	CHECKED	DATE
△2	3	DIS-F-00005614	SE. YOKOYAMA	HS. HIRAHARA	20200611

REMARK

This product is RoHS compliant.

Unless otherwise specified, refer to IEC 60512.

APPROVED	MO. ISHIDA	20140124
CHECKED	HS. SAKAMOTO	20140124
DESIGNED	YS. EBI	20140124
DRAWN	NM. SANPEI	20140121

Note QT:Qualification Test AT:Assurance Test X:Applicable Test

		DRAWING NO.	ELC4-159714-05
HRS	SPECIFICATION SHEET	PART NO.	FH34SRJ-*S-0. 5SH(99)
	HIROSE ELECTRIC CO., LTD.	CODE NO.	CL580 △2 1/2

FORM HD0011-2-1

SPECIFICATIONS

ITEM	TEST METHOD	REQUIREMENTS	QT	AT
DRY HEAT	EXPOSED AT 105±2 °C, 96 h. 2	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS.	×	—
COLD	EXPOSED AT -55±3°C, 96 h.		×	—
SULPHUR DIOXIDE [JIS C 60068-2-42]	EXPOSED AT 40±2 °C , RELATIVE HUMIDITY 80±5% 25±5 ppm FOR 96 h.	① CONTACT RESISTANCE: 100 mΩ MAX. ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. ③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR.	×	—
HYDROGEN SULPHIDE [JIS C 60068-2-43]	EXPOSED AT 40±2 °C , RELATIVE HUMIDITY 80±5% , 10 TO 15 ppm FOR 96 h.		×	—
SOLDERABILITY	SOLDERED AT SOLDER TEMPERATURE, 235±5°C FOR IMMERSION DURATION, 2±0.5 sec.	A NEW UNIFORM COATING OF SOLDER SHALL COVER A MINIMUM OF 95 % OF THE SURFACE BEING IMMERSED.	×	—
RESISTANCE TO SOLDERING HEAT	1) REFLOW SOLDERING : PEAK TMP. 250 °C MAX . REFLOW TMP. OVER 230 °C WITHIN 60 sec. 2) SOLDERING IRONS : TMP. 350 ± 10 °C FOR 5±1 sec .	NO DEFORMATION OF CASE OF EXCESSIVE LOOSENESS OF THE TERMINALS.	×	—

(***note1***)

FASTEN FPC ON PCB OR SOMETHING FIXED IF FORCE IN VERTICAL DIRECTION SHALL BE PREDICTED.
DO NOT CLOSE THE ACTUATOR BEFORE INSERTING FPC EVEN AFTER THE CONNECTOR IS MOUNTED ONTO A PCB. CLOSING THE ACTUATOR WITHOUT FPC COULD MAKE THE CONTACT GAP SMALLER, WHICH INCREASES THE FPC INSERTION FORCE.
THIS CONNECTOR HAS CONTACTS ON THE BOTH TOP AND BOTTOM.
2 THERE'S A CASE WHICH FPC/FFC RETENTION FORCE DOESN'T FULFILL THE VALUE, BECAUSE FPC SPECIFICATION AFFECTS THE RESULT OF FPC/FFC RETENTION FORCE.

Note QT:Qualification Test AT:Assurance Test X:Applicable Test		DRAWING NO.	ELC4-159714-05		
HRS	SPECIFICATION SHEET	PART NO.	FH34SRJ-*S-0.5SH(99)		
	HIROSE ELECTRIC CO., LTD.	CODE NO	CL580	2	2/2

FORM HD0011-2-2

Section 11
Hirose

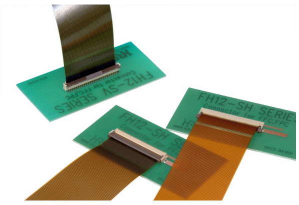
Designators	Part / Value	Qty
J39	FH12-30S-0.5SH	1

RoHS evidence document: **FH12-30S-05SH.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

FH12 Serie

FH12-30S-0.5SH(55) CL0586-0525-1-55

RoHS (EU)2015/863



Tatsächliches Produkt kann sich von der Abbildung unterscheiden

Dokument

Artikelnummerkonfiguration

Artikelnummer	FH12-30S-0.5SH(55)
Steckverbindertyp	Buchse
Bestückungsabstand	0.5 mm
Steckverbinderlänge	19.1 mm
Steckverbinderhöhe	2.0 mm

2D   3D-MODELL (IGES)   3D-MODELL (STEP)

Fußabdruck (Allegro)   Fußabdruck (Altium)

Klicken Sie hier, wenn Sie einen Kauf in Erwägung ziehen. *QA Wie man kauft

INVENTAR/KAUFEN   ARTIKELNUMMERN NACH SERIEN

KONTAKT

Detaillierte Anforderungen

Letztes Update am July. 03, 2024

Produktinformation

Artikelnummer	FH12-30S-0.5SH(55)	CL-Nummer	CL0586-0525-1-55

Grundlegende Informationen

Steckverbindertyp	Buchse	Bestückungsabstand	0.5 mm
Steckverbinderlänge	19.1 mm	Steckverbinderhöhe	2.0 mm
Steckverbinderbreite	6.4 mm	Kontaktorientierung	Unten
ZIF/Non-ZIF/LIF	ZIF	Empfohlenes FFC/FPC	FPC/FFC
FPC/FFC Einsteckrichtung	Horizontal	Anwendbare FPC-Kabelstärke (mm)	
Tiefenmaß (mm) Aktuator OFFEN		Tiefenmaß (mm) Aktuator Zustand SCHLIESSEN	
Aktuatormethode	Vorne	Anzahl der Kontakte	30
Anmerkungen zu der Anzahl der Kontakte		Kontaktabstand	0.5 mm
Anzahl der Reihen (Löt-/Leiterplattenseite)	1	Leistungsmerkmale	
Übertragungsgeschwindigkeit		Betriebstemperatur (max.)	85 ℃
Betriebstemperatur (min.)	-40 ℃	Gleitbereich (XY)(mm)	
Effektive Stecklänge (Z) (mm)		Betriebsspannung (AC)	AC 50.0 V
Betriebsspannung (AC) andere		Betriebsspannung (DC)	DC 50.0 V
Betriebsspannung (DC) andere		Betriebsstrom	0.5 A
Betriebsstrom andere		RoHS (EU)2015/863	Konform
SVHC	31th Nicht enthalten	Steckverbindertyp Detailseite	

Mechanische Angaben

Kontaktbeschichtung	Gold	Beschichtungsstärke	
Steckzyklen	20	Schutz vor dem Aufsteigen des Lötzinns	

Elektrische Spezifikationen

Kontaktwiderstand	50.0 mΩ Max.		

Maßangaben

FFC/FPC-Stärke	0.3 mm	FFC/FPC Anmerkungen	

Verpackungsvorschriften

Abklebeband		Verpackungsart	Spule
Kaufeinheit	2000 PCS	Masse	0.32 g
Empfindlich gegen Aufladungen		Menge / Karton	2000 PCS

Andere

Bemerkung	

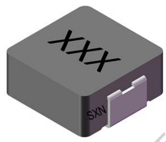
Section 12
SXN

Designators	Part / Value	Qty
L1,L2	2.2uH	2

RoHS evidence document: **2108131530_SXN-Shun-Xiang-Nuo-Elec-SMMS0420-2R2M_C133191.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2 of 28 — RoHS-relevant pages only; the complete document is retained in the project repository.

Molding SMD Power Inductors

SMMS Series Ultra-high current SMT power inductors



◆特征：

- 低直流电阻和超大电流的薄型设计
- 磁屏蔽型抗电磁干扰强适用于高密度安装
- 高可靠性, 通过采用一体成型结构享有卓越的抗震动性
- 由于复合结构, 超低蜂鸣噪声
- 低损耗合金粉末压铸低阻抗, 小寄生电容
- 能效高, 可减少绕线的低直流电阻与磁芯的涡流损耗
- 频率高达 3MHz
- 绝缘最大电压 30VDC
- 符合 RoHS,无卤和 REACH

Features:

- Low RDC and ultra-high current thin design
- Magnetic shielding type, strong anti- electromagnetic Interference, suitable for high- density installation
- High-reliability, High vibration resistance as result of newly developed integral construction
- Ultra Low buzz noise, due to composite construction
- Die-casting by low loss alloy powder low impedance, Small parasitic capacitance
- High efficiency Low DC resistance of winding and low eddy-current loss of the core
- Frequency up to 3MHz
- Absolute maximum voltage 30VDC
- RoHS, Halogen Free and REACH Compliance

◆用途：

- PDA,笔记本,台式机,服务器应用程序
- 大电流 POL 转换器
- 电池供电设备
- 分布式电源系统中的 DC/DC 转换器

Applications:

- PDA , notebook ,desktop ,server applications
- High current POL converters
- Battery powered devices
- DC/DC converters in distributed power systems

◆环境：

- 工作温度：-40℃ 至+125℃

（包括线圈自身温升）

Environmental Data:

- Operating Temperature: -40℃ to +125℃

(Including coils self-temperature rise)

◆试验设备：

- 电感值: WK3260B 或同等仪器
- 电流: WK3260B+WK3265B
- 直流电阻: Chroma 16502 或同等仪器

Test Equipment：

- L: WK3260B LCR meter or equivalent
- Isat & Irms: WK3260B+WK3265B
- DCR:Chroma 16502 or equivalent

◆产品型号：

Product Identification：

SMMS	0420	100	M	T
①	②	③	④	⑤

①

类型 Type	
SMMS	成型贴片功率电感 Molding SMT Power Inductor

②

外形尺寸(L×W×H) (mm) External Dimensions (L×W×H) (mm)	
0420	4.6×4.2×2.0

③

Inductance
10 uH

HENZHEN SHUNXIANGNUO ELECTRONICS CO., LTD. http:// www.sxndz.com 1

Molding SMD Power Inductors

顺翔诺 SXN

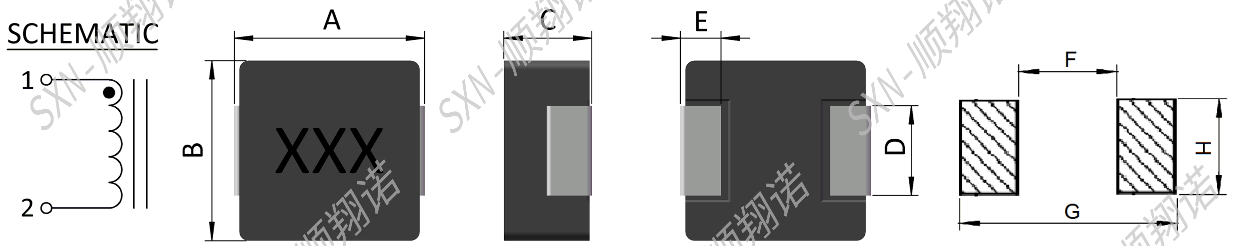
④

公差 Inductance Tolerance
J:±5%,K: ±10%, L: ±15% M: ±20%,P: ±25%, N: ±30%

⑤

包装 Packing	
B	散装Bulk Package
TF	编带Tape & Reel

◆外观尺寸： Shape and Dimensions(dimensions are in mm)：



Recommended
Land Pattern

Part No	ITEM							
	A	B	C	D	E	F	G	H
SMMS0420	4.60±0.30	4.20±0.20	2.00Max	1.50Typ	0.80 Typ	2.20	5.20	2.50
SMMS0520	5.50±0.30	5.20±0.20	2.00Max	2.30 Typ	1.20 Typ	3.00	7.00	2.50
SMMS0530	5.50±0.30	5.20±0.20	3.00Max	2.30 Typ	1.20 Typ	3.00	7.00	2.50
SMMS0624	7.10±0.30	6.60±0.20	2.40Max	3.00 Typ	1.60 Typ	3.70	8.40	3.50
SMMS0630	7.10±0.30	6.60±0.20	3.00Max	3.00 Typ	1.60 Typ	3.70	8.40	3.50
SMMS0650	7.10±0.30	6.60±0.20	5.00Max	3.00 Typ	1.60 Typ	3.70	8.40	3.50
SMMS1040	11.50 Max	10.00±0.30	4.00Max	3.00 Typ	2.00 Typ	4.10	13.60	5.40
SMMS1050	11.50 Max	10.00±0.30	5.00Max	3.00 Typ	2.00 Typ	5.40	13.60	4.10
SMMS1350	13.80±0.50	12.60±0.20	5.50Max	3.70 Typ	2.50 Typ	8.00	14.60	5.00
SMMS1360	13.80±0.50	12.60±0.20	6.50Max	3.70 Typ	2.50 Typ	8.00	14.60	5.00

Section 13
Xinglight

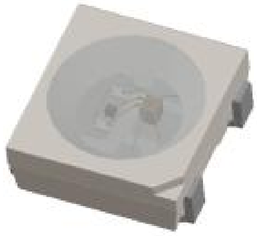
Designators	Part / Value	Qty
LED1-LED36	XL-3030RGBC	36

RoHS evidence document: **2402181502_XINGLIGHT-XL-3030RGBC-WS2812B_C5349958.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2 of 18 — RoHS-relevant pages only; the complete document is retained in the project repository.

XINGLIGHT® Part No.:XL-3030RGBC-WS2812B

XL-3030RGBC-WS2812B **技术数据表** **Technical Data Sheet**

3030 幻彩 贴片式发光二极管



特点（Characteristics）：

＊ 外观尺寸（L/W/H）:3.0*3.0*1.1mm
outline Dimensions (L / w / h): 3.0*3.0*1.1mm

＊ 发光颜色及胶体: 七彩幻彩/雾状胶体
Luminous color and colloid: Colorful magic/mist colloid

＊ 环保工艺符合 ROHS 要求
Environmental protection products Complied With ROHS Directive

＊ 湿气敏感性等级（MSL）:5a 级
Moisture sensitivity level（MSL） : 5a levels

＊ EIA 规范标准包装
EIA standard packaging

＊ 适用于 SMT 贴片自动化生产
Suitable for SMT automatic production

＊ 适用于红外线回流焊制程
Suitable for infrared reflow soldering process

应用领域（Product application）：

＊ LED 全彩发光字灯串
Led full-color luminous word lamp string,

＊ LED 全彩模组
led full-color module

＊ LED 幻彩软硬灯条，LED 护栏管
Led magic color soft and hard light strip, LED guardrail tube

＊ LED 外观，情景照明
Led appearance, scene lighting

＊ LED 异性屏
Led heterosexual screen

＊ 各种电子产品，电器设备跑马灯
All kinds of electronic products, electrical equipment, runninglights

1

XINGLIGHT®　Part No.:XL-3030RGBC-WS2812B

WS2812b 功能特点/Ws2812b Functional characteristics：

1.灯珠内部集成高质量外控单线级联恒流 IC 和优质 RGB LED 芯片，体积小巧，外围简单。

The ball interior integrates high-quality external single-line cascade IC and high-quality RGB LED chips, which are small in size and simple on the periphery.

2.内置 IC 恒流精度高，内部 RGB 芯片预先分光处理。发光高度一致，白光效果纯正。

BUILT-IN IC constant current high precision, internal RGB chip pre-optical processing. High Degree of uniformity of light, white light effect pure.

3.整形转发强化技术，单线数据传输，可无限级联。

Plastic forwarding enhancement technology, single-line data transmission, can be cascaded.

4.数据传输频率 800Kbps/秒，可实现画面刷新速率 30 帧 / 秒时，不小于 1024 点。

The data transmission frequency is 800Kbps per second, and the screen refresh rate can be achieved at 30 frames per second, not less than 1024 points.

5.输出端口 PWM 控制能够实现 256 级灰度调节，端口扫描频率 1.5KHz / s。

The output port PWM control can achieve 256 levels of grayscale adjustment, and the port scan frequency is 1.5 KHz / S.

6.采用优化预置 12mA/通道恒流模式，低压驱动级联数量最大化。高恒流精度，片内误差<1.5%，片间误差<3%。Optimized preset 12mA / channel constant current mode is adopted to maximize the number of low-voltage drive cascades. High constant current accuracy, intraslice error & lt; 1.5 %, interslice error & lt; 3 %.

7.内置低压强化模块，VDD 在 4.5-5.5V 以上 100%正常工作。

With the built-in low-pressure reinforcement module, VDD is 100 % functional above 4.5-5.5 V.

8.超强数据整形能力：接受完本单元数据自动将后续数据整形输出。

Super data shaping ability: accept this unit data automatically will follow the data shaping output.

Section 14
Yageo

Designators	Part / Value	Qty
R1,R2	22 0603	2
R1_1-R1_36,R15,R20,R25	10k 0402	39
R3,R4,R7,R9	5.1K 0402	4
R5,R6,R16,R19,R21	1k 0402	5
R8	5.6K 0402	1
R10,R11,R30	390k 0603 1%	3
R12,R17	150k 0603	2
R13	12k 0603	1
R14	680k 0603	1
R18	10M 0603	1
R22,R23	27 0603	2
R24	2M2 0603	1

RoHS evidence document: **2304140030_YAGEO-RC0603JR-0710ML_C141675.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 10 — RoHS-relevant pages only; the complete document is retained in the project repository.

Innovative Service Around the Globe YAGEO

DATA SHEET

GENERAL PURPOSE CHIP RESISTORS

RC_L series

±0.1%, ±0.5%, ±1%, ±5%

Sizes 0075/0100/0201/0402/0603/0805/ 1206/1210/1218/2010/2512

RoHS compliant & Halogen free

YAGEO Phycomp

Product specification –Mar. 7, 2017 V.7

YAGEO Phycomp

Product specification 2 | 10

Chip Resistor Surface Mount | RC_L | SERIES | 0075 to 2512

SCOPE

This specification describes RC series chip resistors with lead free terminations made by thick film process.

APPLICATIONS

- All general purpose application

FEATURES

- Halogen Free Epoxy
- RoHS compliant
  - Products with lead free terminations meet RoHS requirements
  - Pb-glass contained in electrodes, resistors element and glass are exempted by RoHS
- Reducing environmentally hazardous wastes
- High component and equipment reliability
- Saving of PCB space
- None forbidden-materials used in products/production

ORDERING INFORMATION - GLOBAL PART NUMBER

Global part numbers are identified by the series, size, tolerance, packing type, temperature coefficient, taping reel and resistance value.

GLOBAL PART NUMBER

RC XXXX X X X XX XXXX L
(1) (2) (3) (4) (5) (6) (7)

(1) SIZE

0075/0100/0201/0402/0603/0805/1206/1210/1218/2010/2512

(2) TOLERANCE

B = ±0.1%
D = ±0.5%
F = ±1.0%
J = ±5.0% (for jumper ordering, use code of J)

(3) PACKAGING TYPE

R = Paper taping reel
K = Embossed taping reel
S = ESD safe reel (0075/0100 only)

(4) TEMPERATURE COEFFICIENT OF RESISTANCE

- = Based on spec.

(5) TAPING REEL

07= 7 inch dia. Reel
10=10 inch dia. Reel
13=13 inch dia. Reel
7W = 7 inch dia. Reel & 2 x standard power
7N = 7 inch dia. Reel, ESD safe reel (0075/0100 only)

(6) RESISTANCE VALUE

There are 2~4 digits indicated the resistance value.
Letter R/K/M is decimal point
Example:
97R6 = 97.6Ω
9K76 = 9760Ω
1M = 1,000,000Ω

(7) DEFAULT CODE

Letter L is the system default code for ordering only.(Note)

ORDERING EXAMPLE

The ordering code for a RC0402 0.0625W chip resistor value 100KΩwith ±5% tolerance, supplied in 7-inch tape reel of 10,000 units per reel is: RC0402JR-07100KL.

NOTE

1. All our RSMD products meet RoHS compliant and Halogen Free. "LFP" of the internal 2D reel label mentions "Lead Free Process".
2. On customized label, "LFP" or specific symbol can be printed.

MARKING

RC0075 / RC0100 / RC0201 / RC0402

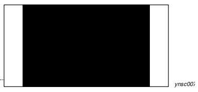


Fig. 1

No Marking

RC0603

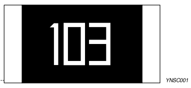


Fig. 2 Value = 10 KΩ

E24 series: 3 digits

First two digits for significant figure and 3rd digit for number of zeros

RC0805 / RC1206 / RC1210 / RC2010 / RC2512

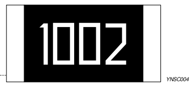


Fig. 3 Value = 10 KΩ

E24/E96 series: 4 digits

First three digits for significant figure and 4th digit for number of zeros

RC1218

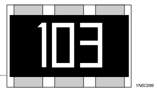


Fig. 4 Value = 10 KΩ

E-24 series: 3 digits, ±5%
First two digits for significant figure and 3rd digit for number of zeros

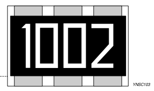


Fig. 5 Value = 10 KΩ

Both E-24 and E-96 series: 4 digits, ±1% & ±0.5%
First three digits for significant figure and 4th digit for number of zeros

For further marking information, please see special data sheet "Chip resistors marking".

CONSTRUCTION

The resistor is constructed on top of a high-grade ceramic body. Internal metal electrodes are added on each end to make the contacts to the thick film resistive element. The composition of the resistive element is a noble metal imbedded into a glass and covered by a second glass to prevent environmental influences. The resistor is laser trimmed to the rated resistance value. The resistor is covered with a protective epoxy coat, finally the two external terminations (matte tin on Ni-barrier) are added, as shown in Fig.6.

Outlines

For dimensions, please refer to Table 1

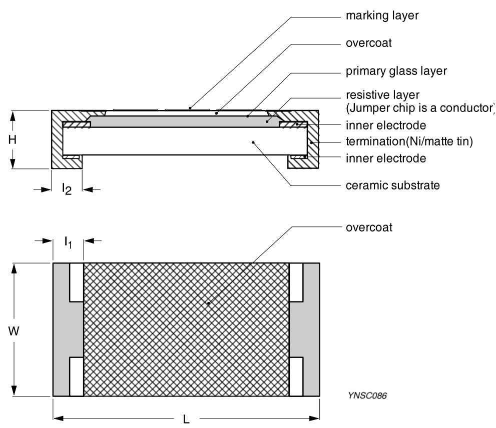


Fig. 6 Chip resistor outlines

Section 15
Panasonic

Designators	Part / Value	Qty
SW2,SW3	EVQPUC02K	2

RoHS evidence document: **panasonic_rohs_2ma_2000379.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

Panasonic
INDUSTRY

120003792 Page:1/2
Issue date:2024/10/11

Panasonic Industry Co., Ltd.

PolyFabriq

Non-use Report for Chemical Substances Contained in Products

Application start time : 7/22/2018

Our company (Panasonic Industry Co., Ltd.) hereby report that our products listed in Item Ⅰ that delivered or to be delivered to your company, subsidiaries or affiliated companies that specified do not contain the substances in Item II below or do not exceed the specified values under RoHS directive as the result of our intentional conduct.
The criteria of exemption to be applied to the subject materials and components shall be conformed to the RoHS directive (2011/65/EU, (EU) 2015/863).

Ⅰ. The Applicable Products: Please refer to the next page.

II. The Applicable Substances (10 substances subject to the RoHS Directive)

No.	Substance name	Regulated concentration/ Threshold (Homogeneous material unit)
1	Cadmium and its compounds	0.01wt%(100ppm)
2	Lead and its compounds	0.1wt%(1,000ppm)
3	Mercury and its compounds	0.1wt%(1,000ppm)
4	Hexavalent chromium compounds	0.1wt%(1,000ppm)
5	Polybrominated biphenyls (PBBs)	0.1wt%(1,000ppm)
6	Polybrominated diphenylethers (PBDEs)	0.1wt%(1,000ppm)
7	Di(2-ethylhexyl)phthalate (DEHP)	0.1wt%(1,000ppm)
8	Butyl benzyl phthalate (BBP)	0.1wt%(1,000ppm)
9	Dibutyl phthalate (DBP)	0.1wt%(1,000ppm)
10	Diisobutyl phthalate (DIBP)	0.1wt%(1,000ppm)

*1 In this report, EU RoHS Directive means the EU RoHS directive applicable as of the date above.
*2 In the event that your company suffers actual damage because restricted substances in the applicable products exceed the threshold by reason attributable to us, we will assume responsibility only as required by the written agreement(s) of sales and purchase with your company, subject to the limitations therein, and, if such written agreement(s) do not exist, only as required by applicable environmental laws and regulations.
In no event will we be liable for indirect, consequential and/or punitive damages.

M. Yamasaki

Masato Yamasaki, Manager

Quality & Environment Planning Section
Quality Assurance Department
Interface Devices Business Unit
Electromechanical Control Business Division
Panasonic Industry Co., Ltd.

Panasonic
INDUSTRY

120003792 Page:2/2
Issue date:2024/10/11

The Applicable Products

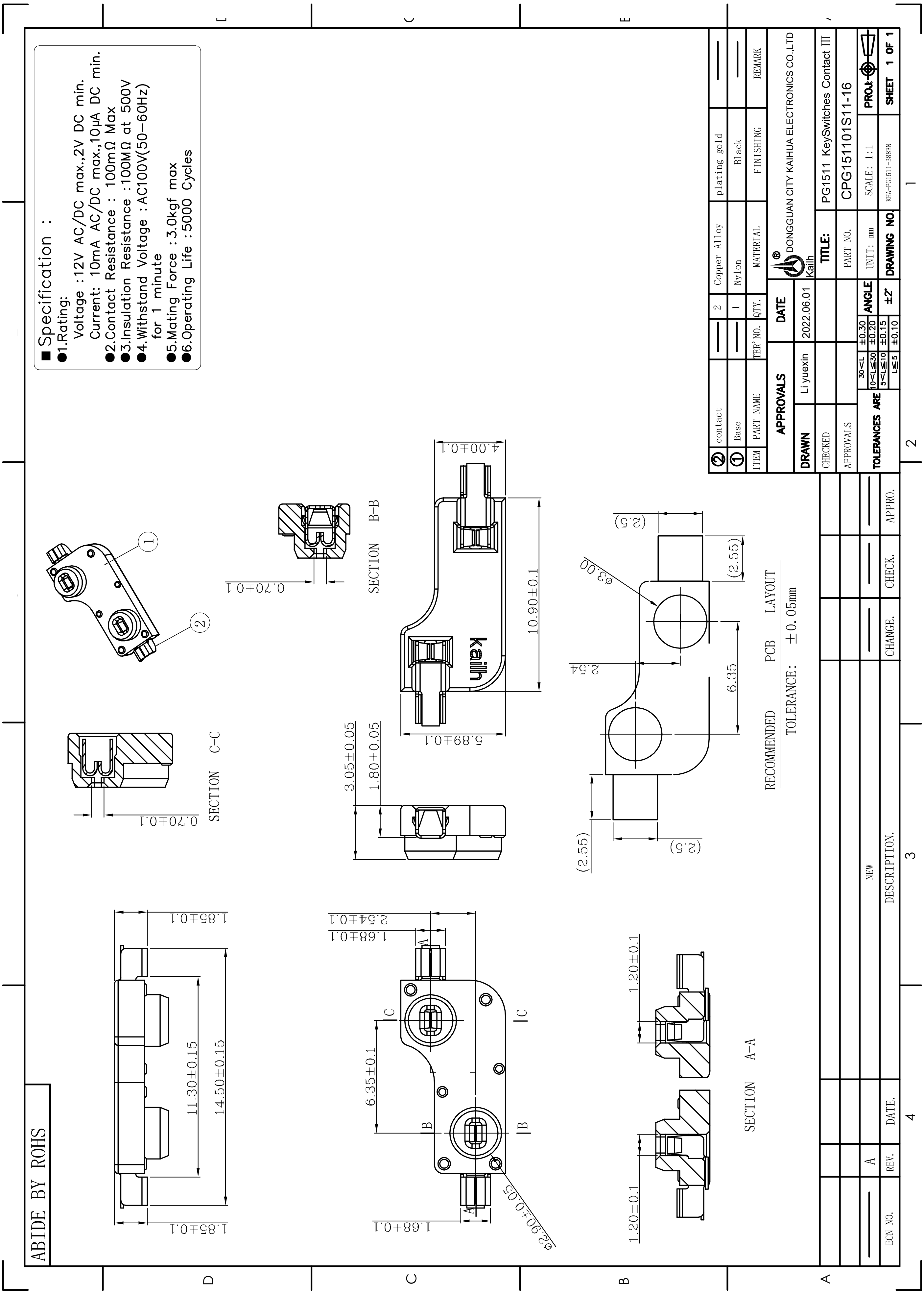
No.	Product Name	Our part No.	Our product No.
1	Small-sized Side-operational SMD	EVQPUC02K	EVQPUC02K

- EOF -

Section 16
Kailh

Designators	Part / Value	Qty
SW_K_1-SW_K_36	Kailh CPG151101S11-16	36

RoHS evidence document: **2208291600_Kailh-CPG151101S11-16_C5156480.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 9, 10 of 10 — RoHS-relevant pages only; the complete document is retained in the project repository.


■ Specification :
●1.Rating:
Voltage :12V AC/DC max.,2V DC min.
Current: 10mA AC/DC max.,10μA DC min.
●2.Contact Resistance : 100mΩ Max
●3.Insulation Resistance :100MΩ at 500V
●4.Withstand Voltage :AC100V(50-60Hz)
for 1 minute
●5.Mating Force :3.0kgf max
●6.Operating Life :5000 Cycles
SECTION B-B
SECTION C-C
SECTION A-A
RECOMMENDED PCB LAYOUT
TOLERANCE: ±0.05mm
ABIDE BY ROHS
kailh
0.70±0.1
4.00±0.1
10.90±0.1
5.89±0.1
3.05±0.05
1.80±0.05
φ3.00
(2.5)
(2.55)
2.54
6.35
1.85±0.1
11.30±0.15
14.50±0.15
2.54±0.1
1.68±0.1
6.35±0.1
1.20±0.1
φ2.90±0.05
② contact 2 Copper Alloy plating gold
① Base 1 Nylon Black
ITEM PART NAME TER' NO. QTY. MATERIAL FINISHING REMARK
APPROVALS DATE
DRAWN Li yuexin 2022.06.01
CHECKED
APPROVALS
DONGGUAN CITY KAIHUA ELECTRONICS CO.,LTD
Kailh
TITLE: PG1511 KeySwitches Contact III
PART NO. CPG151101S11-16
TOLERANCES ARE 30<L ±0.30 10<L≤30 ±0.20 5<L≤10 ±0.15 L≤5 ±0.10
ANGLE ±2°
UNIT: mm
SCALE: 1:1
PROJ.
DRAWING NO. KHA-PG1511-388EN
SHEET 1 OF 1
A NEW
ECN NO. REV. DATE. DESCRIPTION. CHANGE. CHECK. APPRO.

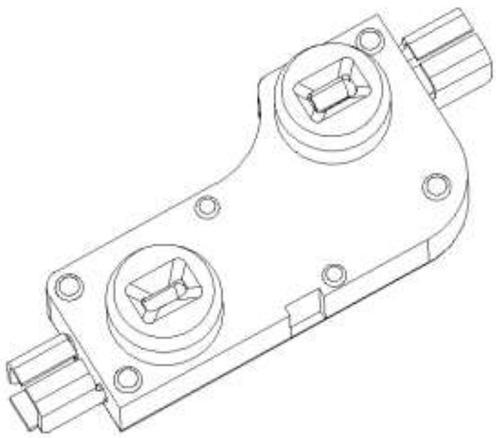
凱華電子
KAIHUA EELETRONICS

Document Number:

KH-PS2206-43

产品规格书

Product Specification



P/N:

CPG151101S11-16

Title :

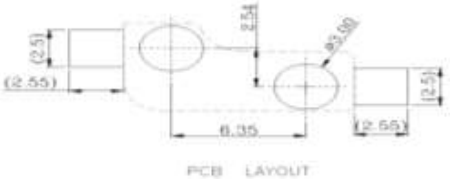
1511 Connector

Rev.	ECN	Release and Revision Description:	Prepared By /Date:	Checked By/Date:	Approved By/Date:
A	————	New releasing 初版发行	王枝轩 2022-06-27	胡远峰 2022-06-27	郑建军 2022-06-27

凱華電子 KAIHUA EELETRONICS	Product Specification			
	P/N: CPG151101S11-16	DOC. No.: KH-PS2206-43	Rev.: A	Page: 8/9

9. Recommended PCB Layout 推荐的 PCB 安装焊盘规格

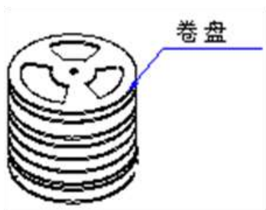
(Top View)
(Single face board T=1.6mm)

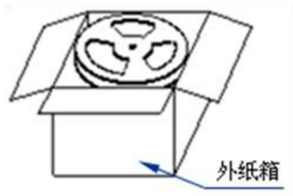


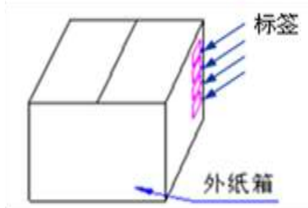
10. Packaging 包装

Packaging type: 13Tray, 26000Pcs/Carton.

包装方式:13 /盘, 26000Pcs/箱.







11.Precaution 注意事项

11.1 Soldering condition 回流焊条件

ITEM 项目		CONDITION 条件
Preheating zone 预热区	Heating zone 升温区	Speed＜2˚C/S, Preheating time15 S Max,temperture150℃ 速度＜2˚C/S, 预热时间 15 S 最多,温度 150℃
	Heatpreservation area 保温区	Speed1.2～3.5˚C/S, Preheating time120 S Max,temperture180℃ 速度 1.2～3.5˚C/S, 预热时间 120 S 最多,温度 180℃
	Fast heating zone 快速升温区	Speed3.5～4.5˚C/S, Preheating time140 S Max,temperture230℃ 速度 3.5～4.5˚C/S, 预热时间 140 S 最多,温度 230℃
Weld area 焊接区		Welding time 40 S Max ,welding temperature peak value, 3 sec Max. 焊接时间 40 S 最多, 焊接温度峰值 260℃ 最大,3 S 最多
Area of flux 助焊剂面积		1/2 Max of PWB Thickness 印刷基板厚度的 1/2 以内
Temperature of solder 焊锡温度		260±5℃ 260±5℃
Number of soldering 焊接次数		2time Max (But should down heat of the first soldering) 2 次以内
Printed wiring board 印刷基板		Single side copper-clad laminates 单面铜箔

(1) After switches were soldered, please be careful not to clean switches with solvent
开关浸焊后,注意不要用溶剂清洗.

(2) Under the condition of using soldering iron, soldering temperature shall be 350±5℃ with 3±0.5s
在使用铬铁的情况下，焊锡温度应在350±5℃,焊接时间3±0.5秒

凱華電子 KAIHUA EELETRONICS	Product Specification			
	P/N: CPG151101S11-16	DOC. No.: KH-PS2206-43	Rev.: A	Page: 9/9

11.2 Notes 注意点

(1) Please be cautious not to give excessive static load connector.
注意不要施加超负荷的压力或晃动连接器.

(2) Connector be careful not to stack up P. W. B. after switches were soldered.
连接器焊接以后,印刷基板注意不要叠放.

(3) Preservation under high temperature and high humidity or corrosive gas should be avoided Especially. When you need to preserve for a long period, do not open the carton.
保管时尤其应注意避开高湿高温和有腐蚀性气体的环境.如需长时间保存,请不要打开包装箱.

(4) The standard storage period is 3 months, with maximum up to 6months, preferably to be used as soon as possible. After opening the package, you should put the remaining switches in a plastic bag to prevent from damp and corrosive gas.
保存标准为 3 个月，限度为 6 个月以内，请尽早使用. 打开包装后，有剩余品时，应将剩余部分以胶袋包装好以同外界隔离，请进行合适的防湿，防腐蚀气体等处理后进行保管.

(5) This Product Specification is considered as the technical agreement on product between the receiving customer and Kailh. Any information on Product Catalogue which is in conflict with or different from the corresponding information of this document is considered as invalid.
该规格书为客户与凯华公司产品在技术方面的共识 ,其他相关数据上与该规格书不一致的内容都是无效的.

(6) If customer issue purchase orders without confirmation by signature of this specification after receipt, such confirmation will be considered as granted upon receipt of the first purchase order.
如果顾客收到规格书后没有信息反馈而直接向我公司订货，我们将认为贵客已接受此规格书.

(7) If there is no order or no request for new specification after 1 year upon this specification is issued, the specification will be regarded as invalid.
本产品规格书从生效日起 1 年后，如果没有订货或再次申请最新规格书时请做无效处理.

(8) Products meet the ROHS & REACH environmental management substances control standards
产品满足 ROHS & REACH 环境管理物质管制标准

Section 17
TI

Designators	Part / Value	Qty
U1,U2	TLV62569DBVR	2

RoHS evidence document: **TexasTLV62569DBVR.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

www.ti.com

PACKAGE OPTION ADDENDUM

15-Dec-2017

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
TLV62569DBVR	ACTIVE	SOT-23	DBV	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 125	16AF	Samples
TLV62569DBVT	ACTIVE	SOT-23	DBV	5	250	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 125	16AF	Samples
TLV62569DRLR	ACTIVE	SOT-5X3	DRL	6	3000	Green (RoHS & no Sb/Br)	CU SN	Level-1-260C-UNLIM	-40 to 125	19D	Samples
TLV62569DRLT	ACTIVE	SOT-5X3	DRL	6	250	Green (RoHS & no Sb/Br)	CU SN	Level-1-260C-UNLIM	-40 to 125	19D	Samples
TLV62569PDDCR	ACTIVE	SOT-23-THIN	DDC	6	3000	Green (RoHS & no Sb/Br)	CU SN	Level-1-260C-UNLIM	-40 to 125	6D9	Samples
TLV62569PDDCT	ACTIVE	SOT-23-THIN	DDC	6	250	Green (RoHS & no Sb/Br)	CU SN	Level-1-260C-UNLIM	-40 to 125	6D9	Samples
TLV62569PDRLR	ACTIVE	SOT-5X3	DRL	6	3000	Green (RoHS & no Sb/Br)	CU SN	Level-1-260C-UNLIM	-40 to 125	19E	Samples
TLV62569PDRLT	ACTIVE	SOT-5X3	DRL	6	250	Green (RoHS & no Sb/Br)	CU SN	Level-1-260C-UNLIM	-40 to 125	19E	Samples

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".
RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.
Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

Section 18
Silergy

Designators	Part / Value	Qty
U3	SY6280AAC	1

RoHS evidence document: **ROHS-2024-SilergyCorp.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 of 14 — RoHS-relevant pages only; the complete document is retained in the project repository.

Test Report

No.: SHAEC23021346804_1 **Date:** Jan 05, 2024 Page 1 of 14

Client Name: Silergy Corp.

Client Address: Oleander Way, 802 West Bay Road,P.O. Box 32052, Grand Cayman KY1-1208,Cayman Islands

Sample Name: Surface Mounted Electric Component

Model No.: DFN4×4

Client Ref. Information: AQFN/BGA/COF/COMBO/CPC/CSP/DFN/DIP/D-SMD/HDIP/GEM/LQFP/LGA/MDFN/MLGA/MQFN/MSOP/ODFN/QFN/PDFN/P-DIP/PLCC/SMB/SMC/SMD/SOD/SOP/SOT/SSOP/TO92/TO220/TO220F/TO252/TO263/TQFP/TSOP/TSSOP/ TSOT See attachment

The above sample(s) and information were provided by the client.

THIS REPORT IS TO SUPERSEDE TEST REPORT NO.SHAEC23021346804, DATE: Jan 03, 2024.

SGS Job No.: SHP23-022732

Sample Receiving Date: Dec 18, 2023

Testing Period: Dec 18, 2023 ~ Jan 03, 2024

Test Requested: Select test(s) as requested by the client.

Test Method(s): Please refer to next page(s).

Test Result(s): Please refer to next page(s).

Test Requirement	Conclusion
EU RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU- Lead, Mercury, Cadmium, Hexavalent chromium, Polybrominated biphenyls (PBBs), Polybrominated diphenyl ethers (PBDEs), Bis(2-ethylhexyl) phthalate (DEHP), Butyl benzyl phthalate (BBP), Dibutyl phthalate (DBP) and Diisobutyl phthalate (DIBP)	Pass
Halogen	See Results
Perfluorooctanesulfonate (PFOS) and its derivatives and Perfluorooctanoic Acid (PFOA).	See Results
AfPS GS 2019:01 PAK-Polycyclic Aromatic Hydrocarbons (PAHs)	See Results

Signed for and on behalf of
SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.

Dora Hu
Approved Signatory

scan to see the report
56E6112B

检验检测专用章
Inspection & Testing Services
SGS-CSTC Standards Technical Services (Shanghai) Co.,Ltd.
Testing Center-Chemical Laboratory

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Test Report

No.: SHAEC23021346804_1 **Date:** Jan 05, 2024

Test Result(s):

Test Part Description：

SN ID	Sample No.	SGS Sample ID	Description
SN1	A1	SHA23-0213468-0001.C001	Black body with silvery pin

Remarks:

(1) 1 mg/kg = 1 ppm = 0.0001%
(2) MDL = Method Detection Limit
(3) ND = Not Detected (< MDL)
(4) “-“ = Not Regulated

EU RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU- Lead, Mercury, Cadmium, Hexavalent chromium, Polybrominated biphenyls (PBBs), Polybrominated diphenyl ethers (PBDEs), Bis(2-ethylhexyl) phthalate (DEHP), Butyl benzyl phthalate (BBP), Dibutyl phthalate (DBP) and Diisobutyl phthalate (DIBP)

Test Method: With reference to IEC 62321-4:2013+AMD1:2017, IEC 62321-5:2013, IEC 62321-7-2:2017, IEC 62321-6:2015 and IEC 62321-8:2017, analysis was performed by ICP-OES, AAS, UV-Vis and GC-MS.

Test Item(s)	Limit	Unit(s)	MDL	A1
Cadmium(Cd)	100	mg/kg	2	ND
Lead (Pb)	1000	mg/kg	2	5
Mercury (Hg)	1000	mg/kg	2	ND
Hexavalent Chromium (Cr(VI))	1000	mg/kg	8	ND
Polybromobiphenyl (PBBs)	1000	mg/kg	-	ND
Monobromobiphenyl (MonoBB)	-	mg/kg	5	ND
Dibromobiphenyl (DiBB)	-	mg/kg	5	ND
Tribromobiphenyl (TriBB)	-	mg/kg	5	ND
Tetrabromobiphenyl (TetraBB)	-	mg/kg	5	ND
Pentabromobiphenyl (PentaBB)	-	mg/kg	5	ND
Hexabromobiphenyl (HexaBB)	-	mg/kg	5	ND
Heptabromobiphenyl (HeptaBB)	-	mg/kg	5	ND
Octabromobiphenyl (OctaBB)	-	mg/kg	5	ND
Nonabromobiphenyl (NonaBB)	-	mg/kg	5	ND
Decabromobiphenyl (DecaBB)	-	mg/kg	5	ND
Polybromodiphenyl ether(PBDEs)	1000	mg/kg	-	ND
Monobromodiphenylether (MonoBDE)	-	mg/kg	5	ND
Dibromodiphenylether (DiBDE)	-	mg/kg	5	ND
Tribromodiphenylether (TriBDE)	-	mg/kg	5	ND
Tetrabromodiphenylether (TetraBDE)	-	mg/kg	5	ND
Pentabromodiphenylether (PentaBDE)	-	mg/kg	5	ND
Hexabromodiphenylether (HexaBDE)	-	mg/kg	5	ND
Heptabromodiphenylether (HeptaBDE)	-	mg/kg	5	ND
Octabromodiphenylether (OctaBDE)	-	mg/kg	5	ND
Nonabromodiphenylether (NonaBDE)	-	mg/kg	5	ND
Decabromodiphenylether (DecaBDE)	-	mg/kg	5	ND
Dibutyl Phthalate(DBP)	1000	mg/kg	50	ND
Benzyl Butyl Phthalate(BBP)	1000	mg/kg	50	ND

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Test Report **No.:** SHAEC23021346804_1 **Date:** Jan 05, 2024 Page 3 of 14

Test Item(s)	Limit	Unit(s)	MDL	A1
Bis-(2-ethylhexyl) Phthalate(DEHP)	1000	mg/kg	50	ND
Diisobutyl Phthalate(DIBP)	1000	mg/kg	50	ND

Notes:
(1) The maximum permissible limit is quoted from RoHS Directive (EU) 2015/863.
(2) IEC 62321 series is equivalent to EN 62321 series.
(3) The restriction of DEHP, BBP, DBP and DIBP shall apply to medical devices, including in vitro medical devices, and monitoring and control instruments, including industrial monitoring and control instruments, from 22 July 2021.

Halogen

Test Method: With reference to EN 14582:2016, analysis was performed by IC.

Test Item(s)	Unit(s)	MDL	A1
Fluorine(F)	mg/kg	20	ND
Chlorine(Cl)	mg/kg	50	ND
Bromine(Br)	mg/kg	50	ND
Iodine(I)	mg/kg	50	ND

Perfluorooctanesulfonate (PFOS) and its derivatives and Perfluorooctanoic Acid (PFOA).

Test Method: With reference to CEN/TS 15968:2010, analysis was performed by HPLC-MS or LC-MS/MS.

Test Item(s)	CAS No.	Unit(s)	MDL	A1
Perfluorooctane sulfonates (PFOS) and its derivatives	-	mg/kg	-	ND
Perfluorooctanesulfonic acid (PFOS)^	1763-23-1	mg/kg	10	ND
N-ethylperfluoro-1-octanesulfonamide (EtFOSA)	4151-50-2	mg/kg	10	ND
N-methylperfluoro-1-octanesulfonamide (MeFOSA)	31506-32-8	mg/kg	10	ND
2-(N-ethylperfluoro-1-octanesulfonamido)-ethanol (EtFOSE)	1691-99-2	mg/kg	10	ND
2-(N-methylperfluoro-1-octanesulfonamido)- ethanol (MeFOSE)	24448-09-7	mg/kg	10	ND
Perfluorooctane Sulfonamide (PFOSA)	754-91-6	mg/kg	10	ND
Perfluorooctanoic Acid (PFOA) and its salts+	-	mg/kg	10	ND

Notes:
(1) + PFOA refer to its salts including PFOA-Na (CAS No.: 335-95-5), PFOA-K (CAS No.: 2395-00-8),PFOA-Ag (CAS No.: 335-93-3), PFOA-F (CAS No.: 335-66-0) and APFO (CAS No.: 3825-26-1);
(2) ^ PFOS including PFOS-K (CAS No.: 2795-39-3), PFOS-Li (CAS No.: 29457-72-5), PFOS-NH4 (CAS No.: 29081-56-9), PFOS-NH(OH)2 (CAS No.: 70225-14-8), PFOS-N(C2H5)4 (CAS No.: 56773-42-3), PFOS-DDA(CAS No.:251099-16-8) and POSF (CAS No.: 307-35-7) .

AfPS GS 2019:01 PAK-Polycyclic Aromatic Hydrocarbons (PAHs)

Test Method: With reference to AfPS GS 2019:01 PAK, analysis was performed by GC-MS.

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Test Report **No.:** SHAEC23021346804_1 **Date:** Jan 05, 2024 Page 4 of 14

Test Item(s)	CAS No.	Unit(s)	MDL	A1
Benzo(a)pyrene(BaP)	50-32-8	mg/kg	0.1	ND
Benzo(e)pyrene(BeP)	192-97-2	mg/kg	0.1	ND
Benzo(a)anthracene(BaA)	56-55-3	mg/kg	0.1	ND
Benzo(b)Fluoranthene(BbF)	205-99-2	mg/kg	0.1	ND
Benzo(j)fluoranthene(BjF)	205-82-3	mg/kg	0.1	ND
Benzo(k)Fluoranthene(BkF)	207-08-9	mg/kg	0.1	ND
Chrysene(CHR)	218-01-9	mg/kg	0.1	ND
Dibenzo(a,h)Anthracene(DBA)	53-70-3	mg/kg	0.1	ND
Benzo(g,h,i)perylene(BPE)	191-24-2	mg/kg	0.1	ND
Indeno(1,2,3-c,d)pyrene(IPY)	193-39-5	mg/kg	0.1	ND
Phenanthrene(PHE)	85-01-8	mg/kg	0.1	ND
Pyrene(PYR)	129-00-0	mg/kg	0.1	ND
Anthracene(ANT)	120-12-7	mg/kg	0.1	ND
Fluoranthene(FLT)	206-44-0	mg/kg	0.1	ND
Sum of Phenanthrene(PHE), Pyrene(PYR), Anthracene(ANT), Fluoranthene(FLT)	-	mg/kg	-	ND
Naphthalene(NAP)	91-20-3	mg/kg	0.1	ND
Sum of 15 PAHs	-	mg/kg	-	ND
Material Category	-	-	-	-

Notes:

AfPS（German commission for Product Safety）: PAHs requirements

Parameter	Category 1	Category 2		Category 3	
	Materials intended to be placed in the mouth, or materials coming into long-term contact with skin (more than 30s) during the intended use -in toys according to Directive 2009/48/EC or -for the use by children[a,b] up to 3 years of age.	Materials not covered by category 1, coming into long-term contact (more than 30s) or short-term repetitive contact[c] with skin during the intended or foreseeable use[d].		Materials covered neither by category 1 nor by category 2, coming into short-term contact (up to 30s) with skin during the intended or foreseeable use.	
		a. use by children	b. other consumer products	a. use by children	b. other consumer products
Benzo(a)pyrene (BaP) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Benzo(e)pyrene (BeP) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1

检验检测专用章
Inspection & Testing Services

SGS-CSTC Standards Technical Services (Shanghai) Co.,Ltd. Testing Center-Chemical Laboratory
3rd Building,No.889 Yishan Road Xuhui District,Shanghai China 200233 t E&E (86–21) 61402553 f E&E (86–21)64953679 www.sgsgroup.com.cn
中国・上海・徐汇区宜山路889号3号楼 邮编: 200233 t HL (86–21) 61402594 f HL (86–21)61156899 e sgs.china@sgs.com

Test Report

No.: SHAEC23021346804_1 **Date:** Jan 05, 2024 Page 5 of 14

Benzo(a)anthracene (BaA) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Benzo(b)fluoranthene (BbF) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Benzo(j)fluoranthene (BjF) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Benzo(k)fluoranthene (BkF)mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Chrysene (CHR) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Dibenzo(a,h)anthracene (DBA) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Benzo(g,h,i)perylene (BPE) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Indeno(1,2,3-cd)pyrene (IPY) mg/kg	< 0.2	< 0.2	< 0.5	< 0.5	< 1
Phenanthrene (PHE), pyrene (PYR), anthracene (ANT), fluoranthene (FLT), mg/kg	< 1 Sum	< 5 Sum	< 10 Sum	< 20 Sum	< 50 Sum
Naphthalene (NAP) mg/kg	< 1	< 2		< 10	
Sum of 15 PAHs	<1	< 5	< 10	< 20	< 50

Notes:
[a] A "Child" is legally defined as a person before reaching the age of 14 years.
[b] Use by children includes both active and passive contact by children.
[c] Definition "short-term repetitive contact" taken from REACH Annex XVII entry 50 amendment (Regulation (EC) No.1272/2013)
[d] According to the definition of the German Product Safety Act (ProdSG) (chapter 1 Article 2 No. 28) "foreseeable use" shall mean the use of a product in a manner that the person placing it on the market, has not intended, but which could be reasonably foreseeable.

Remark:
The German committee on Product Safety (AfPS) published a new PAHs document (AfPS GS 2019:01 PAK) on April 10, 2020, which will be binding for the issue of GS mark certificate from July 1, 2020.

This report updates Client Ref. Information.

Unless otherwise stated, the decision rule for conformity reporting is based on Binary Statement for Simple Acceptance Rule (*w*=0) stated in ILAC-G8:09/2019.

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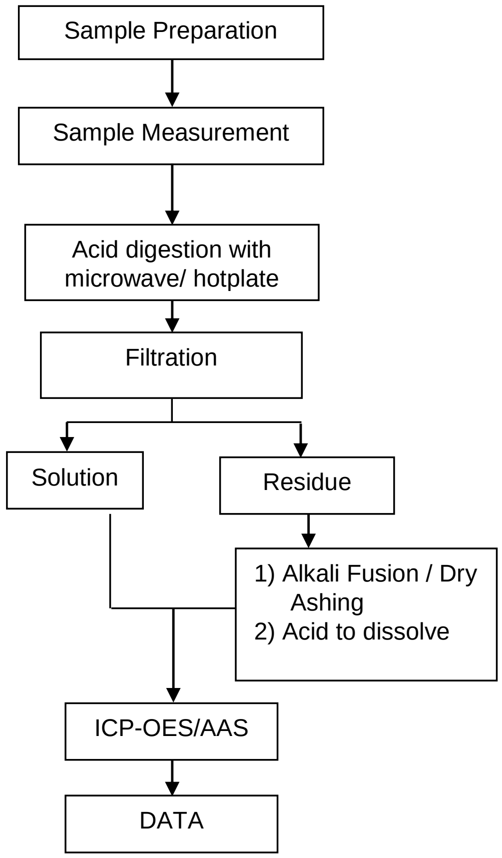
Test Report **No.:** SHAEC23021346804_1 **Date:** Jan 05, 2024 Page 6 of 14

ATTACHMENTS

Elements Testing Flow Chart

These samples were dissolved totally by pre-conditioning method according to below flow chart.



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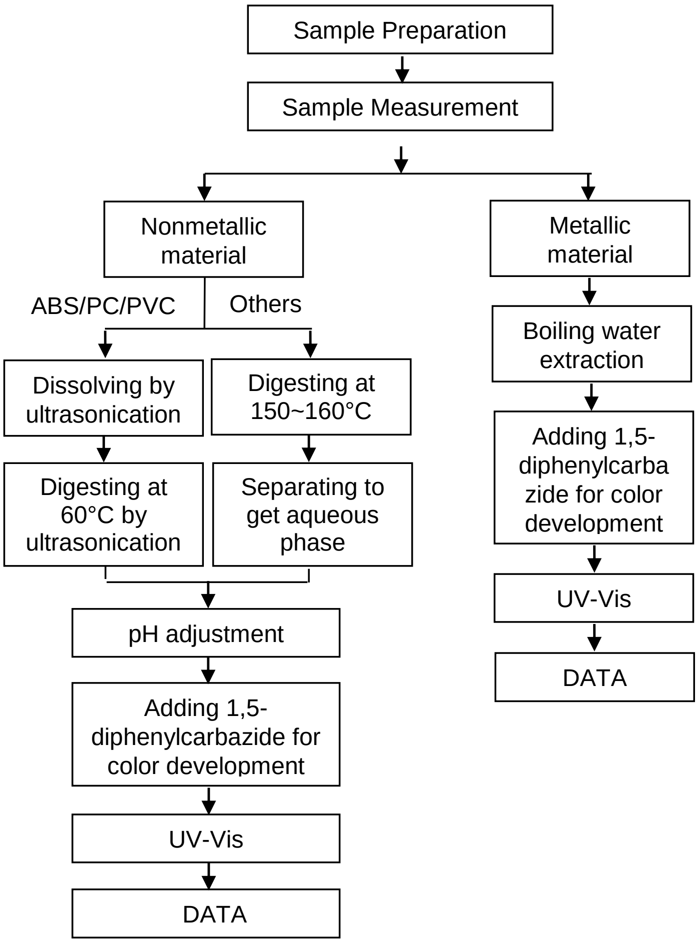
Test Report **No.:** SHAEC23021346804_1 **Date:** Jan 05, 2024

ATTACHMENTS

Hexavalent Chromium (Cr(VI)) Testing Flow Chart



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Test Report

No.: SHAEC23021346804_1 **Date:** Jan 05, 2024

ATTACHMENTS

PBBs/PBDEs Testing Flow Chart

Sample cutting/preparation
↓
Sample measurement
↓
Solvent extraction
↓
Concentration/ Dilution of extraction solution
↓
Filtration
↓
GC-MS
↓
DATA

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Test Report

No.: SHAEC23021346804_1 **Date:** Jan 05, 2024

ATTACHMENTS

Phthalates Testing Flow Chart

Sample cutting/preparation

↓

Sample measurement

↓

Solvent extraction

↓

Concentration/Dilution

↓

Filtration

↓

GC-MS

↓

DATA

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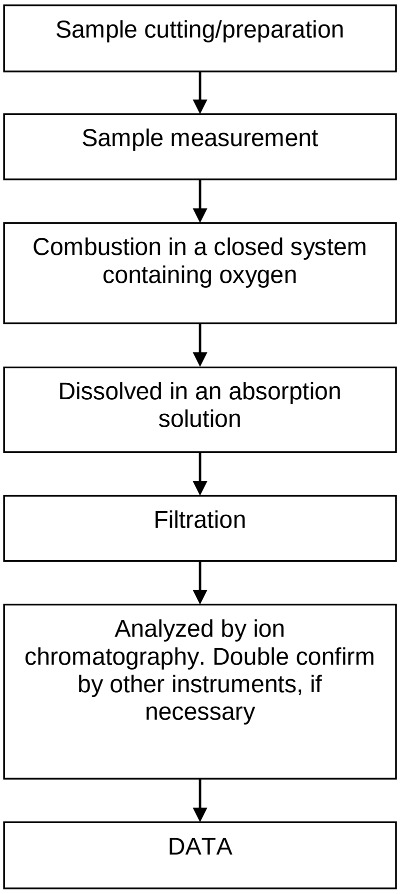
Test Report

No.: SHAEC23021346804_1 **Date:** Jan 05, 2024

ATTACHMENTS

Halogen Testing Flow Chart



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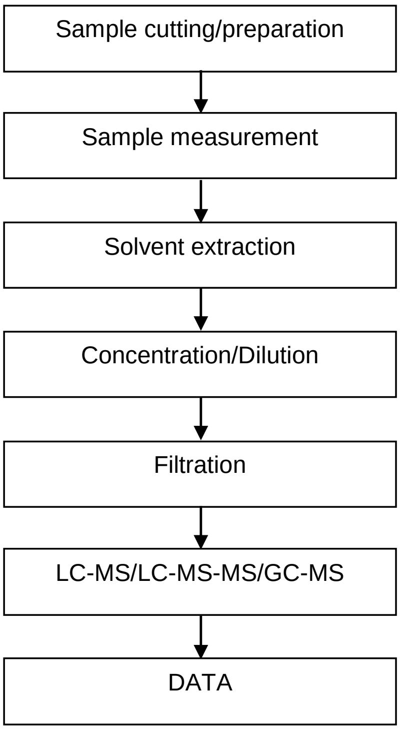
Test Report **No.:** SHAEC23021346804_1 **Date:** Jan 05, 2024

ATTACHMENTS

PFASs/ PFOS/PFOA Testing Flow Chart



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Section 19
Nexperia

Designators	Part / Value	Qty
U4-U8	74HC595BQ	5
U12-U25	74AHC1G125GW,125	14

RoHS evidence document: **Nexperia - Statement on RoHS 20241008.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

October 2024

CERTIFICATE OF COMPLIANCE
– RoHS Declaration –

Nexperia B.V. declares that its semiconductor products, including all homogeneous materials used in their composition, are designed to be RoHS-compliant by meeting the requirements defined under Directive 2011/65/EU of 2011-07-21, amended by Directive (EU) 2015/863 of 2015-03-31, on the restriction of the use of certain hazardous substances in electrical and electronic equipment (EEE):

RoHS-Restricted Substance	Permissible Limit on the Homogeneous Material Level
Cadmium (Cd)	100 ppm (0.01 weight %)
Mercury (Hg)	1000 ppm (0.1 weight %)
Hexavalent chromium (Cr^{6+})	1000 ppm (0.1 weight %)
Lead (Pb)	1000 ppm (0.1 weight %)
Polybrominated biphenyls (PBBs)	1000 ppm (0.1 weight %)
Polybrominated diphenyl ethers (PBDEs) *including decabromodiphenylether (decaBDE)*	1000 ppm (0.1 weight %)
Bis(2-ethylhexyl) phthalate (DEHP)	1000 ppm (0.1 weight %)
Butyl benzyl phthalate (BBP)	1000 ppm (0.1 weight %)
Dibutyl phthalate (DBP)	1000 ppm (0.1 weight %)
Diisobutyl phthalate (DIBP)	1000 ppm (0.1 weight %)

All Nexperia products are EU RoHS-compliant and bear the "RoHS-compliant" logo on the box label. Nexperia devices that contain no more than 0.1 % lead by weight per homogeneous material are assigned the RHF indicators G or D. These products can be recognized by the "Lead-free" logo on the box label. Nexperia products that contain RoHS-exempted lead are assigned the RHF indicators E or H and may make use of the following RoHS exemptions:

RoHS Exemption	RoHS Exemption Description
7(a)	Lead in high melting temperature type solders (i.e. lead-based alloys containing 85% by weight or more lead)
7(c)-I	Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound

The RHF indicator can be found on the respective product page, such as on the product page of BAS16-Q under "Environmental information". Table 1 in the Appendix to this document lists all released packages using RHF indicators E or H and the information on whether the respective package makes use of RoHS exemption 7(a) and/or 7(c)-I, respectively.

Nexperia B.V. • Jonkerbosplein 52 • 6534 AB Nijmegen • 6050 AA Nijmegen • The Netherlands
K.V.K. 66264111 0000 • VAT.NO. NL856469397B01 • Citibank London 18190372 (EUR) • IBAN GB14CITI18500818190372 • BIC CITIGB2L
Citibank London 18190402 (USD) • IBAN GB77CITI18500818190402 • BIC CITIGB2L
nexperia.com

To facilitate compliance checks and inform the public, we make product composition declarations available on our public website. The signature below verifies that the above statements, including any product composition data, are valid and accurate to the best of our knowledge and belief. Nexperia has implemented robust systems to ensure that our products comply with global environmental regulations and laws. In the event of any issues arising from information in this document, Nexperia's standard terms and conditions of sale shall apply, unless alternate contracts have been agreed upon in writing by both parties. For further details, please reach out to your Nexperia contact, get in touch with the nearest Sales office, or fill in the technical support form (after registration).

____

Dr. Timo Stein
Manager ECO-Products
Nexperia B.V.

APPENDIX

Table 1: Packages of Nexperia products that contain lead in concentrations exceeding 0.1 % *w/w* of any homogeneous material along with the applicability of RoHS exemption 7(a) and/or 7(c)-I, respectively.

Package	RoHS exemption 7(a)	RoHS exemption 7(c)-I
SOD1001-1	TRUE	TRUE
SOD1002-1	TRUE	TRUE
SOD1003-1	TRUE	TRUE
SOD123FL	TRUE	FALSE
SOD123HP	TRUE	FALSE
SOD123W	TRUE	FALSE
SOD123W-2	TRUE	FALSE
SOD128	TRUE	FALSE
SOD128FL-1	TRUE	FALSE
SOD323HP	TRUE	FALSE
SOT1023	TRUE	FALSE
SOT1023A	TRUE	FALSE
SOT1205	TRUE	FALSE
SOT1210	TRUE	FALSE
SOT1235	TRUE	FALSE
SOT1289	TRUE	FALSE
SOT1289B	TRUE	FALSE
SOT226C	TRUE	FALSE
SOT404	TRUE	FALSE
SOT404A	TRUE	FALSE
SOT428	TRUE	FALSE
SOT428C	TRUE	FALSE
SOT429	TRUE	FALSE
SOT429-2	TRUE	FALSE
SOT669	TRUE	FALSE
SOT78	TRUE	FALSE
SOT78A	TRUE	FALSE
SOT8000	TRUE	FALSE
SOT8000A	TRUE	FALSE
SOT8005	TRUE	FALSE
SOT8005A	TRUE	FALSE
SOT8017	TRUE	FALSE
SOT8018	TRUE	FALSE
SOT8021	TRUE	FALSE
SOT8022	TRUE	FALSE
SOT8070-1	TRUE	FALSE
SOT8071-1	TRUE	FALSE

Past Use of Exemptions

Glass diodes belonging to packages SOD27, SOD66, SOD68, and SOD80C were making use of RoHS exemption 7(c)-I in the past. On 2024-03-26, Nexperia issued change notice CN-202310015F, titled “Change to Pb-free Glass Diodes”. After the implementation of this change notice on 2024-09-23, Nexperia’s glass diodes no longer require RoHS exemption 7(c)-I.

Section 20
Winbond

Designators	Part / Value	Qty
U9	W25Q64JVXGIM	1

RoHS evidence document: **winbond-W25Q64JV-rohs.png**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

W25Q64JV

winbond

9. ELECTRICAL CHARACTERISTICS

9.1 Absolute Maximum Ratings [1]

PARAMETERS	SYMBOL	CONDITIONS	RANGE	UNIT
Supply Voltage	VCC		–0.6 to 4.6	V
Voltage Applied to Any Pin	V_{IO}	Relative to Ground	–0.6 to VCC+0.4	V
Transient Voltage on any Pin	V_{IOT}	<20nS Transient Relative to Ground	–2.0V to VCC+2.0V	V
Storage Temperature	T_{STG}		–65 to +150	°C
Lead Temperature	T_{LEAD}		See Note [2]	°C
Electrostatic Discharge Voltage	V_{ESD}	Human Body Model[3]	–2000 to +2000	V

Notes:

1. This device has been designed and tested for the specified operation ranges. Proper operation outside of these levels is not guaranteed. Exposure to absolute maximum ratings may affect device reliability. Exposure beyond absolute maximum ratings may cause permanent damage.
2. Compliant with JEDEC Standard J-STD-20C for small body Sn-Pb or Pb-free (Green) assembly and the European directive on restrictions on hazardous substances (RoHS) 2002/95/EU.
3. JEDEC Std JESD22-A114A (C1=100pF, R1=1500 ohms, R2=500 ohms).

9.2 Operating Ranges

PARAMETER	SYMBOL	CONDITIONS	SPEC		UNIT
			MIN	MAX	
Supply Voltage[1]	VCC	F_R = 133MHz, f_R = 50MHz	3.0	3.6	V
		F_R = 104MHz, f_R = 50MHz	2.7	3.0	V
Ambient Temperature, Operating	T_A	Industrial	–40	+85	°C
		Industrial Plus	–40	+105	°C

Note:

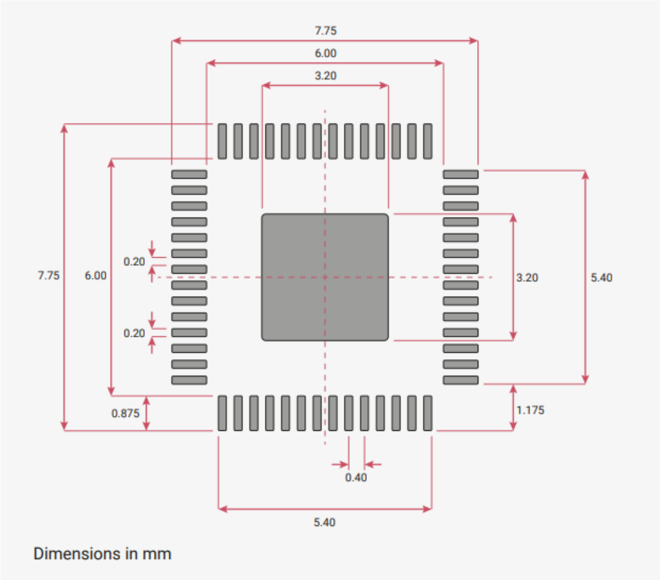
1. VCC voltage during Read can operate across the min and max range but should not exceed ±10% of the programming (erase/write) voltage.

Section 21
Raspberry Pi

Designators	Part / Value	Qty
U10	RP2040	1

RoHS evidence document: **rp2040-rohs.png**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

Figure 165. Recommended PCB Footprint for the RP2040 QFN-56 package



5.1.2. Compliance

RP2040 is compliant to Moisture Sensitivity Level 1.

RP2040 is compliant to the requirement of REACH Substances of Very High Concern (SVHC) that ECHA announced on 25 June 2020.

RP2040 is compliant to the requirement and standard of Controlled Environment-related Substance of RoHS directive (EU) 2011/65/EU and directive (EU) 2015/863.

5.2. Pinout

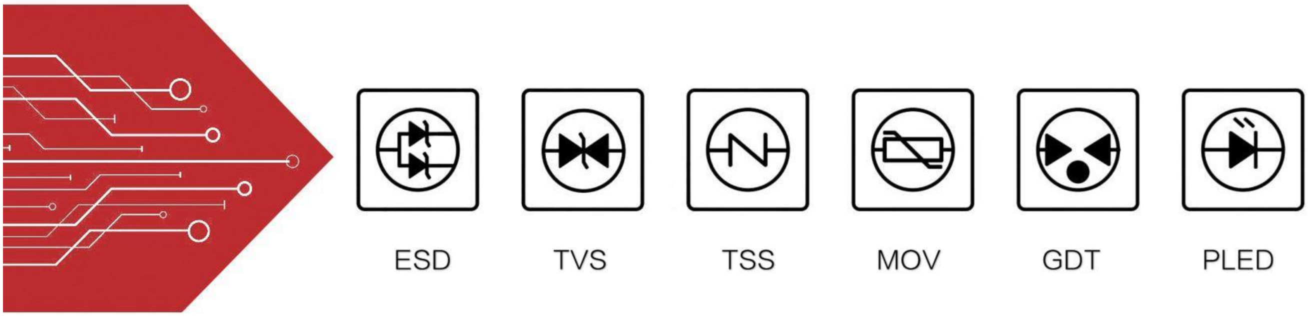
5.2.1. Pin Locations

Section 22
MSKSEMI

Designators	Part / Value	Qty
U11,U26	TPD4E05U06DQAR	2

RoHS evidence document: **C2836386-TPD4E05U06DQAR-MS.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 6 — RoHS-relevant pages only; the complete document is retained in the project repository.

MSKSEMI
SEMICONDUCTOR



Product data sheet

www.msksemi.com

MSKSEMI SEMICONDUCTOR

TPD4E05U06DQAR-MS

Semiconductor Compiance

- Solid-state silicon-avalanche technology
- Low operating and clamping voltage
- Up to four I/O Lines of Protection
- Ultra low capacitance: 0.5pF typical(I/O to I/O)
- Low Leakage
- Low operating voltage:3.3V
- Flow-Through design

IEC COMPATIBILITY (EN61000-4)

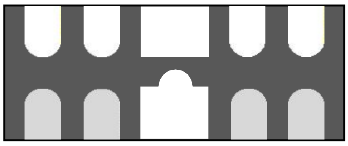
- **IEC 61000-4-2 (ESD) ±15kV (air), ±8kV (contact)**
- **IEC 61000-4-4 (EFT) 40A (5/50ns)**
- **IEC 61000-4-5 (Lightning) 5A (8/20μs)**

Mechanical Characteristics

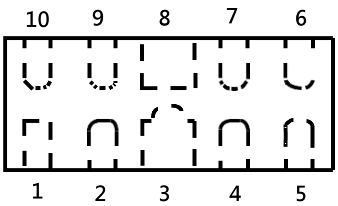
- Molding compound flammability rating: UL 94V-0
- Marking: Marking Code
- Packaging: Tape and Reel
- RoHS/WEEE Compliant

Applications

- Digital Visual Interface(DVI)
- MDDI Ports
- DisplayPortTM Interface
- PCI Express
- High Definition Multi-Media Interface(HDMI)
- eSATA Interfaces



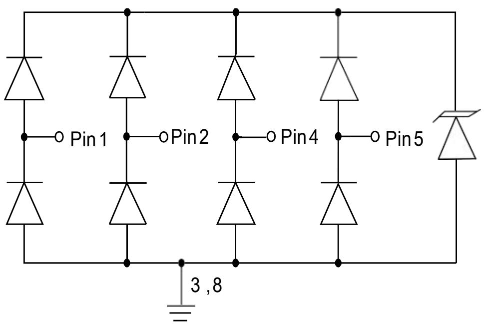
uSON-10



Schematic & PIN Configuration

Pin	Identificaion
1,2,4,5	Input Lines
6,7,9,10	Output Lines (No Internal Connection)
3,8	Ground

Circuit Diagram



4-Line Protection

MSKSEMI SEMICONDUCTOR

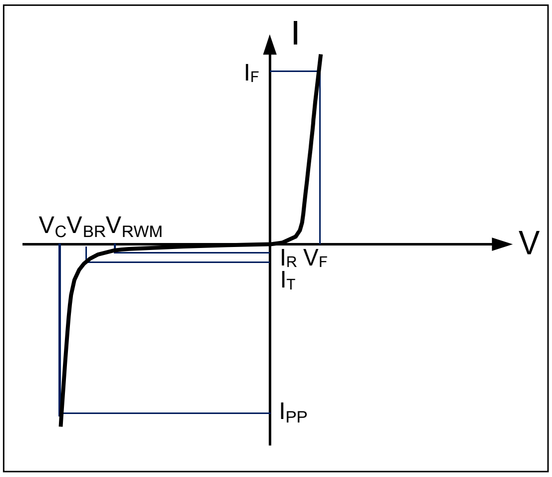
TPD4E05U06DQAR-MS

Semiconductor Compiance

Rating	Symbol	Value	Units
Peak Pulse Power (t_p =8/20μs)	P_{PP}	150	Watts
Peak Pulse Current (t_p =8/20μs)	I_{pp}	5	A
ESD per IEC 61000-4-2(Air) ESD per IEC 61000-4-2(contact)	V_{ESD}	+/-17 +/-12	kV
Operating Temperature	T_J	-55 to + 125	°C
Storage Temperature	T_{STG}	-55 to +150	°C

Electrical Parameters (T=25°C)

Symbol	Parameter
I_{PP}	Maximum Reverse Peak Pulse Current
V_C	Clamping Voltage @ I_{PP}
V_{RWM}	Working Peak Reverse Voltage
I_R	Maximum Reverse Leakage Current @ V_{RWM}
V_{BR}	Breakdown Voltage @ I_T
I_T	Test Current
I_F	Forward Current
V_F	Forward Voltage @ I_F



Electrical Characteristics

Parameter	Symbol	Conditions	Minimum	Typical	Maximum	Units
Reverse Stand-Off Voltage	V_{RWM}	Any I/O pin to ground			3.3	V
Reverse Breakdown Voltage	V_{BR}	I_t = 1mA Any I/O pin to ground	6.0			V
Reverse Leakage Current	I_R	V_{RWM} = 5V, T=25°C Any I/O pin to ground			1	μA
Clamping Voltage	Vc	I_{pp}=5A, t_p=8/20μs Any I/O pin to ground			15	V
Junction Capacitance	C_j	V_R = 0V, f = 1MHz I/O pin to GND			0.8	pF
		V_R = 0V, f = 1MHz Between I/O pins		0.3		pF

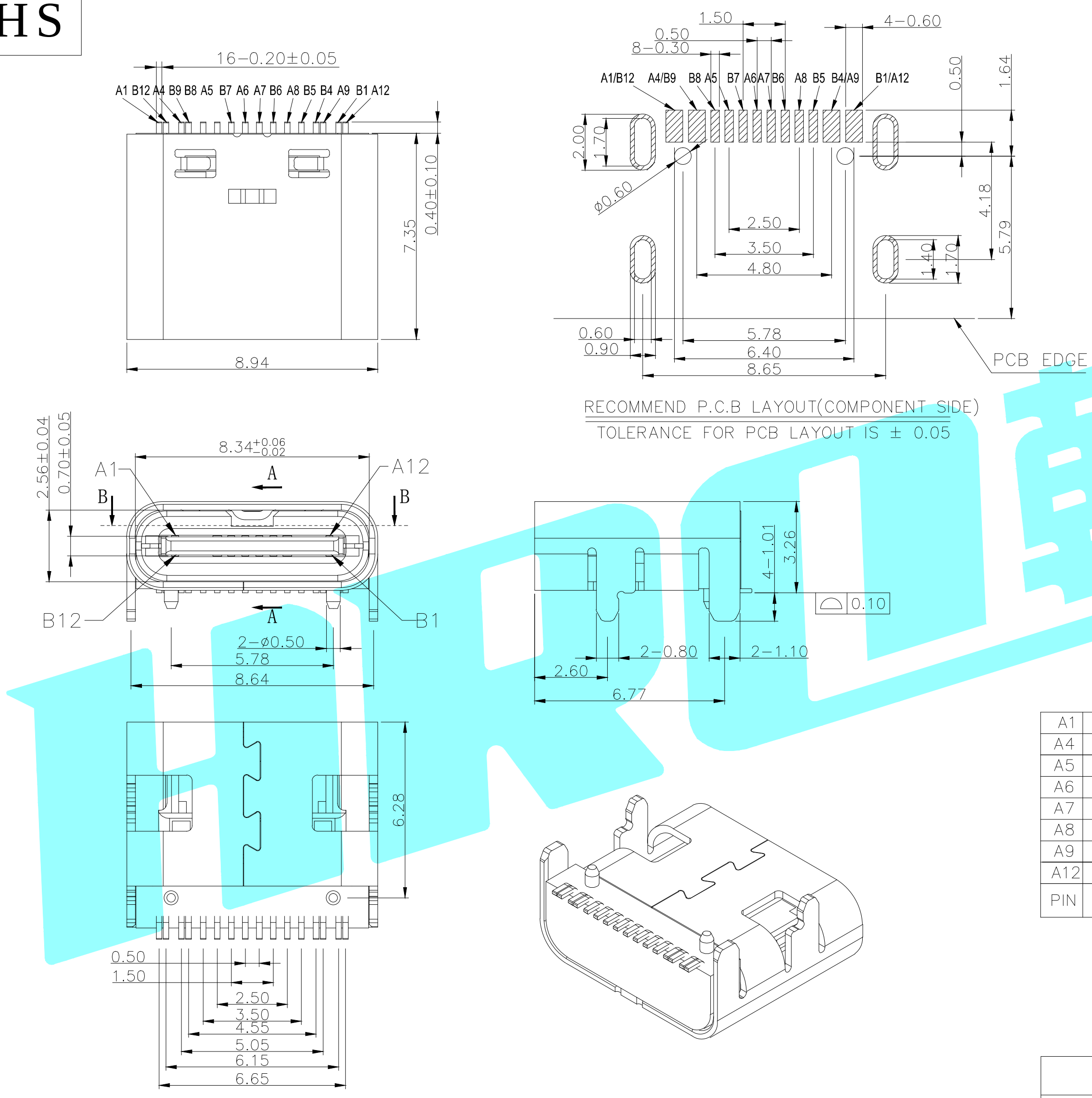
www.msksemi.com

Section 23
Hroparts

Designators	Part / Value	Qty
USB1,USB2	TYPE-C-31-M-12	2

RoHS evidence document: **2410010003_Korean-Hroparts-Elec-TYPE-C-31-M-12_C165948.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

ROHS



NOTE:

1.MATERIAL SPECIFICATION:
- 1.HOUSING: HIGH TEMPERATURE RESISTANT PLASTIC,UL94 V−0.
- 2.CONTACTS: COPPER ALLOY
- 3.MID PLATE: STAINLESS STEEL

2.PLATING SPECIFICATION:
- 2−1.CONTACTS:
  Ni 50u" MIN. UNDER PLATED OVER ALL.
  Au PLATED ON THE FUNCTIONAL AREA OF CONTACT.
  (GOLD PLATING THICKNESS FOLLOW THE P/N)
  GOLD FLASH PLATING ON SOLDER AREA
- 2−2.FRONT SHELL:
  Ni 30u" MIN. UNDER PLATED OVER ALL.

3.MECHANICAL PERFORMANCE,
- 3−1.INSERTION FORCE: 0.5~2.0kgf.
- 3−2.REMOVAL FORCE: 0.8kgf~2.0kgf.
- 3−3.DURABILITY: 10000 CYCLES.

4.ELECTRICAL PERFORMANCE,
- 4−1. CURRENT RATING: 5A
  VOLTAGE RATING: 20V
- 4−2.INSULATION RESISTANCE: 100MΩ MIN
- 4−3.DIELECTRIC WITHSTAND VOLTAGE,AC 100V FOR 1 MINUTE.

5. ENVIRONMENTAL PERFORMANCE:
OPERATING TEMPERATURE: −30°C~+80°C.

6.IR REFLOW:
THE PEAK TEMPERATURE ON THE BOARD SHALL BE MAINTAINED FOR 10 SECONDS AT 260°C.

PIN	SIGNAL NAME	PIN	SIGNAL NAME
A1	GND	B12	GND
A4	VBUS	B9	VBUS
A5	CC1	B8	SBU2
A6	DP1	B7	DN2
A7	DN1	B6	DP2
A8	SBU1	B5	CC2
A9	VBUS	B4	VBUS
A12	GND	B1	GND

ECN NO.	REV.	DATE.	DESCRIPTION.	CHANGE.	CHECK.	APPRO.
	A		NEW			

APPROVALS		DATE	HRO 韩荣® ELECTRONICS CO.,LTD	
DRAWN	LIUCAN	2020.12.08		
CHECKED			TITLE:	DETECTOR SWITCHS
APPROVALS			PART NO.	TYPE−C−31−M−12
TOLERANCES ARE	30~以上 ±0.30; 10~30 ±0.20; 5~10 ±0.15; ~5 ±0.10	ANGLE ±2°	UNIT: mm	SCALE: 1:1 — PROJ:
			DRAWING NO. —	SHEET 1 OF 1

Section 24
TXC

Designators	Part / Value	Qty
Y1	7M12000044	1

RoHS evidence document: **TXC-7M12000044.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 11 — RoHS-relevant pages only; the complete document is retained in the project repository.

TXC CORPORATION

4F, NO. 16, Sec. 2 Chung Yang S Rd., Peitou, Taipei, Taiwan.
TEL : 886-2-2894-1202 , 886-2-2895-2201 FAX : 886-2-2894-1206 , 886-2-2895-6207 www.txccorp.com

SPECIFICATION FOR APPROVAL

CUSTOMER	:	
PRODUCT TYPE	:	SMD SEAM SEALING X'TAL 3.2×2.5
NOMINAL FREQ.	:	12.000000MHz
TXC P/N	:	7M12000044
REVISION	:	A1
CUSTOMER P/N	:	
PM / SALES	:	
DATE	:	

CUSTOMER SIGNATURE & Date

(1) TXC requires one copy returned with signature and title of authorized individual that signifies acceptance of the attached specifications.

(2) Orders received and accepted by TXC after return of signed copy of specification will be produced per these specifications.

(3) Any changes to these specifications must be agreed upon by both parties and new revision of the Product Specification Sheet will be issued.

(4) Any issuance of purchase order prior to consigning back the Approval page of "Specification Sheets" from customers will be regarded as the agreement on the contents of these specifications.

MSL:Level 1
RoHS Compliant

FMT-DOC024 Issue Date: 02.20'14 VER.F

TXC CORPORATION

4F, NO. 16, Sec. 2 Chung Yang S Rd., Peitou, Taipei, Taiwan.

TEL : 886-2-2894-1202 , 886-2-2895-2201 FAX : 886-2-2894-1206 , 886-2-2895-6207 www.txccorp.com

PRODUCT SPECIFICATION SHEET

PRODUCT TYPE	:	SMD SEAM SEALING X'TAL 3.2×2.5
NOMINAL FREQ.	:	12.000000MHz
TXC P/N	:	7M12000044
REVISION	:	A1

PE/RD	QA	MFG
Robin Huang Robin Huang	Samson Xiong Samson Xiong	Jake Liu Jake Liu
4-Aug-14	*4-Aug-14*	*4-Aug-14*

NOTE:

(1)The green product standard set by TXC is based upon the international standards. Related information is publicly described on the TXC's Website, and updated regularly. The document is compliant with the latest green product quality system directives at the time.

(2)Revision "Sx" is for engineering samples only. PE/RD's approval required.

(3)Revision "Ax" is production ready. PE, QA and MFG's approval required

MSL:Level 1
RoHS Compliant

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TXC CORPORATION TXC P/N: 7M12000044 REVISION: A1 PAGE: 1

Rev	Revise page	Revise contents	Date	Ref.No.	Reviser
S1	N/A	Initial released	10-Jul-14	N/A	Xiaoyan Jiang
A1	N/A	S Turn A	4-Aug-14	N/A	Xiaoyan Jiang

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Section 25
JLCPCB

Designators	Part / Value	Qty
PCB / bare board	Printed circuit board (ENIG & lead-free HASL finish); incl. exposed copper of mounting holes & test points	-

RoHS evidence document: **rohs-certificate-of-conformity.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

DECLARATION OF CONFORMITY (DOC)

NO.: S251120003001-2

The following products have been tested by us and found compliance with the RoHS Directive 2011/65/EU Annex II amending Annex (EU) 2015/863 of CE Directive.

Customer	:	SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.
Address	:	27TH FLOOR, OLYMPIC BUILDING,SHANGBAO ROAD,FUTIAN DISTRICT,SHENZHEN CITY,GUANGDONG PROVINCE
Manufacturer	:	SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.
Sample Name	:	PCB
Sample Description	:	Lead-free-HASL
Sample Quantity	:	1
Report No	:	S251120003001-1
Test method	:	EN 62321-4:2014+A1:2017 & EN 62321-5:2014 & EN 62321-7-2:2017& EN 62321-6:2015 & EN 62321-8:2017

Signature:. Frank Wu

Technical director

Signed for and on behalf of

Soar Testing Certification (GuangDong) Co., Ltd.

Dec 01, 2025

Note: This certification is part of the full test report(s) and should be read in conjunction with it. The certificate is based on a single evaluation of one sample of above-mentioned products. It does not imply an assessment of the whole production and does not permit the use of the test lab logo.

Lab name: Soar Testing Certification (GuangDong) Co., Ltd.
Address: Room 1093, No.12, East of Houjie Avenue, Houjie, Dongguan, Guangdong, China
Post Code: 523945
Tel: (86-769) 38937518
E-mail: service@soar-cert.com
Website: http://www.soartestlab.com

DECLARATION OF CONFORMITY (DOC)

NO.: S251120003002-2

The following products have been tested by us and found compliance with the RoHS Directive 2011/65/EU Annex II amending Annex (EU) 2015/863 of CE Directive.

Customer	:	SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.
Address	:	27TH FLOOR, OLYMPIC BUILDING,SHANGBAO ROAD,FUTIAN DISTRICT,SHENZHEN CITY,GUANGDONG PROVINCE
Manufacturer	:	SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.
Sample Name	:	PCB
Sample Description	:	ENIG (Electroless Nickel/ Immersion Gold)
Sample Quantity	:	1
Report No	:	S251120003002-1
Test method	:	EN 62321-4:2014+A1:2017 & EN 62321-5:2014 & EN 62321-7-2:2017& EN 62321-6:2015 & EN 62321-8:2017

Signature:. Frank Wu

Technical director

Signed for and on behalf of

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